Coordination Compounds

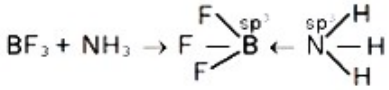
DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Coordination Compounds

Q1 **What will be the hybridisation of nitrogen and boron in the adduct of $H_3N \rightarrow BF_3$ adduct ?**

Sol: Both & N hybridization will be sp^3





Above reaction is known as addition reaction. Where two substances are connected into single entity and the product of addition reaction is known as addition compound.

Addition Compound

When aqueous solution of two sets is subjected to crystallize then new compound is formed which contain ions of both salts then this reaction is known as addition reaction and the product of their addition reaction is known as adduct/addition compound.

$$salt_1(aq.) + salt_2(aq.) \xrightarrow[\Delta]{\text{Crystallisation}} \text{addition compound or adduct}$$

$$KCl(aq.) + MgCl_2(aq.) \xrightarrow{\Delta} \underbrace{KCl.MgCl_2.6H_2O}_{\text{Addition compound or Adduct}}$$

Sometimes addition compound is also formed when salt is mixed in different–different solvent

$$CuSO_4 + NH_3 \;.\; \underbrace{CuSO_4}_{\text{(Salt)}} + \underbrace{NH_3}_{\text{(Solvent)}} \longrightarrow \underbrace{CuSO_4.4NH_3}_{\text{Addition Compound}} .$$

Types of addition compound: (Based upon ionization in water which measured by conductivity measurement)

(i) Double Salt: They are completely ionize in aqueous solution.
All ions show their presence in aqueous medium.
No ions loses its identity in aqueous medium.

Example :

$KCl.MgCl_2.6H_2O$ (carnallite)

$\downarrow H_2O$

Expected			Experimental		
Type of ion		No. of ions	Type of ion		No. of ions
K^+	→	1	K^+	→	1
Mg^{2+}	→	1	Mg^{2+}	→	1
Cl^-	→	3	Cl^-	→	3
		Total ions = 5			Total ions = 5

So, examples of double salt

(i) $KCl.MgCl_2.6H_2O$ (Carnallite)

→ No. of ions ⇒ 5

→ Type of ions ⇒ K^+, Cl^-, Mg^{2+}

(ii) Alums

Example → (Potash alum)

$K_2SO_4.Al_2(SO_4)_3.24H_2O$

No. of ions = 8

Type of ion ⇒ K^+, Al^{3+}, SO_4^{2-}

↓ ↓ ↓

Number → (2) (2) (4)

Important Concept of Alum :

General formula of alums

$M_2SO_4.N_2(SO_4)_3.24H_2O$

M → Monovalent ion

(NH_4^+, Alkali metal (IA) except LI^+)

N → Trivalent ion (Al^{3+}, Fe^{3+}, Cr^{3+}, Sc^{3+})

Alums are also written as $MN(SO_4)_2/2H_2O$ (Empirical formula)

4 metal ion → $24H_2O$

1 metal in → $6H_2O$

$[M(K_2O)_6]^+$ $[N(H_2O)_6]^{3+}$

↓

Li^+ is small, so it can not accommodate 6 water molecules so it does not form alum

(iii) Mohr's salt

$(NH_4)_2\,SO_4FeSO_4.6H_2O$ (Total ions = 5)

Type of ions → $\underset{(2)}{NH_4^+}, \underset{2}{SO_4^{2-}}, \underset{1}{Fe^{2+}}$

Complex salt :

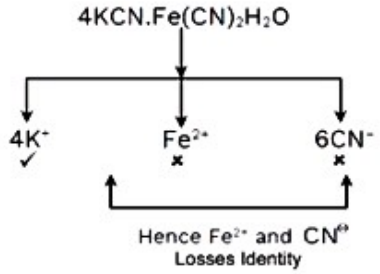
$4KCN + Fe(CN)_2 \rightarrow 4KCN.Fe(CN)_2$

Addition compound

$\downarrow H_2O$

Expected ions			Experimental ions		
Type of ion		No. of ions	Type of ion		No. of ions
K^+	→	4	K^+	→	4
CN^-	→	6	$[Fe(CN)_6]^{4-}$	→	1
Fe^{2+}	→	1			Total ions = 5
		Total ions = 11			

It means this addition compound does not ionize completely because conductivity measurement (Less number of ions as expected)
So complex salt does not ionize completely ion aqueous medium
Some of the ions loses their identity in aqueous medium



Ions which loses their identity are written in square brackets and are known as complex ion and compounds containing such ions are known as complex salt.

Example :

$[Fe(CN)_6]^{4-}$

↓

Complex ion

Complex compound → $K_4[Fe(CN)_6]$

In complex ion at least one metal ion is present which acts as Lewis acid & it is surrounded by neutral molecules/ions which act as Lewis base. Hence complex formation is a Lewis acid-base bond formation takes place. So complex salts are also known as co-ordination compounds.

General Introduction of Complex Salts

The compound which contains acidic(anionic) or basic(cationic) radicals is known as salt.

OR

The compound which is formed by the neutralisation reaction of acid and base is known as salt.

OR

The compound which on hydrolysis gives acid and base is known as salt.

The most appropriate classification in context of present chapter will be as follows:

Simple salt : The salt which contains one type of simple +ve ion and one type of simple -ve ion, Example, NaCl, KCl, Na_2CO_3 etc.

Mixed salt : The salt which contains more than one type of + ve ions or more than one type of -ve ions or both the ions are more than one type. Example, microcosmic salt ($NaNH_4HPO_4.4H_2O$), Bleaching powder Ca(OCl)Cl.

Double salt : When aqueous solution of two simple salts mixed in appropriate proportion are heated to dryness, the residue formed is called double salt.

Example :

Potash alum : $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

Mohr salt : $FeSo_4 \cdot (NH_4)_2 SO_4 \cdot 6H_2O$

Carnallite : $KCl \cdot MgCl_2 \cdot 6H_2O$

Points to remember

Aquestion solution of double salt give the characteristic tests of individual ions from which it is formed.

Complex salts or Coordination compounds or Addition compound or Molecular compounds :-Coordination compounds consists of a central atom (usually a metal atom/ion or sometimes an electropositive non metal like Be, B etc) is bonded with definite no. of species (called ligands) through coordinate bond.

Example :

(1) $K_4\left[\underset{\text{Central atom}}{Fe}\ \underset{\text{These 6 (CN) are ligands}}{(CN)_6}\right]$

(2) $\left[Ni(CO)_4\right].\left[Co(NH_3)_6\right]Cl_3, K_2[PtCl_6]$

$\left[Co(NH_3)_6\right]^{3+} \rightarrow$ Homoleptic complex

$\left[Co(NH_3)_4\right]^{+} \rightarrow$ Heteroleptic complex

Points to remember

Aqueous solution of complex salt does not given the characteristic test of all the ions by which the salt is formed.

Example :

aqueous solution of $K_4(Fe(CN)_6)$ does not given test of Fe^{2+} and CN^-

Basic Terminology:

(i) Charge on complex (charge on square bracket)

Complex $\Rightarrow$ Metal ion (Lewis acid) + Neutral molecule ions (Lewis base)

Example :

(i) Like $Fe^{2+} + 6CN^- \rightarrow [Fe(CN)_6]^{4-}$

(ii) $K_4\left[Fe(CN)_6\right] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$

(iii)

Serial Number	Complex	Charge
1.	$K_2[NiCl_4]$	−2
2.	$K_2[Ni(CN)_4]$	−2
3.	$K[CuCl_3]$	−1
4.	$K_2[NiF_6]$	−2
5.	$[Cr(en)_2Br_2]$	0
6.	$[Cr(gly)(en)_2]^{2+}$	+2
7.	$[Cr(en)_3]^{3+}$	+3
8.	$\left[CrCl_2(NH_3)_4\right]NO_3$	+1
9.	$[Crgly)_3]$	0

10.	$[CrF_6]^{3-}$	−3
11.	$[Cr(H_2O)_6]^{3+}$	+ 3

Counter ion:
Simple ions which neutralize the charge of complex ion are called as Counter ions.

Examples:

		Counter ion
(1)	Brown ring complex → $[Fe(H_2O)_5\text{-}NO]SO_4$	SO_4^{2-}
(2)	Zeise salt → $K[PtCl_3(C_2H_4)]$	K^+
(3)	Fischer's salt → $K_3[CO(NO_2)_6]$	K^+
(4)	$[CO(N_3)_6]Cl_3$	Cl^-
(5)	$[NiBr(H_2O)(en)_2]NO_3$	NO_3^-
(6)	$[CO(N)_2(NH_3)_4]Cl^-$	Cl^-
(7)	$[Fe(H_2O)_5NO]SO_4$	SO_4^{2-}
(8)	$[CrCl_2(NH_3)_4]NO_3$	NO_3^-
(9)	$K_2[NiF_6]$	K^+
(10)	$K_2[Ni(CN)_4]$	K^+
(11)	$K[CuCl_3]$	K^+
(12)	$K_2[NiCN_4]$	K^+

Central metal ion (CMI) :
Metal present in complex ion is Known as central metal ion because It is centre of complex ion surrounded by neutral molecules/ions
Central metal ion is act as Lewis acid.

Examples:

		C.M.I.
(1)	$[Fe(CN)_6]^{4-}$	Fe^{2+}
	$(-4 = x + 6\ (-1)\ x = 2)$	
(2)	$K_2[NiCl_4]$	Ni^{2+}

$2(+1) + x + 4(-1) = 0$

$x = +2$

(3) $K_2[Ni(CN)_4]$ Ni^{2+}

$2(+1) + x + 4(-1) = 0$

$x = +2$

(4) $K[CuCl_3]$ Cu^{2+}

$1(+1) + x + 3(-1) = 0$

$X = +2$

(5) $K_2[NiF_6]$ Ni^{4+}

$2(+1) + x + 6(-1) = 0$

$x = +4$

(6) $[CrCl_2(NH_3)]NO_3$ Cr^{3+}

$x + 2(-1) + 4(0) + 1(-1) = 0$

$x = +3$

(7) $[CrF_6]^{3-}$ Cr^{3+}

$x + 6(-1) = -3$

$x = +3$

(8) $[Cr(H_20)_6]^{3+}$ Cr^{3+}

$x + 6(0) = +3$

$x = +3$

(9) $\left[(Cr(C_2O_4)_3\right]^{3-}$ Cr^{3+}

$x + 3(-2) = -3$

$x = +3$

$\because C_2O_4^{2-} \rightarrow$ charge $\Rightarrow -2$

(oxalate)

(10) $[CrBr(NO_2)(H_2O)_4]^{+}$ Cr^{3+}

$x + 1(-1) + 1(-1) + 4(0) = +1$

$x = +3$

(11) $[Cr(NO)_6]^{-4}$ Cr^{2+}

$x + 6(-1) = -4$

$x = +2$

(12) $[Cr(CO)_6]$ Cr^{0}

$x + 6(0) = 0$

$x = 0$

Ligands ⇒

Example:	**Ligand**
$[Fe(CN)_6]^{4-}$	CN^-
$K[PtCl_3(C_2H_4)]$	Cl^{-1}, C_2H_4 (alkaene)

Ligand

⇓

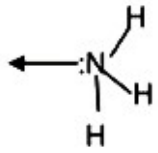
Lewis's base

⇓

Electron pair donor

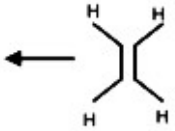
Lone pair donor

Ex: NH_3



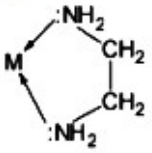
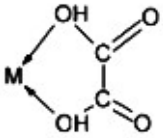
π-electron donor

Ex: C_2H_4, C_6H_6



Definitions

Neutral molecules/ions which acts as Lewis base in complex formation are known as ligands.

	Ligand	**Denticity**
(a)	$:NH_3$	1
(b)	H_2O	1
(c)	F^-	1
(d)	 Ethene-1,2-diamine (en)	2
(e)	 (OX^{2-})	2

Co-ordination Number (C.N.)

Number of lone pair accepted by a Central metal ion in complex formation is called as co-ordination number of Central metal ion.

Definitions

Number of lone pairs donated by a ligand to CMI is known as denticity.

Tips and Tricks :

Denticity is generally defined for those ligands where donation of lone pair occur like NH_3, H_2O

Denticity is property of ligands.

Points to remember

Denticity is property of a ligand C.N. is property of a Central metal ion.

Example:

$[Fe(NH_3)]^{2+}$

Denticity of NH_3 = 1

C.N. of CMI = 6

* **Important coordination Number.**

			C.N.
(a)	+3 ion of 3d series (like Cr^{3+})	→	6
(b)	+2 ion of 3d series (like Cr^{3+}) (like $Fe^{2+}, Mn^{2+}, Ni^{2+}$ etc.)	→	4/5/6

(c) Ni^{2+} (with Cl^-, CN^-) = 4

With NH_3, H_2O = 6

(d) $Zn^{2+}, Cd^{2+}, Hg^{2+}$ = 4

(e) Au^+, Ag^+ = 2

(f) Pd^{2+}, Pt^{2+}, Au^{3+}, Ag^{3+} = 4

(g) Pd^{4+}, Pt^{4+} = 6

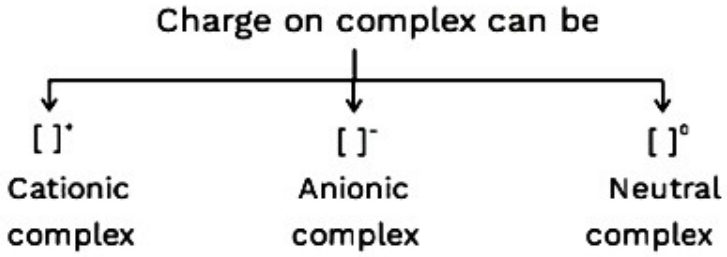
Q2 **Find the formula of Ni^{2+} complexes with CN^-, Cl^-, H_2O, NH_3 individually?**

Sol: (i) $[Ni(CN)_4]^{2-}$ (ii) $[NiCl_4]^{2-}$

(iii) $[Ni(H_2O)_6]^{2+}$ (iv) $[Ni(NH_3)_6]^{2+}$

Points to remember

Maximum C.N. of 3d series = 6
General C.N. of inner transition
=8 + 12 (f-block)



(A) So, on the basis of charge complex can be

1. Simple cation + complex anion

Example :

Sodium nitroprusside $\rightarrow Na_2[Fe(CN)_5NO]$

2. Complex cation + simple anion

Example :

Brain ring complex $\rightarrow [Fe(H_2O)_5NO]SO_4$

3. Complex cation + complex anion

Example :

$[Pt(CH_3)_4][PtCl_4]$, $[Cr(NH_3)_6][Cr(CN)_6]$

In these complexes Charge on complex ions is calculated by hit and trial

	C.N.	**C.M.I.**	**Charge on complex**
$[Pt(NH_3)][PtCl_4]$	4	Pt^{2+}	$[Pt(NH_3)_4]^{3+}[PtCl_4]^{2-}$
$[Cr(NH_3)_6][Cr(CN)_6]$	6	Cr^{3+}	$[Cr(NH_3)_6]^{3+}[Cr(CN)_6]^{3-}$

4. Neutral Complex

Example :

(i) Cis-platin (Anticancer)

Formula $\rightarrow$ Cis-$[Pt(NH_3)_2Cl_2]$

(ii) Wilkinson catalyst $[Rh(PPh_3)_3Cl]$

(B) On the basis of type of donation complex can be

(i) σ-complexes – They have only lone pair donor.

Example :

Cis Platin $\rightarrow$ Cis$[Pt(CH_3)_2Cl_2]$

(ii) π-complexes – They have atleast one π-donor

Example :

Zeise's salt $\rightarrow K[PtCl_3(\pi - C_2H_4)]$

Example :

(1) Fischer's Salt

$\underbrace{K_3}_{\text{Ionisation sphere}}$ $\underbrace{[Co(NO_2)_6]}_{\text{Co-ordination sphere}}$

(2) Brown ring complex

$\underbrace{[Fe(H_2O)_5NO]}_{\text{Coordination spyhere}}$ $\underbrace{SO_4}_{\text{Ionization sphere}}$

Points to remember

Co-ordination sphere	Ionization Sphere
Space where co-ordinate bond is formed	Space where co-ordinate bond is formed
It is non-ionizable	It is ionizable
It is inside the square bracket	It is outer part of square bracket

Q3 **Find the number of co-ordination sphere in $[Pt(NH_3)_4][PtCl_4]$?**

Sol: No. of Co-ordination sphere = 2

$[Pt(NH_3)_6]^{2+}$, $[PtCl_4]^{2-}$

(C) On the basis of type of ligand complexes can be:

Homoleptic Complex (They have only one type of ligand in coordination sphere).

Example :

(i) Fischer's Salt → $K_3[Co(NO_2)_6]$

(ii) Heterolytic Complex: (They have more than one type of ligands)

Example :

(i) Brain ring complex → $[Fe(H_2O)_5NO]SO_4$

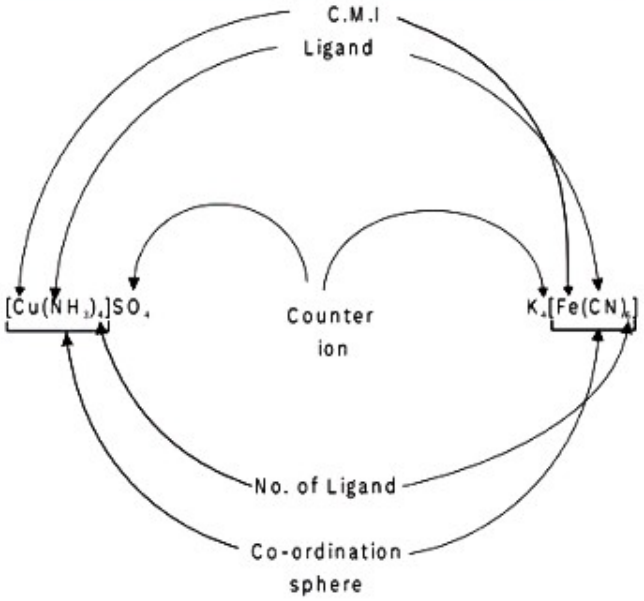
(ii) Zeise salt → → $K[PtCl_3(\pi - C_2H_4)]$

(iii) Cis Platin → $Cis[Pt(CH_3)_2Cl_2]$

(iv) Wilkinson catalyst → $[Rh(PPh_3)_3Cl]$

(v) Sodium nitroprusside → $Na_2[Fe(CN)_5NO]$

Tips and Trick



Points to remember

In $[Pt(NH_3)_4][PtCl_4]$ each coordination sphere is separately homoleptic or neuroleptic.

$\underbrace{[Pt(NH_3)_4]}$ → Homoleptic

$\underbrace{[PtCl_4]}$ → Homoleptic

Coordination Polyhedra:
Geometry around central metal ion in a complex is known as co-ordination polyhedral.

Example :
Linear, square planar, tetrahedral etc.

Heading :Complex on the basis of number of CMI in a co-ordination sphere.

Mononuclear complex

Polynuclear complex

→ It has one CMI
Ex : $[Fe(CO)_5]$

→ It has more than one CMI
Ex : $[Fe_2(CO)_9]$

Points to remember

Co-ordinate bonds are directional bonds as like covalent bonds, so ligands are arranged in particular geometry around CMI

Co-ordination Entity:
Species in square bracket represents coordination entity.

Werner's Theory:
According to Werner's theory CMI shows two types of Valency

(i) Primary Valency(P.V.) : It represents oxidation state(OS) of C.M.I.

(ii) Secondary Valency (S.V) : It represents co-ordination number (C.N.) of CMI.

Example :

(i) $[Cr(NH_3)_6]Cl_3$

$x + 6(0) + 3(1) = 0$

$x = +3$

P.V = 3
O.S. = 3
S.V. = 6
C.N. = 6

(ii) $[Cr(NH_3)_6]Cl_3$ — P.V = 3; O.S. of C.M.I. = +3; S.V. = 6; C.N. of C.M.I. = 6

In Cr^{3+} +3 charge is satisfied by Cl^- (anion)
C.N. = 6 is satisfied by NH_3 (ligand)

(iii) $[Cr(NH_3)_5]Cl_2$

P.V. = 3
S.V. = 3
P.V. is satisfied by Cl^-
S.V. is satisfied by NH_3 & Cl^-

Difference in primary and secondary valency ?

Primary Valency(P.V.)	**Secondary Valency**(S.V.)
1. It represent oxidation State (O.S.) of CMI	**1.** It represents co-ordination number.
2. It is satisfied by anions	**2.** It is satisfied by ligands
3. It is non-directional (Ionic bond)	**3.** It has directional nature (co-ordinate bond)
4. Generally ionizable	**4.** Non ionizable

(i) Neutral ligand satisfy only Secondary Valency
(ii) Counter ion satisfy only Primary Valency.
(iii) Anionic ligand satisfy both Primary Valency and Secondary Valency.

Werner formula	**Modern formula**	**Electrolyte cation : Anion**
$Cr.6NH_3Cl_3$	$[Cr(NH_3)Cl_3$	1 : 3
$Cr.5NH_3Cl_3$	$[Cr(NH_3)_5Cl]Cl_2$	1 : 2
$Cr \cdot 6NH_3Cl_3$	$[Cr(NH_3)_6]Cl_3$	1 : 3
$Cr \cdot 5NH_3Cl_3$	$[Cr(NH_3)_5\ Cl]Cl_2$	1 : 2
$Cr \cdot 4NH_3Cl_3$	$[Cr(NH_3)_4\ Cl_2]Cl$	1 : 1
$Cr \cdot 3NH_3Cl_3$	$[Cr(NH_3)_3\ Cl_3]$	non electrolyte

Werner formula	Modern formula	Electrolyte cation : Anion
$Pt \cdot 6NH_3 \cdot Cl_4$	$[Pt(NH_3)_6]Cl_4$	1 : 4
$Pt \cdot 5NH_3 \cdot Cl_4$	$[Pt(NH_3)_5\,Cl]Cl_3$	1 : 3
$Pt \cdot 4NH_3 \cdot Cl_2$	$[Pt(NH_3)_4]Cl_2$	1 : 2
$Pt \cdot 4NH_3 \cdot Cl_4$	$[Pt(NH_3)_4\,Cl_2]Cl_2$	1 : 2
$Pt \cdot 3NH_3 \cdot Cl_4$	$[Pt(NH_3)_3\,Cl_3]Cl$	1 : 1
$Pt \cdot 2NH_3 \cdot Cl_4$	$[Pt(NH_3)_2\,Cl_4]$	non electrolyte

Example :

Molar Conductivity $\rightarrow [Cr(NH_3)_6]Cl_3 > [Cr(NH_3)_5Cl]Cl_2$

$\downarrow \qquad\qquad \downarrow$

Total ions $\rightarrow \quad 4 \qquad\quad 3$

$[Cr(NH_3)_6]^{3+} + 3Cl^- \quad [Cr(NH_3)_5]^{+} + 2Cl^-$

Points to remember

Complex compound giving higher number of ions in solution will show greater molar conductivity.

How to convert warner's formula in Modern formula :-

(i) By use of C.N. :-

Example :

$Pt \cdot 6NH_3 \cdot Cl_4 \rightarrow [Pt(NH_3)_6]Cl_4$

($\because$ C.N. of Pt^{4+} is 6)

(ii) Conductivity / Number of ions in aqueous medium :-

Example :

If $Pt \cdot 3NH_3 \cdot Cl_2$ gives two ions in aqueous medium then formula of complex will be $[Pt(NH_3)_3Cl]Cl$

(iii) By precipitation of counter ion :-

Example :

If $Cr \cdot 5NH_3 \cdot Cl_3$ on reaction with excess of $AgNO_3$ gives 2 moles of AgCl then formula of complex will be $[Cr(NH_3)_5Cl]Cl_2$

($\because$ Counter ion ($2Cl^-$) will be precipitated as AgCl)

Q4 **$Cr \cdot 5NH_3 \cdot SO_4 \cdot Cl$ complex exist in two isomers A and B, where A gives white precipitate of AgCl when react with $AgNO_3$ and B gives white precipitate of $BaSO_4$ when react with $BaCl_2$ find A and B ?**

Sol: $A \rightarrow [Cr(NH_3)_5 SO_4]Cl$

$B \rightarrow [Cr(NH_3)_5 Cl]SO_4$.

Q5 **If 1 mole of $[Co(NH_3)_5 SO_4]Cl$ and 1 mole $[Co(NH_3)_5 Cl]SO_4$ are passed from cation exchanger then we get acids. Find the number of moles of NaOH required to neutralize the acid obtained from cation exchanger.**

Sol:

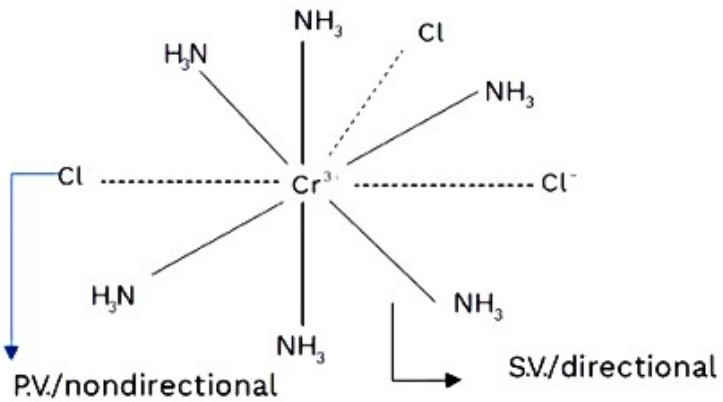
$[CO(NH_3)_5 SO_4]Cl$ ↓ Cation exchanger → 1 mole HCl ⇓ So 1 mole NaOH is required to neturalized 1 mole HCl

1 mole $[CO(NH_3)_5 Cl]SO_4$ ↓ Cation exchanger → 1 mole H_2SO_4 ⇓ So, 2 mole NaOH is required to neturalized 1 mole H_2SO_4

werner's representation:

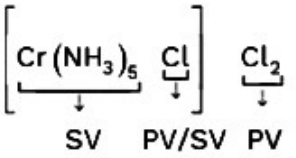
1. Primary valency is represented by dotted line(.....)
2. Secondary valency is represented by solid line(_____)
3. Ligand which satisfy both primary valency as well as secondary valency are represented by solid with dotted line (..... / _____)

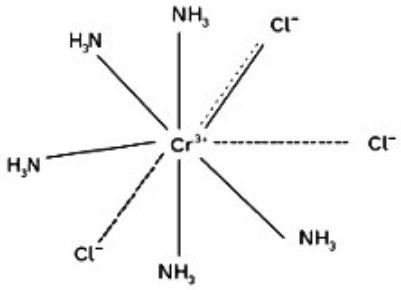
Example: $Cr.6NH_3.Cl_3 \longrightarrow [Cr(NH_3)_6]Cl_3$



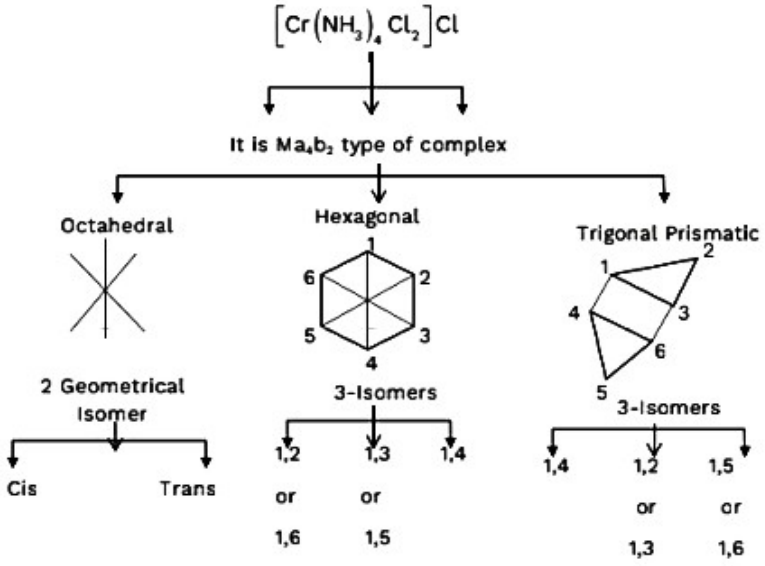
Example :

$Cr.5NH_3Cl_3$





Werner at that time stated that geometry of $[Cr(NH_3)_4Cl_2]Cl$ is octahedral as two isomers were obtained from this complex which can be justified by Octahedral geometry as follows:



Conclusion : As only two isomers were obtained in $[Cr(NH_3)_4Cl_2]Cl$, so it should have octahedral geometry

Fill the table according to WENER'S THEORY-

S.No.	Werner Formula	Modern Formula	No. of Ions in Aqueous Medium	Electrolyte
1.	$Cr.6NH_3.Cl_3$			
2.	$Cr.6NH_3.Cl_3$			
3.	$Cr.4NH_3.Cl_3$			
4.	$Cr.3NH_3.Cl_3$			
5.	$Pt.6NH_3.Cl_4$			
6.	$Pt.5NH_3.Cl_4$			
7.	$Pt.4NH_3.Cl_4$			
8.	$Pt.3NH_3.Cl_4$			
9.	$Pt.2NH_3.Cl_4$			
10.	$Pt.4NH_3.Cl_2$			
11.	$Pt.3NH_3.Cl_2$			
12.	$Pt.2NH_3.Cl_2$			
13.	$NiCl_2.6H_2O$			
14.	$PtCl_4.2HCl$			

Solutions-

S.No.	Werner Formula	Modern Formula	No. of Ions in Aqueous Medium	Electrolyte
1.	$Cr.6NH_3.Cl_3$	$[Cr(NH_3)_6]Cl_3$	4	1 : 3
2.	$Cr.5NH_3.Cl_3$	$[Cr(NH_3)_5Cl]Cl_2$	3	1 : 2
3.	$Cr.4NH_3.Cl_3$	$[Cr(NH_3)_4Cl_2]Cl$	2	1 : 1
4.	$Cr.3NH_3.Cl_3$	$[Cr(NH_3)_3Cl_3]$	0	Nonelectrolyte
5.	$Pt.6NH_3.Cl_4$	$[Pt(NH_3)_6]Cl_4$	5	1 : 4
6.	$Pt.5NH_3.Cl_4$	$[Pt(NH_3)_5Cl]Cl_3$	4	1 : 3
7.	$Pt.4NH_3.Cl_4$	$[Pt(NH_3)_4Cl_2]Cl_2$	3	1 : 2

S.No.	Werner Formula	Modern Formula	No. of Ions in Aqueous Medium	Electrolyte
8.	$Pt.3NH_3.Cl_4$	$[Pt(NH_3)_3Cl_3]Cl$	2	1 : 1
9.	$Pt.2NH_3.Cl_4$	$[Pt(NH_3)_2Cl_4]$	0	Nonelectrolyte
10.	$Pt.4NH_3.Cl_2$	$[Pt(NH_3)_4]Cl_2$	3	1 : 2
11.	$Pt.3NH_3.Cl_2$	$[Pt(NH_3)_3Cl]Cl$	2	1 : 1
12.	$Pt.2NH_3.Cl_2$	$[Pt(NH_3)_2Cl_2]$	0	Nonelectrolyte
13.	$NiCl_2.6H_2O$	$[Ni(H_2O)_6]Cl_2$	3	1 : 2
14.	$PtCl_4.2HCl$	$H_2[PtCl_4]$	3	2 : 1

Classification of Ligands

(1) On the basis of charge:

(A) Cationic Ligands:

Example : NO^+, $N_2H_5^+$

(B) Anionic Ligands:

Example : $Cl^-, SO_4^{2-}, S_2O_4^{2-}$

(C) Neutral Ligands:

Example : $NH_3, H_2O, en, CO, NO, PPh_3, C_2H_4$

Points to remember

$NH_4^{\oplus}$ is not a ligand (as it does not have lone pair)

Naming of Ligands:

(i) Cationic Ligands

Here suffix 'ium' is used.

$NO \rightarrow$ Nitrosyl $\qquad N_2H_4 \rightarrow$ Hydrazine

$NO^+ \rightarrow$ Nitrosylium $\qquad N_2H_5^{\oplus} \rightarrow$ Hydrazinium

(ii) Anionic Ligand

Counter ion

(1) ide suffix $\rightarrow$ Ido

Example :

Cl^-

Chloride $\rightarrow$ Chlorido

(2) ate $\rightarrow$ ato

Example :

SO_4^{2-}

(Sulphate) Sulphato

(3) ite → ito

SO_3^{2-}

Sulphite Sulphito

Points to remember

Only { ide, ite, ate } are charged to { ido, ito, ato }

CH_3^- — Counter ion → methyl; Ligand → methyl

iii) Neutral Ligand:

Common name is always used (Except)

(a) H_2O →aqua

(b) NH_3 → ammine

(c) CO → carbonyl

(d) NO →Nitrosyl

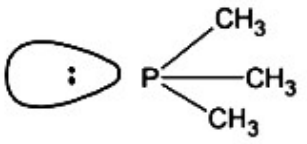
Example (Neutral ligands)

Ligand	**Name**
$CH_3 - NH_2 \longrightarrow$	Methylamine
$C_6H_6 \longrightarrow$	Benzene
$PH_3 \longrightarrow$	Phosphine
$PPh_3 \longrightarrow$	Triphenylphosphine
$PMe_3 \longrightarrow$	Trimethylphosphine
$NMe_3 \longrightarrow$	Trimethylamine

Where,

Me = CH_3

Like = $P(Me)_3$



(Trimethylphosphine)

On the basis of denticity:

Denticity → Total no of lone pair donated by a ligand to a central metal ion.

(1) Monodentate Ligand:

(a) (Having one donor site:)

* Remember these monodentate ligands

Ligand		**Name of ligand**
$F^{\ominus}$	⟶	fluorido

$Cl^{\ominus}$	⟶	chlorido
$Br^{\ominus}$	⟶	Bromido
$I^{\ominus}$	⟶	Iodido
$H^{\ominus}$	⟶	Hydrido
O^{2-}	⟶	Oxido
OH^{-}	⟶	hydroxide
H_2O	⟶	aqua
NH_3	⟶	ammine
NH_3^-	⟶	amido
NH_2^-	⟶	Imido
S^{2-}	⟶	Sulphido
PH_3	⟶	Phosphine
Me_2S	⟶	Dimethylthioether
PPh_3	⟶	Triphenylphosphine
PMe_3	⟶	Trimethyl phosphine
NMe_3	⟶	Trimethylamine
$NMeH_2$	⟶	Methylamine
CH_3^-	⟶	Methyl
$C_2H_5^-$	⟶	Ethyl
$C_6H_5^-$	⟶	Phenyl

(b) (Having two donor site:)
Form 3-membered ring with CMI:

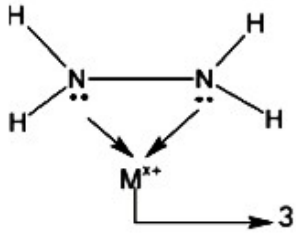


Angle strain high so ring unstable

⇓

So donate only one l.p. at a time
(Monodentate ligand)

Example : N_2H_4



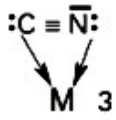
3 membered ring

↓

unstable → So donate only one l.p. at a time

So N_2H_4 is Monodentate ligand

(2)

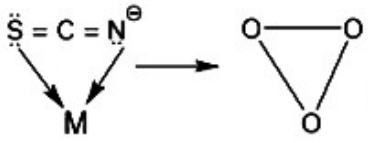


3 membered ring

↓

unstable → So, $CN^{\ominus}$ is also monodentate ligand

(3)



3 – corners

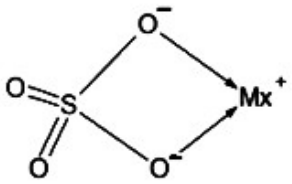
↓

counted as 3 membered ring

$SCN^{\ominus}$ is also monodentate ligand

(4) Form 4 -membered ring with CMI

Example: SO_4^{2-}



4 membered ring is stable in comparison to 3 -membered ring but still it is Unstable due to bond angle (90°) they are generally monodentate ligands but as per requirement of Co-ordination number of CMI they Can act as bidentate and such ligands are called as flex dentate ligands. .

Points to remember

If 5 membered/6 membered/7 membered ring is formed with CMI by ligands they can easily at as bidentate ligand.

Example:

$[Cr(NH_3)_5SO_4]^+$ ⇓ Here Cr^{3+} ⇒ C.N. = 6

5 (by NH_3) 1 ($bySO_4^{2-}$) ↓ Act as Monodentate ligand

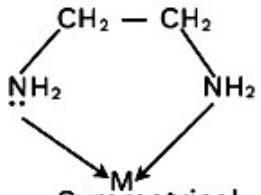
$[Cr(NH_3)_5SO_4]^+$ ⇓ Here Cr^{3+} ⇒ C.N. = 6

4 (by NH_3) 2 ($bySO_4^{2-}$) ↓ Act as bidentate ligand

Bidentate/Didentate ligand: Here a ligand donates two lone pair to CMI

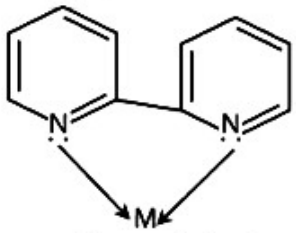
(1) ethan-1, 2-diamine (en)

or Ethylenediamine



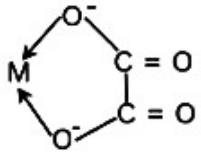
Symmetrical
(5-membered ring N, N-donor)

(2) dipyridine(dipy)



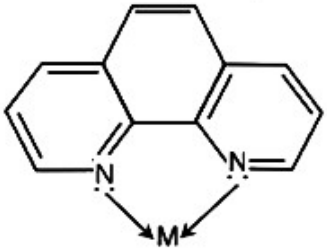
Symmetrical
(5-membered ring N, N-donor)

(3) Oxalato (OX^{2-}) or $C_2O_4^{2-}$



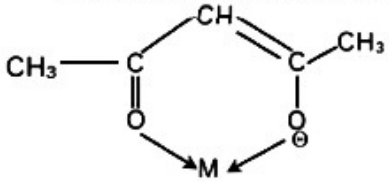
Symmetrical
(5-membered ring O, O-donor)

(4) Phenanthroline (Phen)



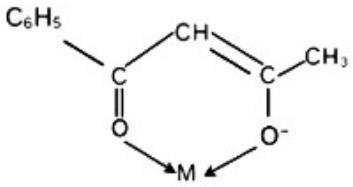
Symmetrical
(5-membered ring N, N-donor)

(5) acetylacetonato ($acac^-$)



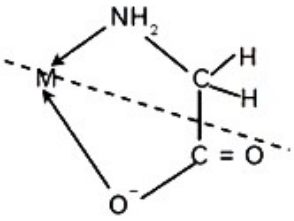
(6-membered ring 0,0-donor smmetrical)

(6) Benzylacetonato $(bcac^-)$



(6-membered ring o,o -donor Unsymmetrical)

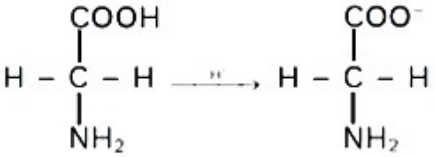
(7) Glycinato (gly^{-1})



(5-membered ring N, O-donor unsymmetrical)

Glycine $\xrightarrow{OH^\ominus}$ glycinate

Gly (aminoacid) gly



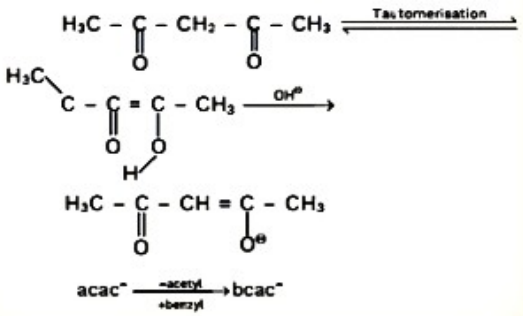
Glycine glycinate

↓

It is only amino acid which is optically inactive

Points to remember

Acetylacetone



Q6 **Find the value of [(a + b – c)d] ÷ e**
Where
a = number of symmetrical ligands
b = number of unsymmetrical ligands
c = number of N, N-donor
d = number of N, O-donor
e = number of O, O-donor
among gly^{-1}, en, OX^{-2}, dipy, phen, $acac^{-1}$, $bcac^{-1}$

Sol: a = 5 → en, OX^{2-}, dipy, phen, $acac^{-1}$
b = 2 → gly^{-1}, $bcac^{-1}$
c = 3 → en, dipy, phen
d = 1 → gly^{-1}
e = 3 → OX^{2-}, $acac^{-1}$, $bcac^{-1}$

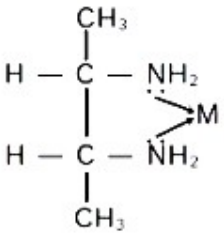
$$\Rightarrow \frac{(5+2-3)1}{3} = \frac{4}{3}$$

Ans. $\frac{4}{3}$

Q7 **Find the value of 'x' where x = number of ligand having sp^2 hybridized carbon-atoms among**
en, gly^{-1}, OX^{-2}, dipy, phen, $acac^{-1}$, $bcac^{-1}$?

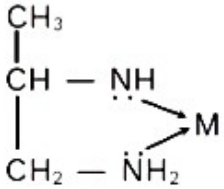
Sol: 6(gly^{-1}, OX^{-2}, dipy, phen, $acac^{-1}$, $bcac^{-1}$)

8. Butane-2,3-diamine (bn)



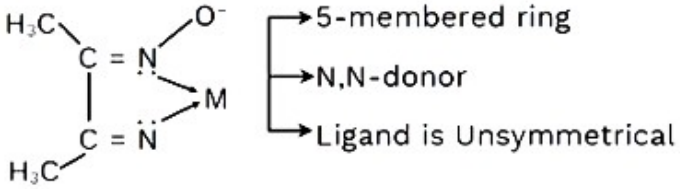
(N, N-donor, 5-numbered symmetrical)

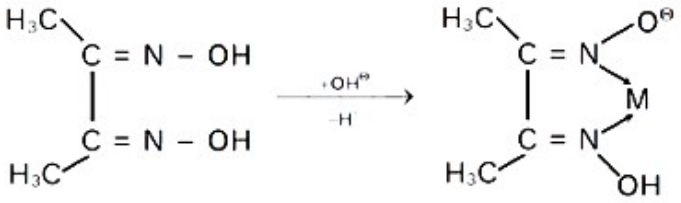
9. Propane-1, 2-diamine(pn)



(N, N-donor, unsymmetrical, 5-membered)

10. Dimethylglyoximato (dmg^{-1})

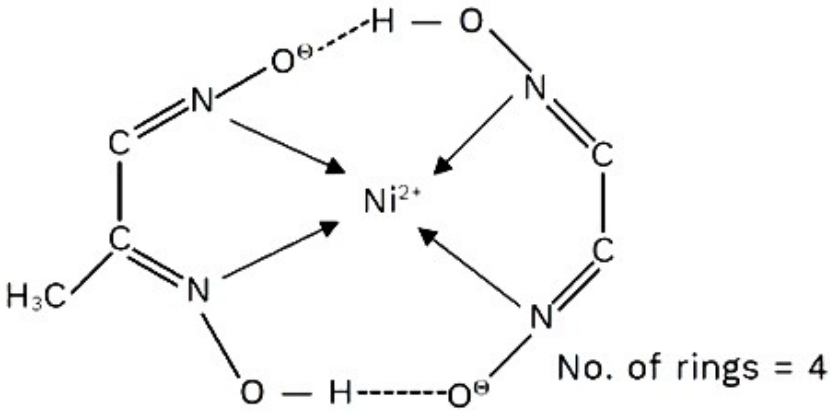




Dimethylglyoxime (DMG)

N,N-donor, dmg^{-1} No. of rings = 1

Detection of Ni^{2+} (specific test) $Ni^{2+} \xrightarrow[\text{or } (dmg+OH^-)]{dmg^{-1}} [Ni(dmg)_2]\downarrow$ Rosy Red ppt.

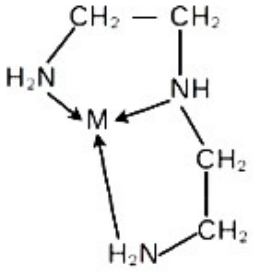


No. of rings = 4

due to H-bond Stability increases, so in dmg^{-1} N, N–atoms act as donor site.

Ligand	No. of member in ring	Donor site	Symmetrical or Unsymmetrical
dmg^{-1}	5	N,N	Symmetrical
pn	5	N,N,	Unsymmetrical
bn	5	N,N	Unsymmetrical
dipy	5	N,N	Symmetrical
OX^{2-}	5	O,O	Symmetrical
phen	5	N,N,	Symmetrical
$acac^{-1}$	6	O,O	Symmetrical
$bcac^{-1}$	6	O,O	Unsymmetrical
gly^{-1}	5	N,O	Unsymmetrical
en	5	N,N	Symmetrical

Tridentate:
Diethylenetriamine (dien)
No. of rings = 2

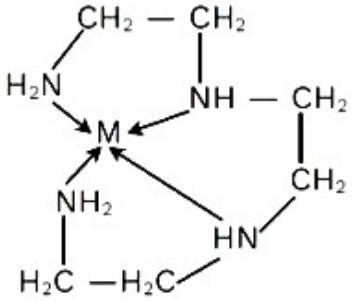


(N, N, N-donor)

Points to remember

Number of rings formed by a ligand = Dencity – 1

Tetradentate:
triethylenetetramine (trien)

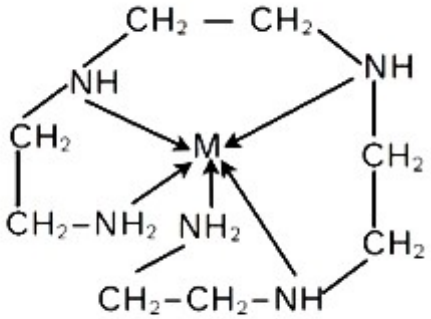


No. of rings = 3

Q8 **What will be number of rings in $[Cr(en)_3]^{3+}$?**

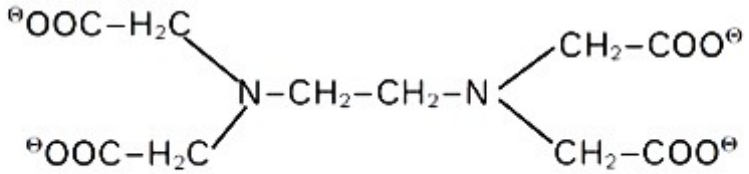
Sol: en → bidentate ⇒ denticity = 2
No. of rings formed by one en = 1
So total number of rings in $[Cr(en)_3]^{3+}$ = 3 × 1 = 3

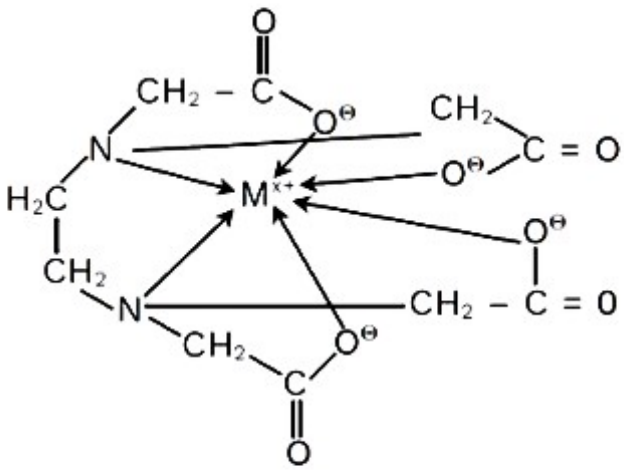
Pentadentate legend
tetraethylenepentaamine (teten)



No. of rings = 4

Hexadentate:
edta^{4-} (ethylenediaminetetracetate ion)





Ambidentate Ligand

Anionic legends having two or more donor site but form co-ordinate bond by only one site at a time with CMI is know as ambidentate ligand.

$:AB:^{\ominus}$ **Ex:** $:CN:^{\ominus}$

$A \leftarrow AB^-$ $M \leftarrow CN^{\ominus}$

Or or

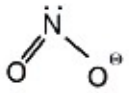
$M \leftarrow BA^-$ $M \leftarrow NC^{\ominus}$

Points to remember

- 1st letter of legend represent donor site
- In the naming of such ligands, donor site first letter is written as suffix.

Example :

Ligand		**Name of legend**
$CN^{\ominus}$	→	Cyanido – C
$NC^{\ominus}$	→	Cyanido – N
$OCN^{\ominus}$	→	Cyanato – O $(\dot{O} = C = \dot{N}^{\ominus})$
$NCO^{\ominus}$	→	Cyanato – N
$SCN^{\ominus}$	→	thiocyanato – S $(S = C = \dot{N}^{\ominus})$
$NCS^{\ominus}$	→	thiocyanato – N
$NO_2^{\ominus}$	→	Nitro / Nitrito – N Monodentate
$ONO^{\ominus}$	→	Nitrito – O



$O_2N^{\ominus}$ → Nitro(bidentate) (NO_2^-)

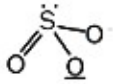
$NO_2^{\ominus}$ is a flexidentate legend

$NOS^{\ominus}$ → thionitro-N

$ONS^{\ominus}$ → thionitro-O

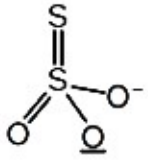
$SNO^{\ominus}$ → thionitro-S

SO_3^{2-} → Sulphato-S SO_3^{2-} (Sulphate)



OSO_2^{2-} → Sulphato-O

$S_2O_3^{2-}$ → thiosulphato-S thiosulphato $(S_2O_3^{2-})$



$OS_2O_2^{2-}$ → thiosulphato-O

Flexidentate Ligand

Ligands which form 4-membered ring with CMI can act as flexidentate ligand

Example: All oxoanions having single central atom

$NO_2^-, CO_3^{2-}, NO_3^-, PO_4^{3-}, SO_4^{2-}, SO_3^{2-}, ClO_4^-, CH_3COO^{\ominus}$

Chelating Ligand:

Didentate/Bidentate or polydentate legends (donate more than 2*l*.p.) when co-ordinate with a CMI, then they form ring like structure, such rings are called as chelate rings and phenomenon is called as chelation.

Effective atomic number (E.A.N.)

E.A.N. is number of electrons in central metal ion after complex formation.

$K_4[Fe(CN)_6]$

(total electron on Fe = 26 – 2 + 6 × 2 = 36 from each legend (CN^-) monodentate)

Here atomic number of Fe is 26 So Fe has 26 electron ⇒ But Fe has + 2 charge (O.S.) ⇒ So we have subtracted 2 (26 – 2)

⇒ Also 1$CN^{\ominus}$ has donated 2 electrons (1 l.p.) and here total 6$CN^{\ominus}$ are present, so added 12e (6 × 2)

So total electrons on Fe is 26 – 2 + 6 × 2 = 36

$k_3[Fe(CN)_6]$ 26 – 3 + 6 × 2 = 35

$[Ni_1Cl_4]^{2-}$ 28 – 2 + 4 × 2 = 34

$K_3[Cu(CN)_4]$ 29 – 1 + 4 × 2 = 36

$[Cr\ (NH_3)_4\ (H_2O)_2]Cl_3$

↑ ↑ ↑

24–3+(4×2)+(2×2) = 33

$[\ Cr(en)\ (H_2O)_2(NH_3)_2]Cl_3$

↑ ↑ ↑

24–3+(1×4)+(2×2)+(2×2) = 33

↓

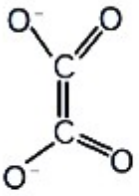
Bidentate

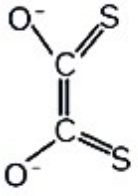
$[Co(C_2O_4)(en)\ (gly)]^0$

Points to remember

Generally ambidentate ligands are monodentate

$C_2O_4^{2-} \xrightarrow[+2S]{-2O}$ dithiooxalato $(C_2S_2O_2^{2-})$





$O_2C_2S_2^{2-}$ → dithiooxalato-O, O'

$S_2C_2O_2^{2-}$ → dithiooxalato–S, S'

$27 - 3 + \underbrace{4+4+4}_{\text{Bidentate}} = 36$

Charge on Co^{3+}

	l.p.	e^-
Monotentate	1	2
bidentate	2	4
tridentate	3	6
2 bidentate	2×2	4×2

Classification of complex on the basis of stability

$Fe^{2+}_{(aq.)} + 6CN^- K_f \rightleftharpoons [Fe(CN)_6]^{4-}$

$[Fe(H_2O)_6]^{2+} + CN^- \overset{K_1}{\rightleftharpoons} [Fe(H_2O)_5CN]^+$

$[Fe(H_2O)_5CN]^+ + CN^- \overset{K_2}{\rightleftharpoons} [Fe(H_2O)_4(CN)_2]$

$\vdots \qquad\qquad \vdots$

$[Fe(H_2O)(CN)_5]^{3-} + CN^- \overset{K_6}{\rightleftharpoons} [Fe(CN)_6]^{4-}$

$\boxed{K_f = k_1.k_2....k_6}$

$k_1, k_2.....,k_6$ are equilibrium constant

$k_f \rightarrow$ overall equilibrium constant or formation constant

some of the complexes have high value of k and some of the complex have low value of k

If $\underbrace{K \text{ is high} \Rightarrow [\text{product}] >> [\text{Reactant}]}_{\Downarrow}$ $\Rightarrow$ Low availability of ion

Here stability of complex is high

$\Rightarrow$ If $\underbrace{K \text{ is less} \Rightarrow [\text{Product}] << [\text{Reactant}]}_{\Downarrow}$ $\Rightarrow$ High availability of ion

stability of complex is less

($\because$ stability of complex α k)

$Cu^{2+} + CN^- \rightarrow \overset{+2-1}{Cu(CN)_2}\downarrow \longrightarrow \overset{+1}{Cu}CN + \overset{0}{(CN)_2}\uparrow$

$\downarrow CN^-$

(High value of K_f for complex) $Cu(CN)_4^{3-} \xleftarrow{H_2S}$ No(ppt)

$Cd^{2+} + CN^- \rightarrow Cd(CN)_2\downarrow \xrightarrow{CN^-} [Cd(CN)_4]^{2-}$

$\downarrow H_2S$

$CdS\downarrow$

ppt

(Low value of K_f for complex)

(i) Perfect complex:

They have high value of K_f

They do not dissociate in presence of particular reagent

Example: $CN(CN)_4^{3-}$ is perfect w.r.t. H_2S

(ii) Imperfect complex

They have low value of K_f

They dissociate in presence of particular reagent

Example: $Cd(CN)_4^{2-}$ is imperfect w.r.t. H_2S

All ammonia complex are imperfect in presence of acid

$Cu^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+}$

If H^+ is added here

⇓

It reacts with NH_3 and form NH_4^+ (Acid base reaction)

So concentration of NH_3 decreased

⇓

Equilibrium shift backward as per Le-chatelier principle

⇓

So complex of ammonia become unstable and become imperfect

Points to remember

- There is no. sharp line between perfect and imperfect salt.
- They are defined in presence of particular reagent

To write electronic configuration of CMI (Central Metal Ion) (transition) always consider all electrons in d-subshell

Example :

$[Ni(CN)_4]^{4-}$ $\quad Ni^0 \rightarrow \underbrace{3d^8 4s^2}_{\text{Free atom}}$

In complex

$Ni \rightarrow 3d^{10}$

(Electron move from 4s to 3d)

How to distinguish NO & NO^+

Ex. $[Fe(H_2O)_5 NO]SO_4$

		Charge difference
$[Fe(NO)]^{+2}$ → NO^+	Fe^+	0
→ NO	Fe^{+2}	+2

Species having less charge difference are more stable :

So,In $[Fe(H_2O)_5NO]SO_4$

Fe^{+1} NO^+

(Charge of Fe & NO)

(2) Sodium Nitroprusside

$Na_2[Fe(CN)_5NO]$ Charge difference

$[FeNO]^{+3}$ ✓ NO^+ Fe^{2+} 1

x NO Fe^{3+} 3

So ,In $Na_2[Fe(CN)_5NO]$

Fe^{2+} NO^+ (Correct)

$[Fe(NO)_4]$ x NO^+ Fe^{-4}

✓ NO^0 Fe^0

If charge on complex [FeNO] is positive than NO is NO^+

If charge on Complex [FeNO] is zero or negative then NO is NO^+

Innocent ligands	**Non innocent ligands**
• Ligands in which charge is predictable Ex : Cl^-, NH_3, H_2O, en etc.	• Charge can't be predictable directly (Some hint is required) Ex : (i) O_2 → O_2, O_2^-, O_2^{2-} (ii) NO → NO^+, NO

Q9 $\underset{\text{I}}{\left[Zn(CN)_2(O_2)_2\right]} \xrightarrow{\text{Reduction}} K_2\underset{\text{II}}{\left[Zn(CN)_2(O_2)_2\right]}$

Find charge on O_2 in first and second complex.

(A) 0, -1 **(B) 0, -2** **(C) -1, -2** **(D) -1, 0**

Sol: **Ans. A**

In complex – I common oxidation state of Zn is Zn^{2+}, So to satisfy change of complex.

O_2 in complex – I should have zero charge $\left[\overset{+2}{Zn}\overset{-2}{(CN)_2}\overset{0}{(O_2)_2}\right]$

Similarly in complex – II O_2 should be O_2^{-1} to satisfy net charge of complex

$K_2^{+2}\left[\overset{+2}{Zn}\overset{-2}{(CN)_2}\overset{-2}{(O_2)_2}\right]$

So option (A) is correct.

Q10 **Find out EAN of the following compounds-**

$\left[Cr(H_2O)_6\right]^{3+}$, $\left[Cr(C_2O_4)_3\right]^{3-}$, $\left[CrBr(NO_2)(H_2O)_4\right]^{+}$, $\left[Cr(CN)_6\right]^{4-}$, $\left[Cr(en)_3\right]^{2+}$, $\left[Cr(CO)_6\right]$, $\left[Cr(CN)_6\right]^{3-}$, $\left[Cr(NH_3)_6\right]^{2+}$, $\left[Cr(H_2O)_6\right]^{2+}$, $[MnF_6]^{3-}$, $\left[Mn(CN)_6\right]^{4-}$, $\left[Mn(H_2O)_6\right]^{2+}$, $\left[Mn(NH_3)_6\right]^{2+}$, $[MnF_6]^{4-}$, $\left[Fe(NO_2)_2(en)_2\right]$, $\left[Fe(CN)_6\right]^{3-}$, FeF_6^{3-}, $\left[Fe(H_2O)_5NO\right]SO_4$, FeF_6^{4-}, $\left[Fe(EDTA)\right]^{-}$

Sol:

1. $\left[Cr(H_2O)_6\right]^{3+}$

   $Cr^{+3} \rightarrow 3d^3, C.N. = 6, H_2O$ Denticity $\rightarrow 1$

   $24 - 3 + (6 \times 2) = 33$

2. $\left[Cr(C_2O_4)_3\right]^{3-}$

   $Cr^{+3} \rightarrow 3d^3, CN = 6, C_2O_4^{2-}$ Denticity $\rightarrow 2$

   $24 - 3 + 3 \times 2 \times 2 = 33$

3. $\left[CrBr(NO_2)(H_2O)_4\right]^+$

$Cr^{3+} \rightarrow 3d^3$, C.N. = 6, Br^-, NO_2^-, H_2O Denticity $\rightarrow$ 1

$24 - 3 + 2 + 2 + 4 \times 2 = 33$

4. $\left[Cr(CN)_6\right]^{4-}$

$Cr^{+2} \rightarrow 3d^4$, CN = 6, CN^- $\rightarrow$ Denticity = 1

$24 - 2 + 6 \times 2 = 34$

5. $\left[Cr(en)_3\right]^{2+}$

$Cr^{+2} \rightarrow 3d^4$, C.N. = 6, en Denticity $\rightarrow$ 2

$24 - 2 + 3 \times 2 \times 2 = 34$

6. $\left[Cr(CO)_6\right]$

$Cr \rightarrow 3d^5 4s^1$ C.N. = 6, CO, Denticity $\rightarrow$ 1

$Cr \rightarrow 3d^6$

$24 - 0 + 12 = 36$

7. $\left[Cr(CN)_6\right]^{3-}$

$Cr^{3+} \rightarrow d^3$, C.N. = 6, CN^- Denticity $\rightarrow$ 1

$24 - 3 + 12 = 33$

8. $\left[Cr(NH_3)_6\right]^{2+}$

$Cr^{2+} \rightarrow 3d^4$, C.N. = 6 NH_3 Denticity $\rightarrow$ 1

$24 - 2 + 12 = 34$

9. $\left[Cr(H_2O)_6\right]^{2+}$

$Cr^{+2} \rightarrow 3d^4$, C.N. = 6, H_2O Denticity $\rightarrow$ 1

$24 - 2 + 12 = 34$

10. $\left[MnF_6\right]^{3-}$

$Mn^{+3} \rightarrow 3d^4$, C.N. = 6, F^- Denticity $\rightarrow$ 1

$25 - 3 + 12 = 34$

11. $\left[Mn(CN)_6\right]^{4-}$

$Mn^{2+} \rightarrow 3d^5$, C.N. = 6, CN^- Denticity $\rightarrow$ 1

$25 - 2 + 12 = 35$

12. $\left[Mn(H_2O)_6\right]^{2+}$

$Mn^{2+} \rightarrow 3d^5$ C.N. = 6, H_2O Denticity $\rightarrow$ 1

$25 - 2 + 12 = 35$

13. $\left[Mn(NH_3)_6\right]^{2+}$

$Mn^{2+} \rightarrow 3d^5, NH_3$ Denticity $\rightarrow$ 1

C.N. = 6

$25 - 2 + 12 \rightarrow 35$

14. $[MnF_6]^{4-}$

$Mn^{+2} \rightarrow d^5$, C.N. = 6, F^- Denticity $\rightarrow$ 1

CN = 6 D $\rightarrow$ 1

$25 - 2 + 12 = 35$

15. $\left[Fe(NO_2)_2(en)_2\right]$

$Fe^{2+} \rightarrow 3d^6$, C.N. = 6, NO_2^- Denticity $\rightarrow$ 1, en Denticity $\rightarrow$ 2

$26 - 2 + 2 \times 2 + 2 \times 2 \times 2 = 36$

16. $\left[Fe(CN)_6\right]^{3-}$

$Fe^{3+} \rightarrow d^5$, CN^- Denticity $\rightarrow$ 1

C.N. = 6

$26 - 3 + 12 = 35$

17. FeF_6^{3-}

$Fe^{+3} \rightarrow 3d^5$, C.N. = 6, F^-, Denticity $\rightarrow$ 1

$26 - 3 + 12 = 35$

18. $\left[Fe(H_2O)_5 NO\right]SO_4$

$Fe^+ \rightarrow 3d^7$, C.N. = 6, H_2O Denticity $\rightarrow$ 1, NO^+ Denticity $\rightarrow$ 1

$26 - 1 + 12 = 37$

19. FeF_6^{4-}

$Fe^{+2} \rightarrow 3d^5$, C.N. = 6, F^- Denticity $\rightarrow$ 1

$26 - 2 + 12 = 36$

20. $\left[Fe(EDTA)\right]^-$

$Fe^{+3} \rightarrow 3d^5$, C.N. $\rightarrow$ 6, $EDTA^{-4}$ Denticity $\rightarrow$ 6

$26 - 3 + 12 = 35$

Q11 **Find out EAN of the following compounds-**

(i) $[Fe(NH_3)_6]^{2+}$, (ii) $[Fe(CN)_6]^{4-}$, (iii) $[Fe(NH_3)_6]^{3+}$, (iv) $[Fe(en)_3]^{2+}$, (v) $[Fe(NO_2)_6]^{4-}$, (vi) $[Fe(NCS)_6]^{4-}$, (vii) $[Fe(H_2O)_2(NH_3)_4]^{2+}$, (viii) $[Fe(en)_3]^{3+}$, (ix) $[Fe(H_2O)_6]^{2+}$, (x) $[Fe(H_2O)_6]^{3+}$, (xi) $[Co(CO)_4]^-$, (xii) $[CoCl_3(NH_3)_3]$, (xiii) $[CoCl_4]^{2-}$, (xiv) $[Co(H_2O)_2F_4]^-$, (xv) $[CO(NH_3)_6]^{2+}$, (xvi) $[Co(CN)_2(NH_3)_4]Cl$, (xvii) $[Co(OX)_3]^{4-}$, (xviii) $[Co(H_2O)_3(NH_3)_3]^{+3}$, (xix) $[Co(H_2O)_4F_2]^+$, (xx) $[Co(Ox)_3]^{3-}$

(i) $Fe^{2+} \rightarrow d^6$, C.N. = 6, NH_3, Denticity $\rightarrow$ 1

$26 - 2 + 12 = 36$

(ii) $Fe^{2+} \rightarrow d^6$, C.N. $\rightarrow$ 6, CN^- Denticity = 1

$26 - 2 + 12 = 36$

(iii) $Fe^{3+} \rightarrow d^5$, C.N. = 6, NH_3 Denticity $\rightarrow$ 1

$26 - 3 + 12 = 35$

(iv) $Fe^{+2} \rightarrow 3d^6$, C.N. = 6, en Denticity $\rightarrow$ 2

$26 - 2 + 12 = 36$

(v) $Fe^{+2} \rightarrow d^6$, C.N. = 6, NO_2^- Denticity $\rightarrow$ 1

$26 - 2 + 12 = 36$

(vi) $Fe^{2+} \rightarrow d^6$, C.N. = 6, NCS^- Denticity $\rightarrow$ 1

$26 - 2 + 12 = 36$

(vii) $Fe^{2+} \rightarrow d^6$, C.N. = 6, $\underbrace{H_2O \rightarrow 1, NH_3 \rightarrow 1}_{\text{Denticity}}$

$26 - 2 + 12 = 36$

(viii) $Fe^{3+} \rightarrow d^5, C.N. = 6, en\ Denticity \rightarrow 2$

$26 - 3 + 12 = 35$

(ix) $Fe^{2+} \rightarrow 3d^6, C.N. = 6, \underbrace{H_2O \rightarrow 1}_{Denticity}$

$26 - 2 + 12 = 36$

(x) $Fe^{+3} \rightarrow d^5, C.N. = 6, H_2O\ \ Denticity \rightarrow 1$

$26 - 3 + 12 = 35$

(xi) $Co \rightarrow 3d^7, 4s^2 .....$

$Co^{-1} \rightarrow 3d^{10}, C.N. = 4, CO, Denticity \rightarrow 1$

$27 + 1 + 8 = 36$

(xii) $Co^{+3} \rightarrow d^6, C.N. = 6, \underbrace{Cl^- \rightarrow 1, NH_3 \rightarrow 1}_{Denticity}$

$27 - 3 + 12 = 36$

(xiii) $Co^{+2} \rightarrow d^7, C.N. = 4, Cl^-\ Denticity \rightarrow 1$

$27 - 2 + 8 = 33$

(xiv) $Co^{+3} \rightarrow d^6, C.N. = 6, \underbrace{H_2O \rightarrow 1, f^- \rightarrow 1}_{Denticity}$

$27 - 3 + 4 + 8 = 36$

(xv) $Co^{+2} \rightarrow d^7, C.N. = 6,\ NH_3\ Denticity \rightarrow 1$

$27 - 2 + 12 = 37$

(xvi) $Co^{+3} \rightarrow 3d^6, C.N. = 6, \underbrace{CN^- \rightarrow 1, NH_3 \rightarrow 1}_{Denticity}$

$27 - 3 + 4 + 8 = 36$

(xvii) $Co^{+2} \rightarrow 3d^7, C.N. = 6, ox^{-2} \rightarrow Denticity \rightarrow 2$

$27 - 2 + 12 = 37$

(xviii) $Co^{+3} \rightarrow d^6, C.N. = 6, \underbrace{H_2O \rightarrow 1, NH_3 \rightarrow 1}_{\text{Denticity}}$

$27 - 3 + 12 = 36$

(xix) $Co^{+3} \rightarrow d^6, C.N. = 6, \underbrace{I^- \rightarrow 1, H_2O \rightarrow 1}_{\text{Denticity}}$

$27 - 3 + 12 = 36$

(xx) $Co^{+3} \rightarrow 3d^6, C.N. = 6, ox^{-2} \rightarrow \text{Denticity} \rightarrow 2$

$27 - 3 + 12 = 36$

IUPAC Naming:-

(i) Cations are written before anions

Example :

$NaCl \rightarrow$ Sodium chloride

$MgO \rightarrow$ Magnesium oxide

(ii) Never consider number of ions

Example :

$MgCl_2 \rightarrow$ Magnesium chloride

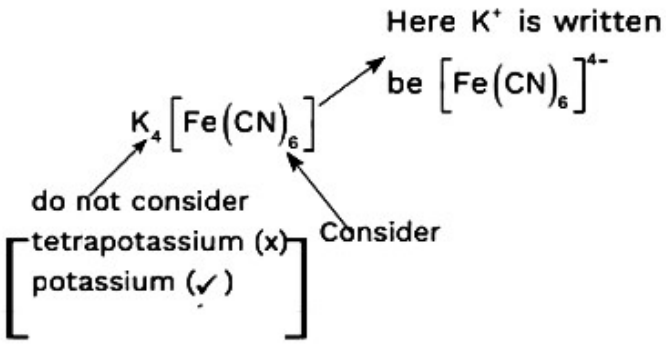
$Na_2O \rightarrow$ Sodium oxide

(iii) In case of covalent bond always consider number of atoms

$CO_2 \rightarrow$ Carbon dioxide

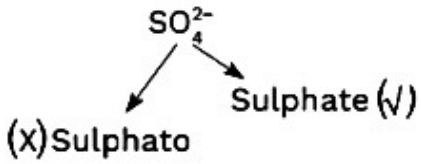
$CO \rightarrow$ Carbon monoxide

Similarly :



$\Rightarrow \left[Cr(NH_3)_6\right]SO_4$.

Here complex part is written before



$\left(\because \text{Here } SO_4^{2-} \text{ is not ligand}\right)$

$\Rightarrow \left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$

Here $\left[Cr(NH_3)_6\right]^{+}$ is written before $\left[Co(CN)_6\right]^{3-}$

While writing naming of complex part (Co-ordination sphere)

(i) Ligands (donor) are written before CMI

Example :

In $\left[Fe(CN)_6\right]^{3-} \Rightarrow CN^-$ is written before Fe^{3+}

The ligands can be neutral, anionic or cationic.

The neutral ligands are named as the molecule Ex. (C_5H_5N) pyridine, $(C_6H_5)_3P$ Triphenyl phosphine, $(H_2N - CH_2 - CH_2 - NH_2)$ ethylene diamine.

The neutral ligands which are not named as the molecule are CO carbonyl, NO nitrosyl, H_2O Aqua, NH_3 ammine.

Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	**Name as ligand**	**Symbol**	**Name as ligand**
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo/Peroxido
CN^-	Cyano/Cyanido	O_2H^-	Perhydroxo/Perhydroxido
O^{2-}	Oxo/Oxido	S^{2-}	Sulphido
OH^-	Hydroxo/Hydroxido	NH^{2-}	Imido
H^-	Hydrido/Hydrido	NH_2^-	Amido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

Symbol	Name as ligand	Symbol	Name as ligand
CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$C_2O_4^{2-}$	Oxalato	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	ONO^-	(bonded through oxygen) nitrito
NO_3^-	Nitrato	NO_2^-	(bonded through nitogen) nitro/nitrito-N
$S_2O_3^{-2}$	Thiosulphato		

Positive ligands naming ends in 'ium' $NH_2—NH_3^+$ Hydrazinium, NO_2^+ nitronium, NO^+ nitrosonium/nitrosylium.

To represent number of ligands prefixes are used like –

2 → di
3 → tri
4 → tetra
5 → penta
6 → hexa

$K_4[Fe(CN)_6]$ → Potassium hexacyanido .....

$[Cu(NH_3)_4]SO_4$ → tetra ammine ..... sulphate

If such prefixes are present in name of ligand, for organic ligands, oxoanions new prefixes are used.

2 → Bis
3 → tris
4 → tetrakis
5 → pentakis
6 → hexakis

en → ethane-1,2-diamine

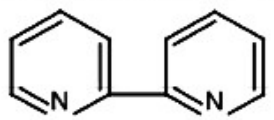
$(en)_2$ ✗ → di(ethane-1,2-diamine)

$(en)_2$ ✓ → bis(ethane-1,2-diamine)

$(Py)_2$ ✗ → dipyridine

$(Py)_2$ ✓ → Bis Pyridine

(It will confuse with dipy



)

Naming of CMI

Complex part or Co-ordination sphere is cationic or neutral

$[\ ]^{0}$ or $[\ ]^{+x}$

Complex part or Co-ordination sphere is Anionic

$[\ ]^{-x}$

General name is used along with oxidation state of CMI in Roman number in small bracket	Latin name is used with ate suffix along with oxidation state of CMI in roman number in small bracket.
$Fe^{3+} \rightarrow Iron(III)$	$Fe^{3+} \rightarrow Ferrate(III)$
$Cu^{+2} \rightarrow Copper(II)$	$Cu^{+2} \rightarrow Cuperate(II)$
$Zn^{2+} \rightarrow Zinc(II)$	$Zn^{2+} \rightarrow Zincate(II)$
$Ag^{+} \rightarrow Silver(I)$	$Ag^{+} \rightarrow Argentate(I)$

$[\ ]_x\ [\ ]_y$

↑ Cation ↑ Anion (do not consider x,y)

Example : $-K_3\left[Al(C_2O_4)_3\right]$

Potassium trioxalatoaluminate (III)

$\left[Co(NH_3)_5\,CO_3\right]Cl$

Penta ammine carbonato cobalt (III) chloride

$\left[Co(NH_3)_4\,(H_2O)Cl\right]Cl_2$

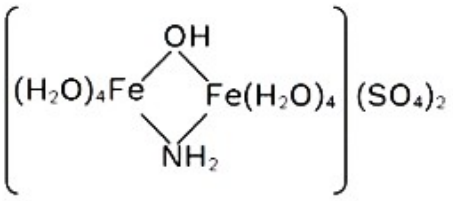
Tetra ammine Aquachloro cobalt (III) chloride

Some examples of IUPAC naming of coordination compounds-

Complex Compounds		**IUPAC Name**
(i)	**$K_4[Fe(CN)_6]$** (anionic complex) So suffix 'ate' is added with metal name	Potassium hexacyanidoferrate(II)
(ii)	**$K_2[PtCl_6]$**	Potassium hexachloridoplatinate(IV)
(iii)	**$[Co(NH_3)_6]Cl_3$** (Cationic complex) So metal name will be written without any suffix	Hexamminecobalt(III) chloride
(iv)	**$[Cr(H_2O)_4Cl_2]Cl$**	Tetraaquadichloridochromium(III) chloride
(v)	**$[Pt(NH_3)_2Cl_4]$**	Diamminetetrachloridoplatinum(IV)
(vi)	**$[Co(NH_3)_3Cl_3]$** (Neutral complex) So no suffix is used with name of metal ion	Triamminetrichloridocobalt(III)
(vii)	**$K_3[Co(NO_2)_6]$**	Potassium hexanitrocobaltate(III)
(viii)	**$Na_3[Fe(CN)_5NO]$**	Sodium pentacyanidonitrosoniumferrate(II)
(ix)	**$[NiCl_4]^{-2}$**	Tetrachloridonickelate(II) ion

Complex Compounds		IUPAC Name
(x)	**$[Ru(NH_3)_5Cl]^{+2}$**	Pentamminechloridoruthenium(III) ion
(xi)	**$[Fe(en)_3]Cl_3$**	Tris-(ethylenediamine)iron(III) chloride
(xii)	**$[Ni(Gly)_2]$**	Bis-(glycinato)nickel(II)

(i) If a complex ion has two metal atoms then it is termed polynuclear. The ligand which connects the two metal ions is called as **Bridging ligand or Bridge group.** A prefix of Greek letter μ, is repeated before the name of each different kind of bridging group.



Tetraaquairon(III)-μ-amido-μ-hydroxidotetraaquairon(III) sulphate

Q12 **Write the systematic names of the following co-ordination entities and compounds:**
(i) $[CoCl_2(NH_3)_3]^+$ (ii) $[CrCl_3(NH_3)_3]$ (iii) $K_3[Cr(C_2O_4)_3]$
(iv) $K_4[Fe(CN)_6]$ (v) $[PtCl(NH_3)_5]Cl_3$

Sol: (i) triaamminedichioridocobalt(III)
(ii) triamminetrichloridochromium(II)
(iii) potassium trioxalatochromate(III)
(iv) potassium hexacyanidoferrate(II)
(v) pentaamminechloridoplatinum(IV) chloride

Q13 **Write I.U.P.A.C name for the following:**

1. $[Co(NCS)(NH_3)_5]Cl_2$ ⟶
2. $K_2[HgI_4]$ ⟶
3. $[CoBr(ONO)(en)_2]^{+1}$ ⟶
4. $K_2[Mo(CN)_6]$ ⟶
5. $K[Au(CN)_2]$ ⟶
6. $[Fe(en)_3]Cl_3$ ⟶
7. $[Co(NH_3)_6]Cl_3$ ⟶
8. $[Co(H_2O)(NH_3)_5]Cl_2$ ⟶
9. $K_4[Fe(NO_2)_6]$ ⟶
10. $LiAlH_4$ ⟶

Sol:

1. Pentaamminethiocyanato–N–cobalt (III) chloride
2. Potassium tetraiodomercurate (II)
3. Bromobis(ethylenediamine) nitrito cobalt (III) ion (nitro-O)
4. Potassium hexacyanomolybdate (IV)
5. Potassium dicyanoaurate (I)
6. Tris (ethylenediamine) iron (III) chloride
7. Hexaamminecobalt (III) chloride
8. Pentaammineaquacobalt (II) chloride
9. Potassium hexanitroferrate (II) (nitro-N)
10. Lithium tetrahydridoaluminate (III)

Q14 **Write I.U.P.A.C name for the following:**

1. $[CuCl_2(Et_2NH)_2]$ ⟶
2. $[CoCl_2(en)_2]Cl$ ⟶
3. $K[PtCl_3(C_2H_4)]$ ⟶
4. $K_2[OsCl_5N]$ ⟶
5. $Na_3[Ag(S_2O_3)_2]$ ⟶
6. $Na_2[Fe(CN)_5(NO)]$ ⟶

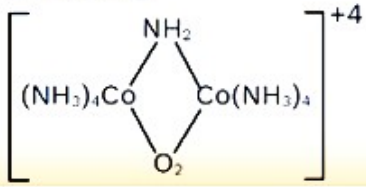
7. $K_3[Fe(CN)_5(CO)]$ ⟶
8. $K_2[Ni(CN)_4]$ ⟶
9. $K_3[Co(NO_2)_6]$ ⟶
10. $K_3[Cr(C_2O_4)_3]$ ⟶

Sol:

1. Dichlorobis (diethylamine) copper (II)
2. Dichlorobis (ethylenediamine) cobalt (III) chloride
3. Potassium trichloro (η^2 – ethylene) platinate (II)
4. Potassium pentachloronitridoosmate (VI)
5. Sodium bisthiosulphatoargentate (I)
6. Sodium pentacyanonitrosonium ferrate (II)
7. Potassium carbonyl pentacyanoferrate (II)
8. Potassium tetracyanonickelate (II)
9. Potassium heranitro–N cobaltate (III) (nitrito)
10. Potassium trisoxalatochromate (III)

Q15

Write I.U.P.A.C name for the following:

1. $[PtBr(NH_3)_3]NO_3$ ⟶
2. trans-$[PtCl_2(NH_3)_2]$ ⟶
3. $Na_3[Ag(S_2O_3)_2]$ ⟶
4. $NH_4[Cr(NCS)_4(NH_3)_2]$ ⟶
5. $[(Ph_3P)_2]Ir(CO)Cl$ ⟶
6. $RhCl(PPh_3)_3$ ⟶
7. $[CrCl_2(H_2O)_4]NO_3$ ⟶
8. $[Cu(NH_3)_4]^{2+}$ ⟶
9. $[Ni(dmg)_2]$ ⟶
10.  ⟶

Sol:

1. Triamminebromoplatinum (II) nitrate
2. Trans–diammine dichloroplatinum (II)
3. Sodium bisthiosulphatoargentate (I)
4. Ammonium diamminetetrathiocyanato–N chromate (III)
5. Carbonyl bis (triphenylphosphine) iridium (I) chloride
6. Chlorotris (triphenylphosphine) rhodium (I)
7. Tetraaquodi chlorochromium (III) nitrate
8. Tetraamminecopper (II) ion
9. B is (dimethylglyoximato) nickel (II)
10. μ–amido μ–superoxo bis tetraamminedicobalt

Classification of complexes

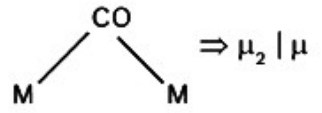
Naming of polynuclear complexes :

In polynuclear complexes two type of ligands are present

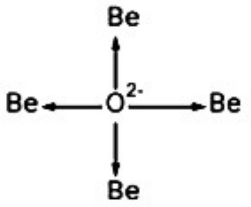
(1) Terminal ligands: They are boarded with one CMI

(2) Bridging Ligands : They are bonded with more then one CMI → Bridging ligands are represent with prefix 'μ_n' where n → number of CMI to which bridging ligand is attached.

Example :

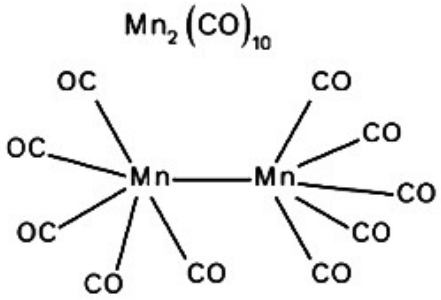


$\Rightarrow \mu_2$ | μ



$\Rightarrow \mu_4$

Example :



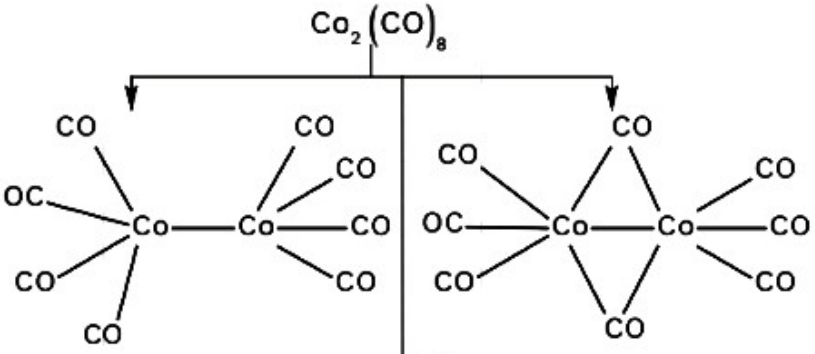
Name:

(i) Penta carbonyl manganese (0)

(ii) bis (Penta carbonyl manganese (0))

(iii) Deca carbonyl dimanganese (0)

2.

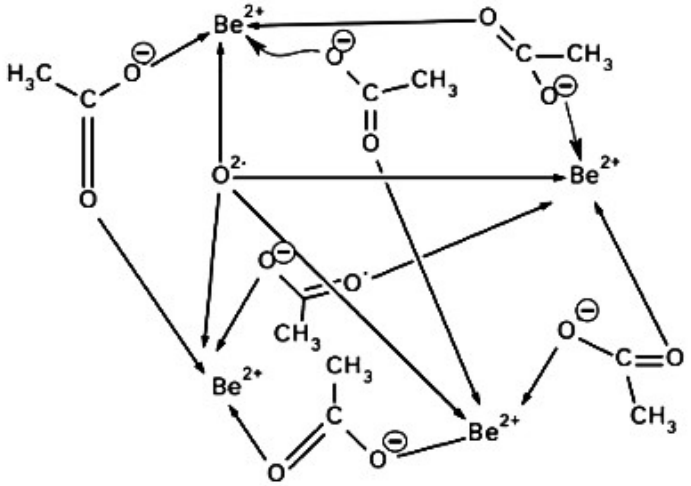

$Co_2(CO)_8$
CO
OC
Co
Co
CO
CO
CO

(1) Tricarbonyl iron (0)-tri-μ -carbonyl tricarbonyl iron (0)

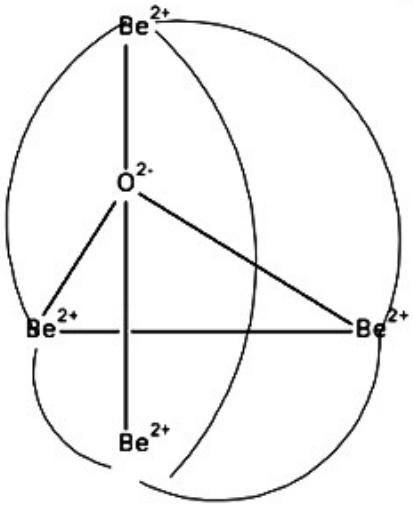
(2) Tri-μ -carbonyl bis (tricarbonyl iron (0))

(3) Tri-μ -carbonyl hexacarbonyl diiron (0)

$\left[Be_4O(CH_3COO)_6\right]$ → Hexa-μ_2 -acetato-μ_4 -oxidotetraberyllium (II)



or



→ These all are CH_3COO^-

Extra Information:

Sometimes oxidation state of CMI is written in Arabic number which represent charge on complex.

$K_4\left[Fe(CN)_6\right]$ → Potassiumhexacyanidoferrate(II)

Or

Potassium hexacyanidoferrate(-4) → Change on complex ion

Writing formula of compound from its name :-

$[CMI\ \underbrace{L_1, L_2, L_3}_{\text{alphabetical order}} ----]$ L_1, L_2, L_3 are ligands

CMI → central metal ion

VBT

Valence bond theory for complex compounds:-

According to valence bond theory CMI and ligand is bonded with Co-ordinate bond in which ligand provide lone pair and central metal ion provide Vacant orbital.

Vacant orbitals of CMI hybridize and configured in 3-D at minimum repulsion.

Number of hybrid orbital = number of vacant orbitals = Co-ordination number = σ – bond only.

Q16 **Why only σ – bond is considered in hybridization of transition metals, while in hybridization of p-block elements lone pair $+\sigma$ – bond is consider ?**

Sol: In p-block elements lone pair is present in outer shell, so they also participate in geometry.

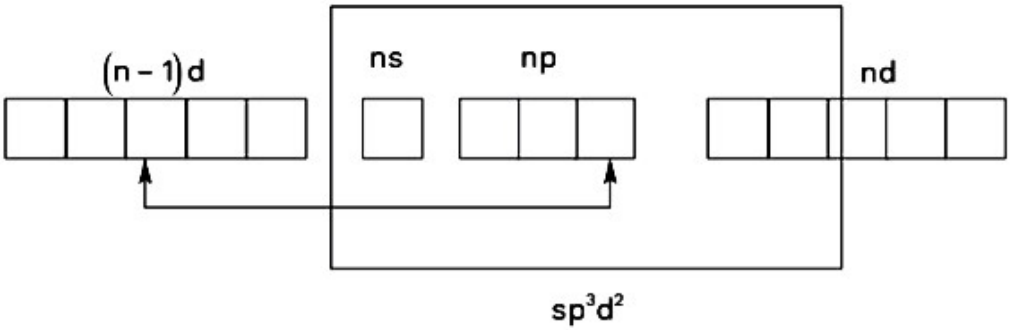
Hybrid orbital in p-block = σ + lone pair

While in transition metals, lone pair is present in inner shell having high nuclear change, so there will be greater attrition of nucleus and hence lone pair is stereo chemically inert in d-block elements.

$\therefore$ No of hybrid orbitals = No. of σ – bond.

C.N.	Hybridisation	Geometry
2	sp	Linear
3	sp^2	Trigonal planar
4.	sp^3	Tetrahedral
	dsp^2	Square planar
5.	dsp^3, sp^3d Here d_{z^2} orbital is used	Trigonal bipyramidal
5.	dsp^3, sp^3d Here $d_{x^2-y^2}$ orbital is used	Square Pyramidal
6.	sp^3d^2, d^2sp^3	Octahedral

C.N = 6 $\Rightarrow$ sp^3d^2 / d^2sp^3



When $(n-1)$ orbitals are used in hybridization then the complex is called as inner orbital complex.

When nd orbital are used in hybridization then the complex is called as outer orbital complex.

Inner orbital complex	Outer orbital complex
d^2sp^3	sp^3d^2
dsp^3	sp^3d
dsp^2	

Example :

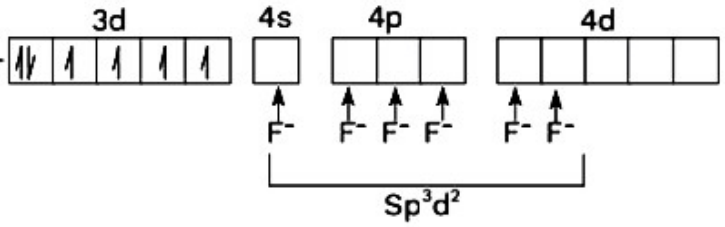
$K_4[FeF_6] \rightarrow 4K^+ + [FeF_6]^{4-}$

$\rightarrow Fe^{2+}$ (CMI)

$\rightarrow$ C.N = 6

sp^3d^2 or d^2sp^3

$Fe^{2+} \rightarrow .....3d^6$



So $K_4[FeF_6]$

$\rightarrow$ Hybridization $\rightarrow Sp^3d^2$

$\rightarrow$ Outer orbital complex

$\rightarrow$ Paramagnetic ($\because$ Unpaired electrons are present)

$\rightarrow$ Spin magnetic moment $(\mu) = \sqrt{n(n+2)}$ B.M

$= \sqrt{4(4+2)}$

$= \sqrt{24}$

($\because$ n $\Rightarrow$ no. of unpared electrons=4)

Example :

$K_4[Fe(CN)_6]$

$\rightarrow$ C.N. = 6

$\rightarrow Fe^{2+} \rightarrow [Ar]3d^6$



It is complex shluld be sp^3d^2, paramagnetic but practically the complex is diamagnetic, so here ligends paly important role.

Types of ligand :

Ligands

Strong field ligard (SFL)

- It form strong bond with CMI
- They force electrons of CMI to pair up as per requirement of d-orbitals in hypridisation

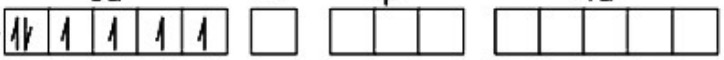
  Ex:-CN^-

Weak field ligand (WFL)

- It form weak bond with CMI and Can't force electrons of CMI to pair up

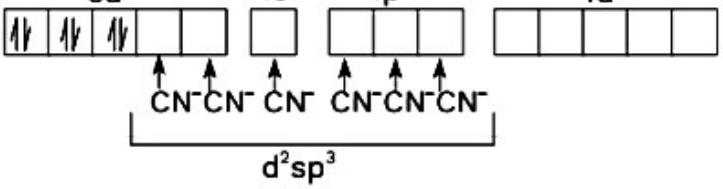
  Ex:-F^-

$\therefore$ Fe^{2+} →



CN^- is SFL so it can pair up to electrons of CMI

Fe^{2+} →



So,

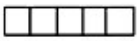
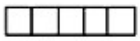
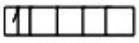
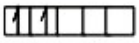
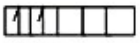
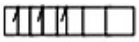
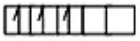
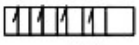
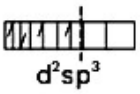
$K_4\left[Fe(CN)_6\right]$

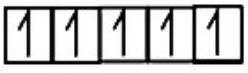
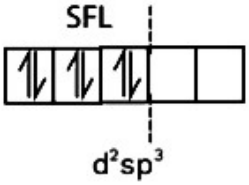
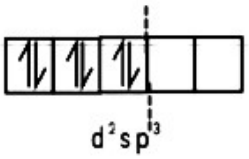
- d^2Sp^3 hybridization
- inner orbital complex
- Diamagnetic $(\because$ all electron are paired up$)$

CN = 6

→ Hybridization d^2sp^3 | sp^3d^2

→ requirement of d – orbitals = 2

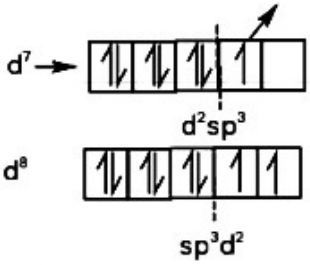
Configuration of CMI	WFL	SFL
d^0	 d^2sp^3	 d^2sp^3
d^1	 d^2sp^3	 d^2sp^3
d^2	 d^2sp^3	 d^2sp^3 (paring not occur as already two d-orbitals are vacant)
d^3	 d^2sp^3	 d^2sp^3
d^4	 sp^3d^2 (as two inner d-orbitals are not vacant)	 d^2sp^3

	WFL	SFL
d^5	 sp^3d^2	 d^2sp^3
d^6	 sp^3d^3	 d^2sp^3

Similarly

d^7
d^8
d^9
d^{10}
sp^3d^2 (withWFL)

One electron can be excited to higher orbital by SFL



Similarly

$d^9 \rightarrow sp^3d^2$

$d^{10} \rightarrow sp^3d^2$

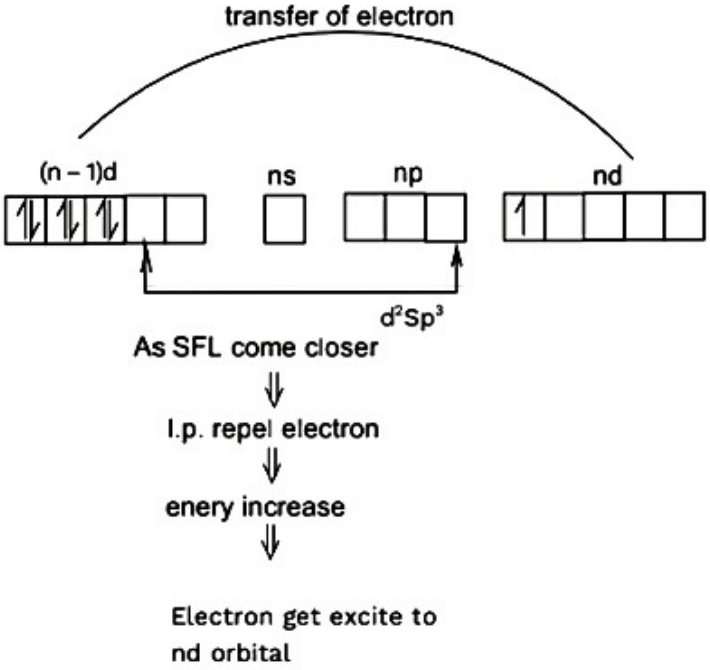
So, In C.N. = 6

- $d^{0,1,2,3}$ → d^2sp^3 (always)
- $d^{4,5,6,7}$ → WFL → sp^3d^2 / SFL → d^2sp^3
- $d^{8,9,10}$ → always sp^3d^2

Diamagnetic → d^0, d^{10}

(all electrons are paired) d^6 – with SFL

d^7 → SFL(C.N. = 6)

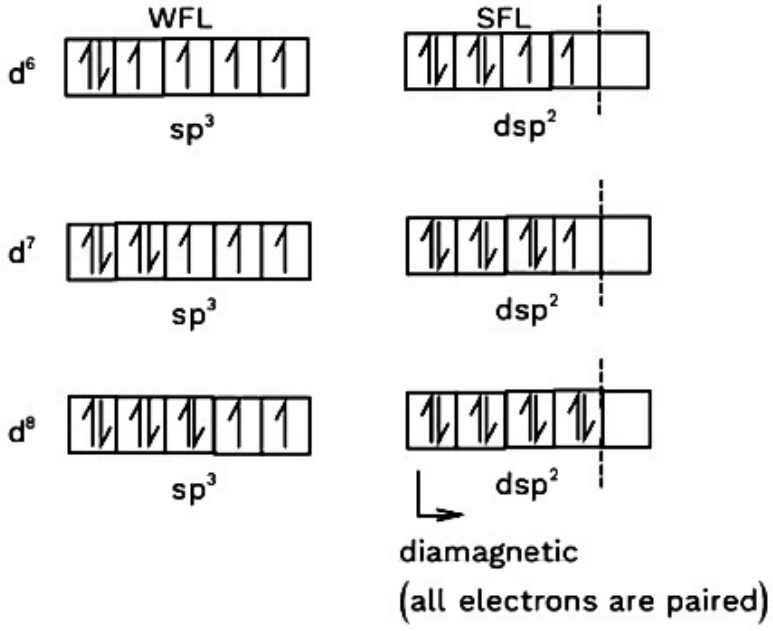


This is called transference (excitation) Condition : for transference:

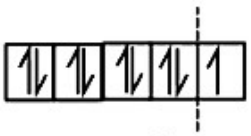
(i) SFL should be there

(ii) single elector is present in orbital which is required for hybridization:

C.N. = 4 → Sp^3 / dsp^2



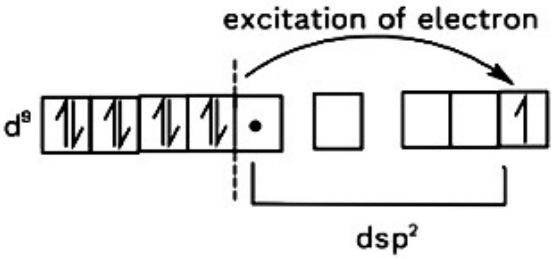
diamagnetic
(all electrons are paired)

$d^9 \rightarrow sp^3$ $d^9 \rightarrow$ 

(This electron get excited to higher np orbital)

$d^{10} \rightarrow sp^3$ $d^{10} \rightarrow sp^3$

$d^9 \rightarrow$ SFL(C.N. = 4)



C.N. = 4

- diamagnetic (d^0d^{10}, d^8(SFL))
- transference (from (n-1)d to np transfer of electron occur)

C.N. = 5 → dsp^3 / sp^3d

- diamagnetic (d^0, d^{10}, d^6(SFL))
- transference (d^9 – SFL) (from (n-1)d to nd transfer of electron occur)

According to V.B.T.

→ SFL ⇒ C-donor, N-donor

↳WFL ⇒ halogen donor, O-donor

Ligand

⇓

Behave as Lewis base

⇓

Provide electron pair for sharing

⇓

Lewis base strength ⇒ $\underbrace{C > N > O}_{\text{as per Electronegativity}} > \underbrace{F^- > Cl^- > Br^- > I^-}_{\text{As per size}}$

Conclusion:

Generally

C-donor, N-donor → SFL

$\left.\begin{matrix} O-\text{donor}, \\ \text{halogen}-\text{donor} \end{matrix}\right] \rightarrow$ WFL

If we go from 3d → 4d → 5d series elements

$\xrightarrow{Ni^{2+} \rightarrow Pd^{2+} \rightarrow Pt^{2+}}$

Z_{eff} ↑

attraction ↑

So all ligands are SFL for 4d & 5d series elements and +4 ions(high positive charge)

Example :

(i) $Cl^- \rightarrow \underset{WFL}{Ni^{2+}}$ $\underset{SFL}{Pd^{2+}}$ $\underset{SFL}{Pt^{2+}}$

(ii) $F^- \rightarrow \underset{WFL}{Ni^{2+}}$ $\underset{SFL}{Ni^{4+}}$

(1) $\xrightarrow{Mn^{2+} \quad Fe^{2+} \quad Co^{+2} \quad Ni^{2+} \quad Cu^{2+}}$

d electron↑(poor shielding by d electron)

Z↑

Z_{eff} ↑

attraction ↑

Generally NH_3 is SFL but act as WFL for $Mn^{2+}, Fe^{+2}, Co^{+2}$

(2) $Co^{+2} \xrightarrow[Z_{eff}\uparrow]{} Co^{+3}$

(∵ positive charge ↑)

Generally O-donor WFL but act as SFL for Co^{+3} & Cu^{2+}

Spectrochemical Series -

(Arrangement of ligands based on their strength)

$I^- < Br^- < SCN^- < Cl^- < S^{-2} < NO_3^- < F^- < OH^- < CH_3COO^- < C_2O_4^{2-} < H_2O$ .....

(itna bura tha kalu samu Nayi fulle hath ese uske (pani bhara ho jaise)

NCS^- (Naveen chadra saxena) $< EDTA^{4-}$ (en dino) $< NH_3 \approx Py <$ aman pane ke

always SFL

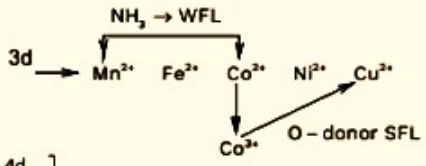
en liye $\approx$ dipy (Double double) $< NO_2^- < CN^-$ Nosena $<$ CO court ke chakkar laga raha hain

Low spin and High spin complex

Low spin: Low value of spin only magnetic moment

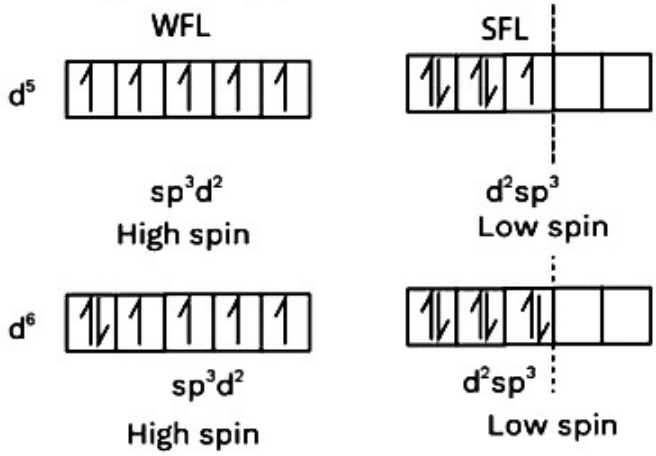
High spin:High value of spin only magnetic moment.

Points to remember



Example :

C.N. $\rightarrow 6(sp^3d^2 / d^2sp^3)$



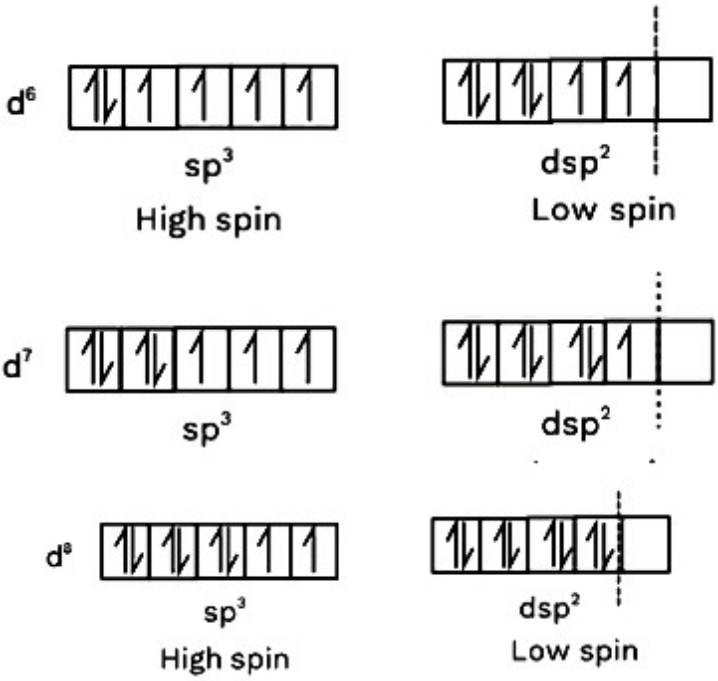
Points to remember

gly⁻ strength is near to en

	Spin (Low/High)
$\Rightarrow d^{0,1,2,3} \longrightarrow$ always d^2sp^3	–
$\Rightarrow d^{4,5,6,7}$ SFL $\rightarrow d^2sP^3$ WFL $\rightarrow sp^3d^2$	low spin High spin
$d^{8,9,10} \rightarrow$ always sp^3d^2	–

C.N. = $4(sp^3 / dsp^2)$

WFL SFL



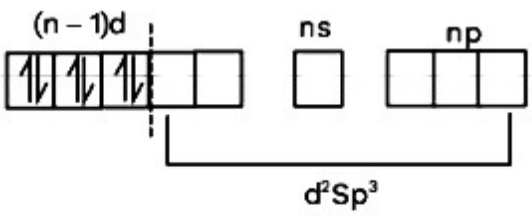
d^9, d^{10} → spin (No comment) for both WFL & SFL

$K_4[Fe(CN)_6]$

C.N. = $6(sp^3d^2 / d^2sp^3)$ ✓

$Fe^{2+} \rightarrow d^6$

$CN^- \rightarrow$ SFL



↳ Diamagnetic

↳ Inner orbital complex

↳ $\mu = 0$

($\because n = 0$)

↓ no. of unpaired e^-

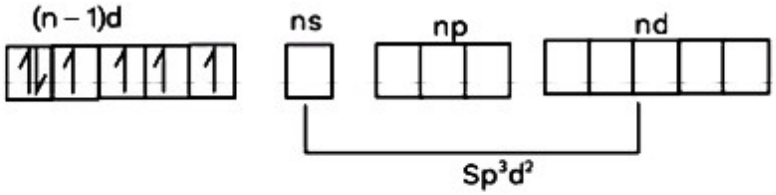
& $\mu = \sqrt{n(n+2)}$

$K_4[FeF_6]$

C.N. = $6(sp^3d^2 / d^2sp^3)$ ✓

$Fe^{2+} \rightarrow d^6$

($F^- \rightarrow$ WFL)



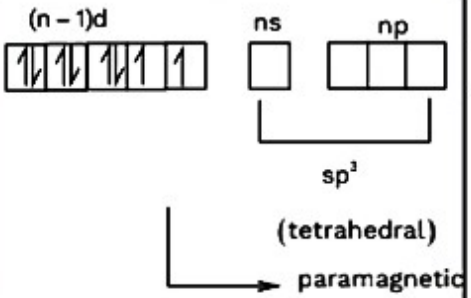
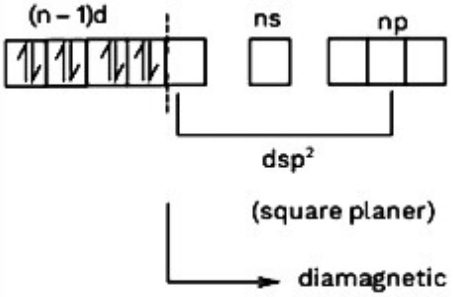
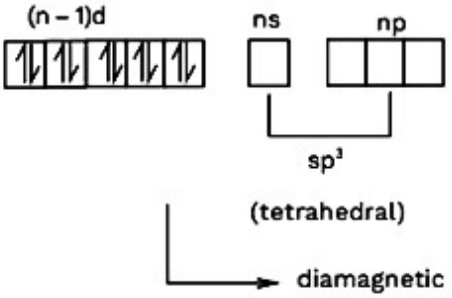
↳ Paramagnetic

↳ $\mu = \sqrt{4(4+2)}$

↳ $\mu = \sqrt{24}$ B.M.

↳ Outer orbital complex

↳ High spin complex

$NiCl_4^{2-}$	$Ni(CN)_4^{2-}$	$Ni(CO)_4$
C.N. = $4(sp^3 / dsp^2)$	C.N. = $4(sp^3 / dsp^2)$	C.N. = $4(sp^3 / dsp^2)$
$Ni^{2+} \rightarrow d^8(Cl^- \rightarrow WFL)$	$Ni^{2+} \rightarrow d^8(CN^- \rightarrow SFL)$	$Ni^{\circ} \rightarrow d^{10}(CO \rightarrow SFL)$
		

(1) Identity transition series of CMI

(a) If it belongs to 3d series find out its atomic number then add digit of tis atomic number

$Ni^{\circ} \longrightarrow 28 \Rightarrow 2 + 8 = 10$

$Ni^0 :\Rightarrow d^{10}$

If charge is present then add or remove electrons from above written result.

$Ni^{2+} \Rightarrow 10 - 2 = 8 \Rightarrow d^8$

$Ni^{\circ} \Rightarrow 10 - 0 = 10 \Rightarrow d^{10}$

$Ni^{4+} \Rightarrow 10 - 4 = 6 \Rightarrow d^6$

Tips and Trick

Configuration of transition metal (Sc to Cu) ion in complex (not applicable for Zn, Cd, Hg)

(b) If metal belongs to 4d and 5d series then find atomic number of its group 3d series metal and follow rule (a)

Like for $Pt^{2+} \Rightarrow$ Follow Ni^{2+} (3d series of Pt group element)

$\downarrow$

$Ni^{\circ} \rightarrow 28 \rightarrow 2 + 8 = 0$

$Ni^{+2} \rightarrow 10 - 2 = 8 \Rightarrow d^8$

So; Pt^{2+} also $\rightarrow d^8$

Points to remember

In case of $Al_2O_3 \Rightarrow Al^{3+}, O^{2-}$

Similarly

Case- I

$$\left[Pt(NH_3)_6\right]_1\left[PtCl_6\right]_2$$

$\Downarrow$

$$\left[Pt(NH_3)_6\right]^{2+}\left[PtCl_6\right]^-$$

$\rightarrow$ C.N. = 6 for Pt but charge on Pt is +2 (not possible)

∴ Pt should be Pt^{4+} to satisfy C.N. 6

⇓

So, complex should be

$\underbrace{[Pt(NH_3)_6]^{4+}[PtCl_6]^{2-}}_{\text{Correct}}$

(∴ Pt here is Pt^{4+} and C.N. is 6)

Case-II:

$[Pt(NH_3)_4][PtCl_6]$

↳ Use hit & trail for C.N.

C.N. of Pt is 4 → Pt^{2+}

$[Pt(NH_3)_4]^{2+}[PtCl_4]^{2-}$

(C.N. and charge are satisfied in both part)

Some examples-

$[Fe(NH_3)_6]_3[CO(CN)_6]_2 \Rightarrow [Fe(NH_3)_6]^{2+}[Co(CN)_6]^{3-}$

$[Fe_2(Fe(CN)_6] \Rightarrow Fe^{x+}[Fe(CN)_6]^{2+}$

⇓

(In this part charge on Fe ⇒ 6–2x)

So, if

$Fe = 6 - 2x$

Fe^{+x} → If x = 1 → +1; x = 2 → +2

$[Fe(CN)_6]^{2x-}$ → +4

+2 common oxidations state

$\Rightarrow \overset{+2}{Fe_2}\left[\overset{+2}{Fe}(CN)_6\right]$

$Fe_3[Fe(CN)_6]_2 \Rightarrow \overset{+2}{Fe_3}\left[\overset{+3}{Fe}(CN)_6\right]$

$Fe_4[Fe(CN)_6]_3 \Rightarrow \overset{+3}{Fe}_4\left[\overset{+2}{Fe}(CN)_6\right]$

$Fe[Fe(CN)_6] \Rightarrow Fe^{x+}[Fe(CN)_6]^{x-}$

⇓

So, charge on Fe = 6–x

	Fe^{x+}	$[Fe(CN)_6]^{x-}$
If x = 1	+1	+5
	+2	+4
	+3	+ 3 (Satisfy common O.S. of Fe)

$\Rightarrow \overset{+3}{Fe}\left[\overset{+3}{Fe}(CN)_6\right]$

C.N. = 5 ⇒ dsp^3/ sp^3d

⇓ geometry

Trigonal bipyramidal	Square pyramidal
(d_{z^2} orbital is used)	$d_{x^2-y^2}$ orbital is used
⇓	⇓
With CO & WFL Trigonal bipyramidal	With SFL geometry is square pyramidal(Except CO)

Example :

$Fe(CO)_5$

↓

C.N. = 5

$Fe^0 \rightarrow 3d^6 4s^2$

In complex (Fe^0) → $3d^8$

Points to remember

In heteroleptic complex

(a) If number of SFL ≥ number of WFL

⇓

Effect of SFL will be considered (dominates)

Exception

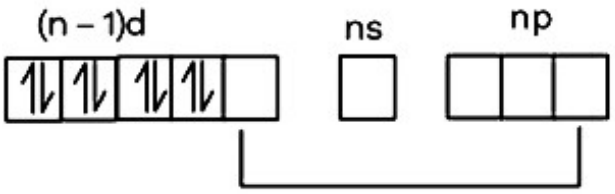
(i) $[CO(H_2O)_3F_3]$

(ii) $[Ni(NH_3)_2Cl_2]$

(b) If number of SFL < WFL

⇓

Effect of WFL to be considered.



dSp^3, diamagnetic

Generally

3d series CMI
- d^1 to d^6 → Generally show C.N. = 6
- d^7 to d^{10} → Generally show C.N. = 4

Also
- $Pt^{2+} \longrightarrow d^8 \Rightarrow$ C.N. = 4
- $Pt^{4+} \longrightarrow d^6 \Rightarrow$ C.N. = 6

Points to remember

∵ From d^1 to d^6
They have less electrons and to satisfy EAN rule they need more lone pair hence show greater C.N.

Drawbacks of V.B.T : -

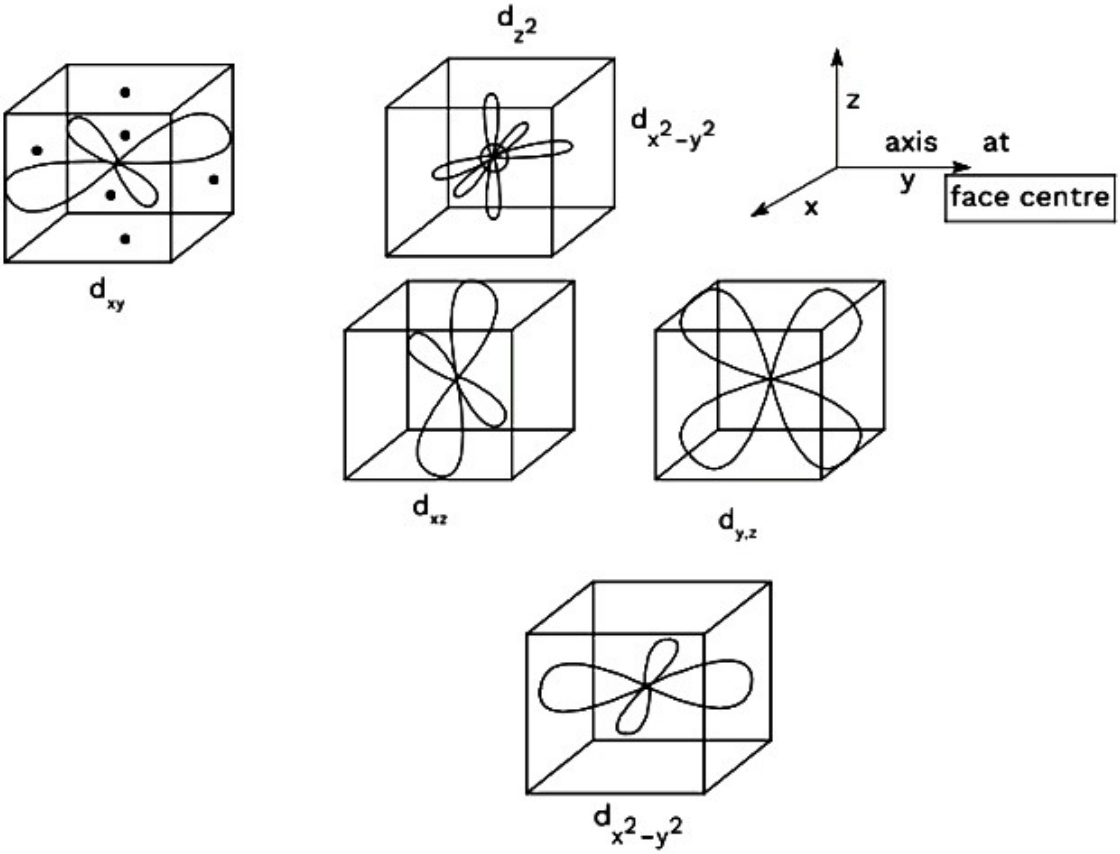
While the VBT theory, to a larger extent the formation, structures and magnetic behaviour of coordination compounds, it suffers from the following shortcoming.

1. It involves a number of assumptions $\left(\underset{\text{donor}}{C,N} \rightarrow SFL\right)(Halogen, o-donor \rightarrow WFL)$
2. It does not give quantitative interpretation of magnetic data.
   (∵ Here in VBT hybridization is calculated from unpared electrons
   Unpaired electrons $\xrightarrow{\checkmark}$ hybridization
   Hybridization $\xrightarrow{\times}$ unpaired electrons)
3. It does not explain colour of complex
4. It does not explain stability of complex
5. It does not distinguish between SFL and WFL
6. It does not make exact predictions regarding tetrahedral and square planar structures of 4-cordination number complexes.

Type of d-orbitals

t_2-orbiatls (d_{xy}, d_{yz}, d_{xz}) → electron density between axis

e-orbitals $(d_{x^2-y^2}, d_{z^2})$ → along the axis



Crystal Field Theory

Crystal Field theory (CFT)

- Used to explain drawbacks of V.B.T
- Crystal word is used for ionic compounds so base of crystal field theory is ionic interaction.
- Acc. to this theory CMI act as positive point charge
- Ligands
  - anionic → negative point charge
  - Neutral → dipole (+ −)

When ligand opproaches CMI then two type of forces will be present

1. force of attraction between nucleus of CMI (The charge) and lone pair of ligand (-ve charge)
2. force of repulsion between d-orbitals of CMI and lone pair of ligand (-ve charge)

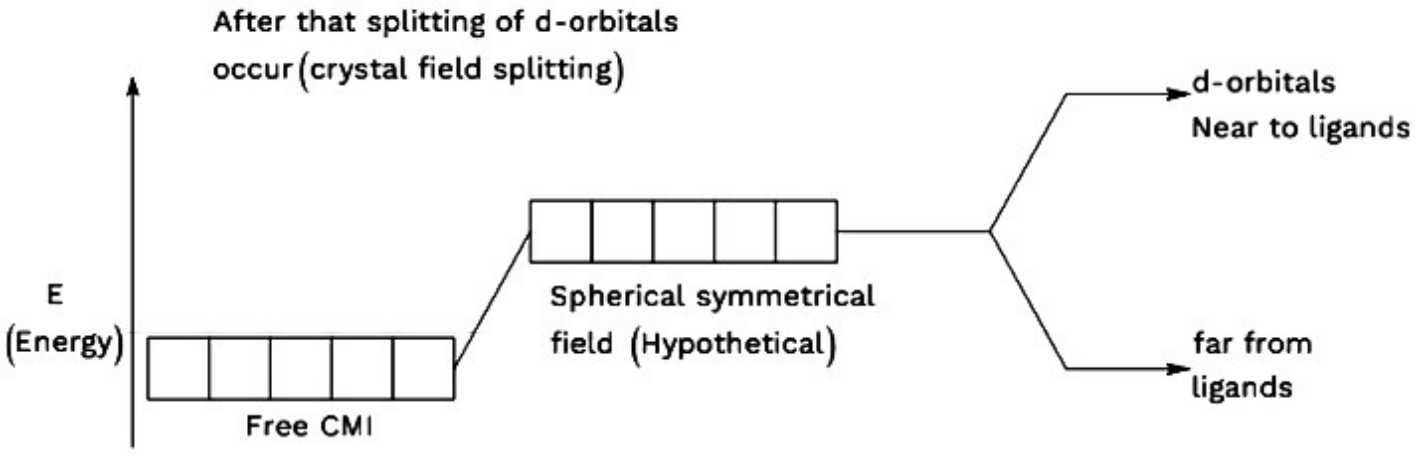
As attraction between CMI & ligands ↑ (increase)

⇓

Repulsion between d-orbital and ligands ↑

⇓

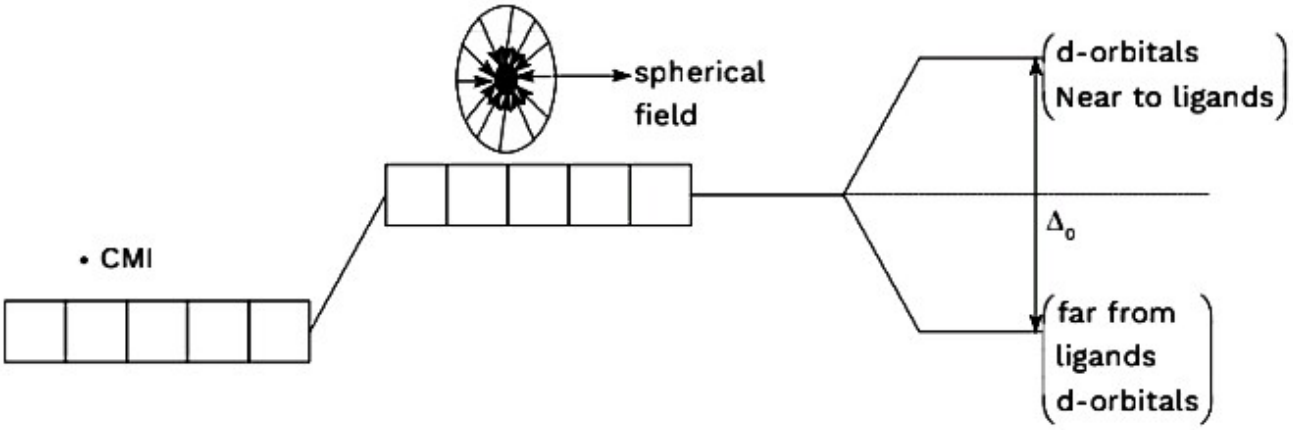
So, Energy of d-orbital of CMI ↑



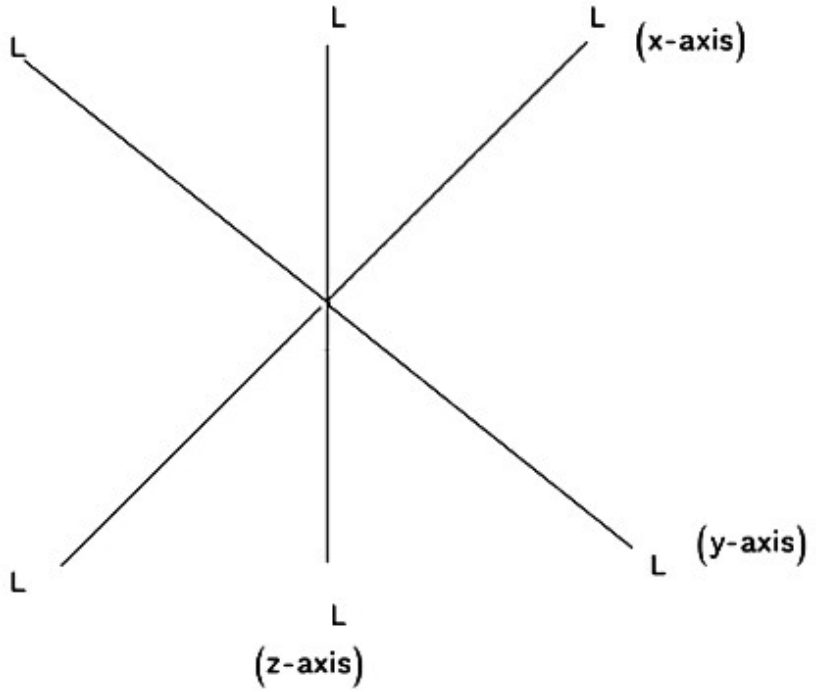
In free CMI d-orbitals are degenerate

In hypothetical spherical field ligands are equally near to all d-orbitals and d-orbitals remain degenerate (orbitals of same energy are degenerate orbitals) but practically some of the d-orbitals are near to ligand and other are far from ligand due to geometry. So d-orbitals experienced different-different repulsion, so they splits into different energy set which is called as crystal field splitting.

Energy different between d-orbitals is known as splitting energy (Δ)



Splitting in octahedral :-



In octahedral ligands are near to e-orbitals $\left(d_{x^2-y^2}\ \&\ d_{z^2}\right)$ because they are present on the axis and also ligands are far from t_2 orbitals $\left(d_{xy}, d_{yz}, d_{xz}\right)$

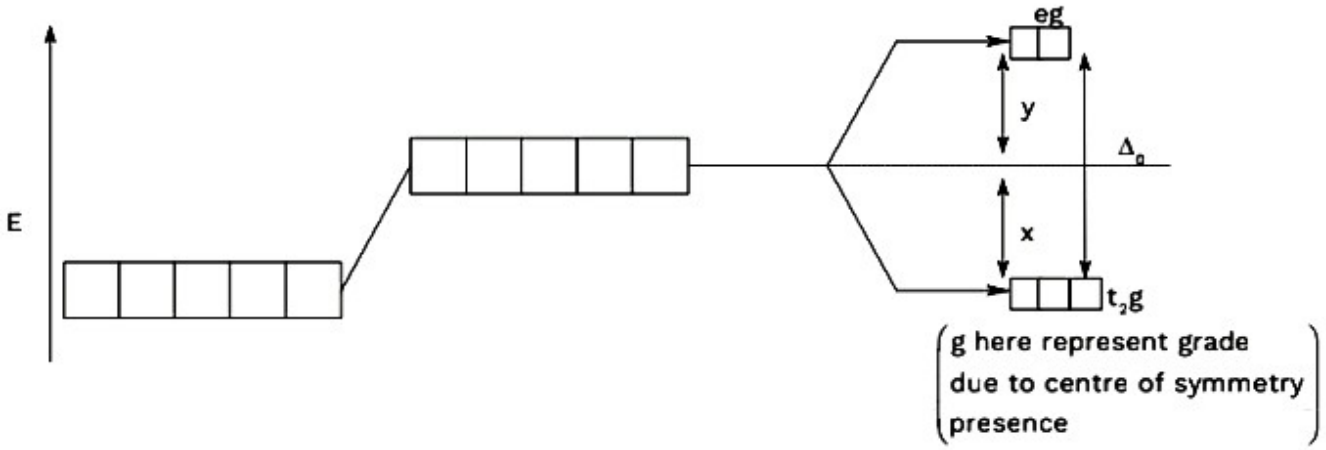
So energy of e-orbitals increases (as they feel more repulsion)

$x+y=\Delta_0$

$-6x+4y=0$ (in $t_{2g}\rightarrow$ 6 electron can come)

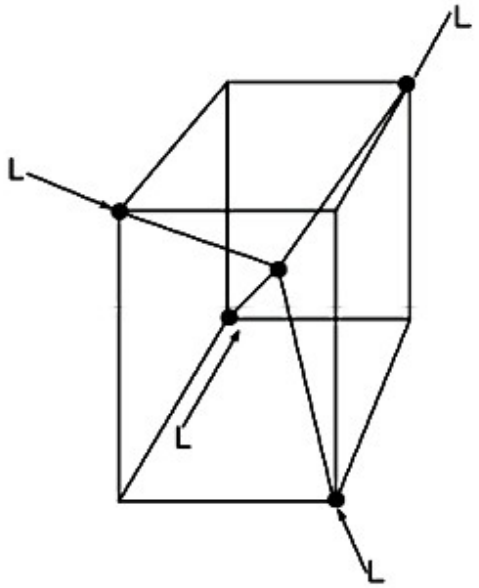
$\Rightarrow x=\frac{2}{5}\Delta_0=-0.4\Delta_0$ (–ve sign because w.r.t bary centre x energy ↓)

$\Rightarrow y=\frac{3}{5}\Delta_0=+0.6\Delta_0$ (+ve sign as w.r.t bary can be y energy ↑)



(g here represent grade due to centre of symmetry presence)

Tetrahedral splitting :

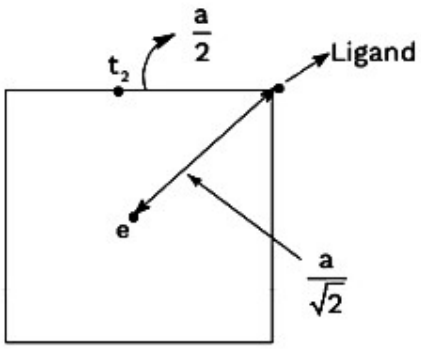


1 ligand → 3edge centre

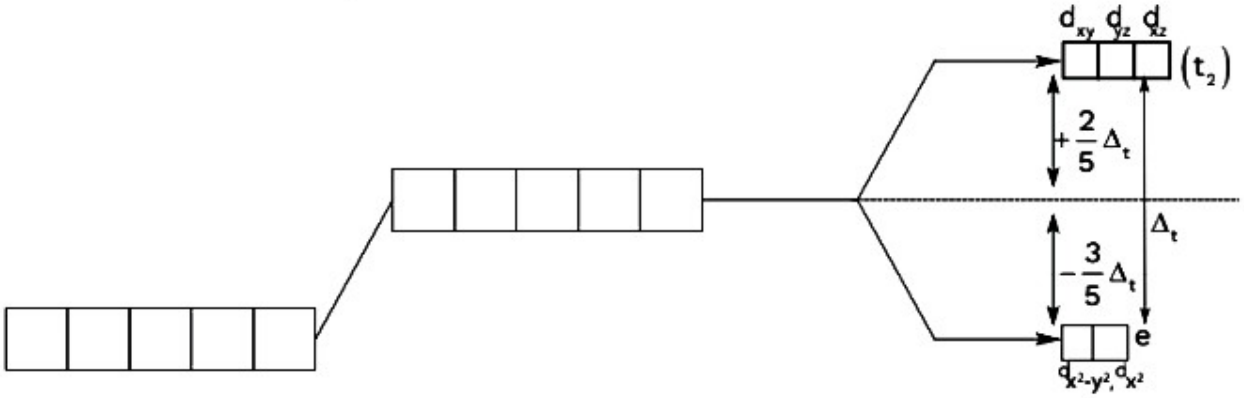
4 ligand → 12 edge centre

dxy,dyz,dxz

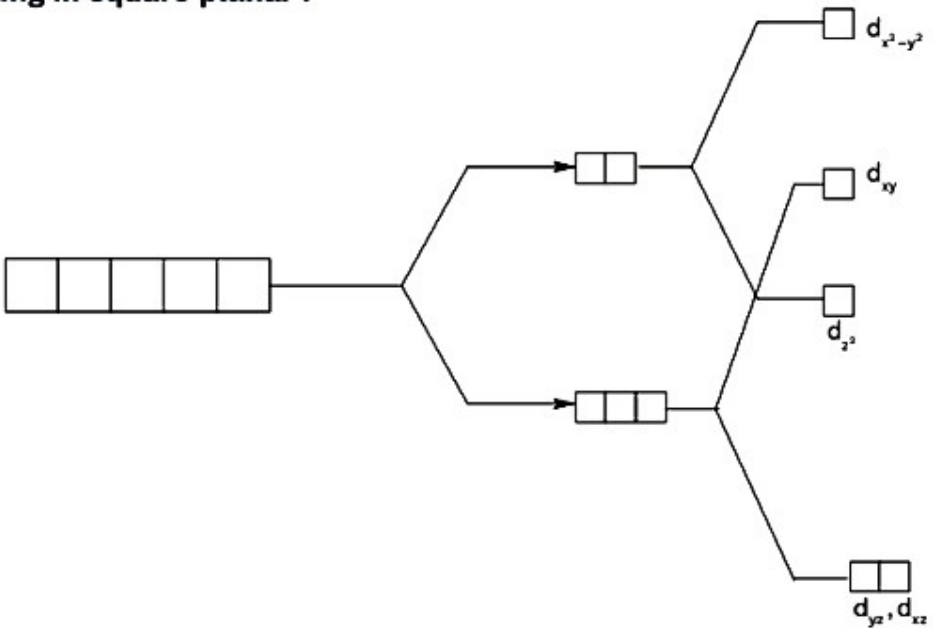
Ligands are not perfectly present in direction of any d-orbital

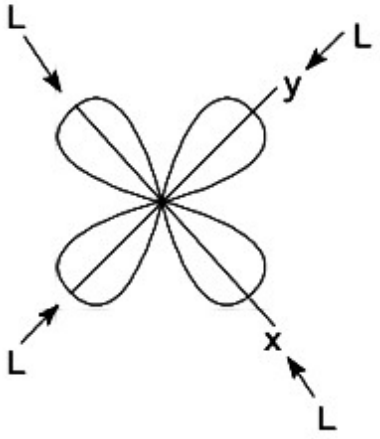


So ligands are near to t_2 set of orbitals and far from e-set of orbitals.

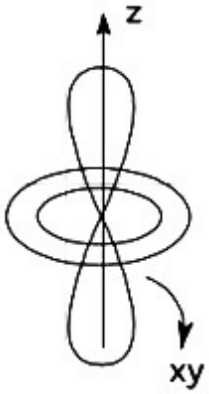


Splitting in square planar :





Ligand approaches through x & y axis.



Energy order:-

(Some electron cloud is present in xy plane)

$d_{x^2-y^2} > d_{xy} > d_{z^2} > \underbrace{d_{yz} = d_{xz}}$

perpendicular to xy plane

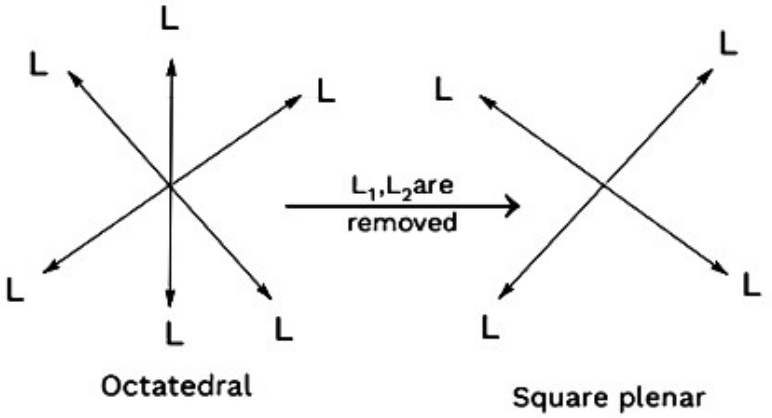
Orbital is between the x,y axis (approching axis of ligand)

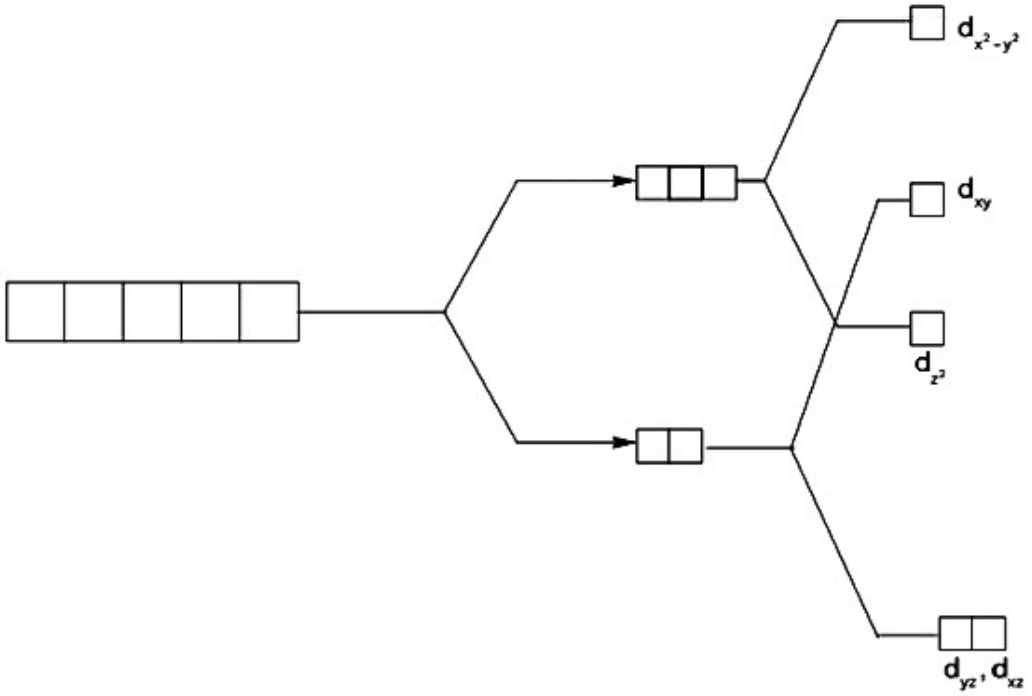
maximum repulsion

(∵ Ligards approaches along x,y axis)

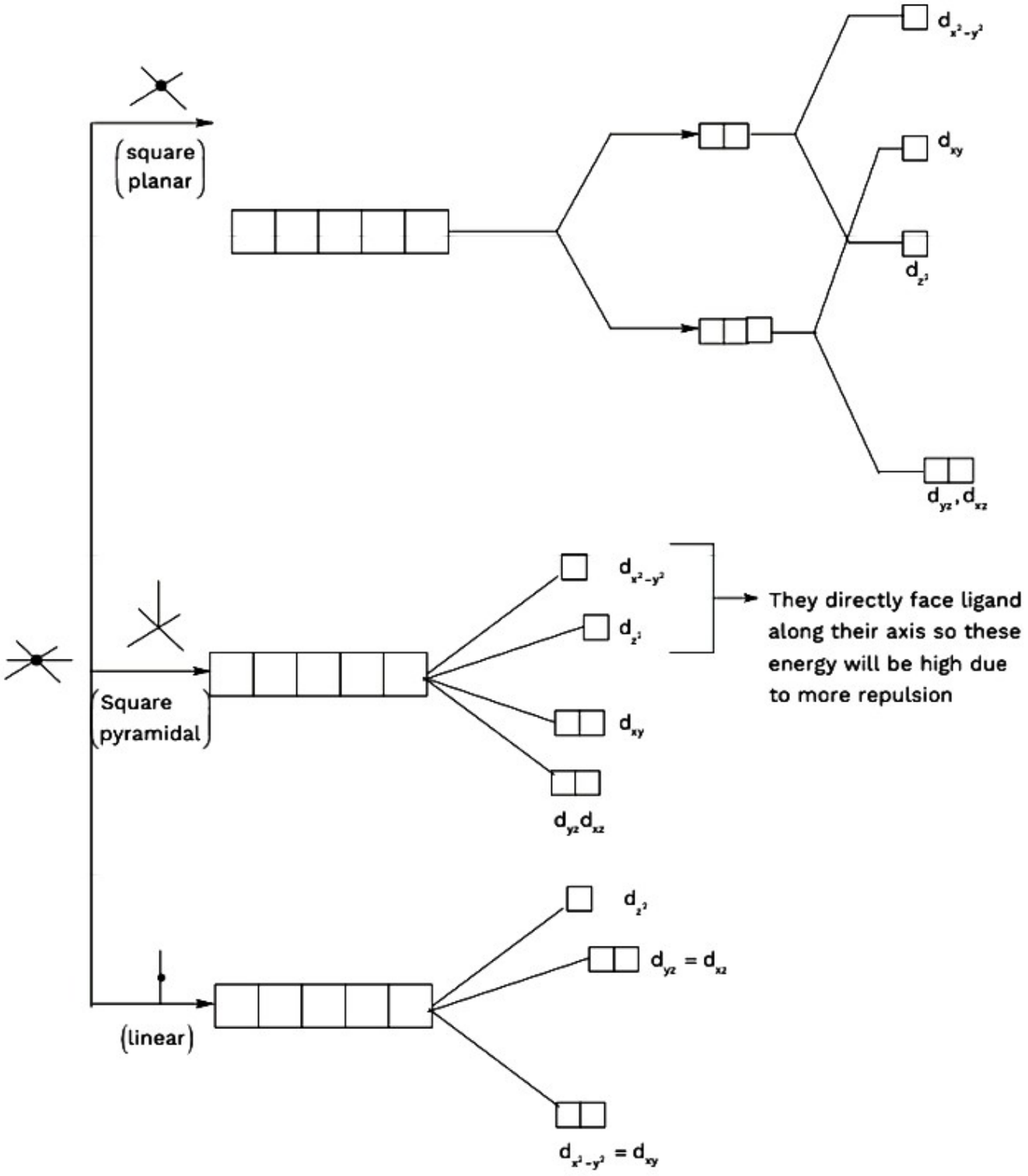
Q17 **If in octahedral complex ligands are removed from one axis, (z-axis) then what will be the energy order of orbitals finally?**

Sol:





Energy order → $d_{x^2-y^2} > d_{xy} > d_{z^2} > d_{yz} = d_{xz}$

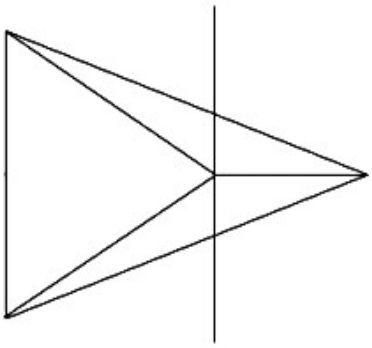


(Here ligands are removed from x & y axis, so repulsion will be minimum in $d_{x^2-y^2}$ & d_{xy} orbitals along the z-axis where ligand approaches, so here repulsion will be maximum & there for energy highest.)

Trigonal planar $(sp^2) \Rightarrow$ xy plane

Energy order $\rightarrow$ $\underbrace{d_{x^2-y^2} = d_{xy}}_{\text{They faces similar repulsion so they have same energy}} > d_{z^2} > d_{yz} = d_{xz}$

Triangle bipyramidal :-



$\underbrace{d_{z^2}}_{\downarrow} > \underbrace{d_{x^2-y^2} = d_{xy}}_{\text{xy plane}} > d_{yz} = d_{xz}$

d_{z^2} → Z-axis → 2 lobe & 2 ligand → more repulsion

$d_{x^2-y^2} = d_{xy}$ → xy plane → 8 lobe → 3 ligand (Comparative less repulsion than d_{z^2})

Note

As attraction between CMI and ligand increases than repulsion between d-orbital of CMI and ligand also increase.

Factors affecting splitting energy (Δ)

(1) Splitting energy $\propto$ positive charge on CMI

Example :

(i) $\left[Fe(NH_3)_6\right]^{2+} < \left[Fe(NH_3)_6\right]^{3+}$

(ii) $\underset{\substack{\uparrow \\ \text{Act as WFL here}}}{\left[Co(H_2O)_6\right]^{2+}} < \underset{\substack{\uparrow \\ \text{Act as SFC here}}}{\left[Co(H_2O)_6\right]^{3+}}$

(2) Splitting energy $\propto Z_{eff}$

$$3d \xrightarrow{(\Delta)} 4d \longrightarrow 5d$$

$Z_{eff} \uparrow$

$\Delta \uparrow$

Example :

$NiCl_4^{2-} < PdCl_4^{2-} < PtCl_4^{2-}$

↓	↓	↓
Act as	Act as	Act as
WFL	SFL	SFL

Also for 3d series :

$\underbrace{Mn^{2+} \rightarrow Fe^{2+} \rightarrow Co^{2+}}_{NH_3 \text{ act as WFL here}} \longrightarrow \underbrace{Ni^{2+} \rightarrow Cu^{2+}}_{NH_3 \text{ act as SFL}}$

$Z_{eff} \uparrow$

3. $\Delta \propto$ strength of ligand.

Some Important Points:

(1) $Fe_2[Fe(CN)_6] \rightarrow Fe^{2+}[Fe(CN)_6]^{4-}$

(+2)

- Ferroferrocyanide
- White ppt.
- 2, 2-complex

(2) $Fe_4[Fe(CN)_6]_3 \rightarrow Fe^{3+}[Fe(CN)_6]^{4-}$

(+2)

- Prussian blue
- 3, 2-complex
- Ferriferrocyanide

(3) $Fe_3[Fe(CN)_6]_2 \rightarrow Fe^{2+}[Fe(CN)_6]^{3-}$

(+3)

- Turn bull's blue
- 2, 3-complex
- Ferroferricyanide

(4) $Fe[Fe(CN)_6] \rightarrow Fe^{3+}[Fe(CN)_6]^{3-}$

(+3)

- Ferroferricyanide
- 3, 3-complex
- Red colouration

(5) $H_2[PtCl_6] \rightarrow H^+[PtCl_6]^{2-}$

↓

IUPAC → hydrogenhexachloridoplatinate (IV)

or

dihydrogenhexachloridoplatinate

Q18 **100 ml of 0.01 m solution of complex compound $IrBr_3 5NH_3$ was treated with excess of $AgNO_3$ solution and precipitate of AgBr was obtained. Then find the number of moles of AgBr obtained?**

Sol: Complex would be $[Ir(NH_3)_5Br]Br_2$

Now number of moles of complex = M × V

$$= 0.01 \times \frac{100}{1000} = 0.001$$

$\therefore$ Number of moles of AgBr precipitated = 2 × 0.001 = 0.002

** $\overset{\oplus}{N}O$ → Nitrosonium or Nitrosylium

Geometry :

Square planar	**Octahedral**	**Tetrahedral**
Here 4 ligands approaches/orbital $\left(d_{x^2-y^2}\right)$	Here 6 ligands approaches 2-orbitals $\left(d_{x^2-y^2}, d_{z^2}\right)$	Here 4 ligands approaches 3 orbitals $\left(d_{xy}, d_{yz}, d_{xz}\right)$

$\Delta_{sp} > \Delta_0 > \Delta_t$

↓

Average repulsion

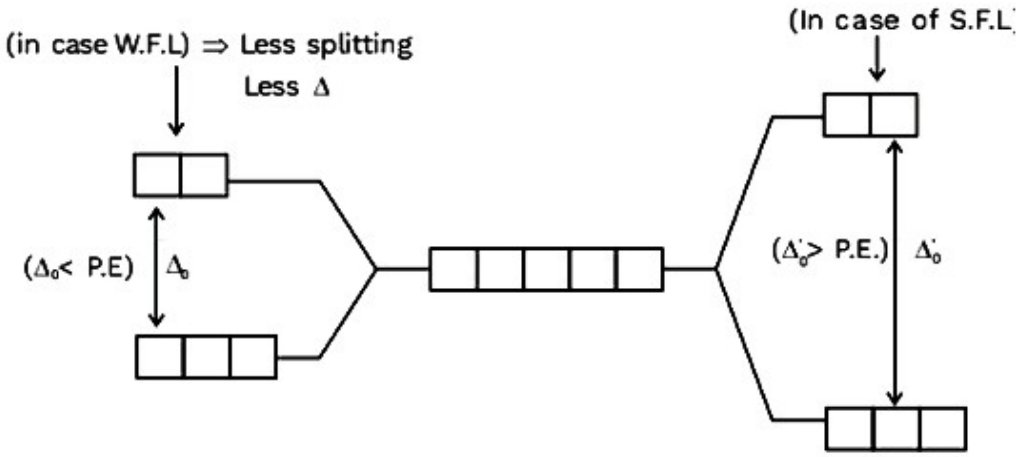
high(4 ligands w.r.t 1 orbital)

Per orbital average repulsion $\Rightarrow \frac{1}{4}\Delta_{sp} = \frac{2}{6}\Delta_0 \approx \frac{3}{4}\Delta_t$

$\therefore \quad \Delta_{sp} = \frac{4}{3}\Delta_0$

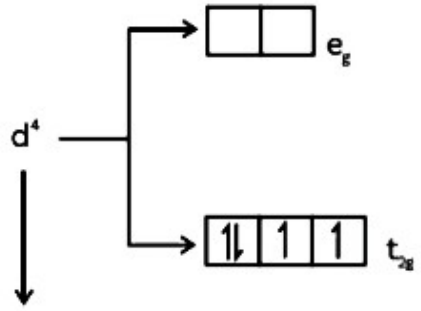
$$\boxed{\Delta_t \approx \frac{4}{9}\Delta_0}$$

Why low spin and high spin complex are formed



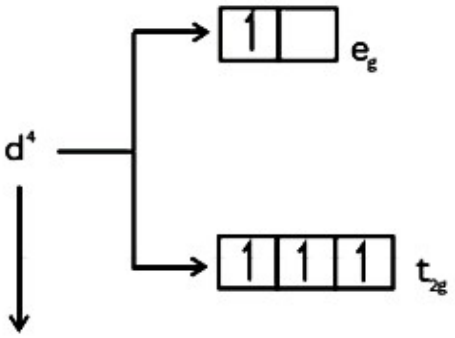
When 4th electron enter in e_g orbital it require Δ_0 energy but when 4th electron enter in t_{2g} orbital it requires pairing energy (P.E.)

If, Δ_0 > P.E. : it means effect of ligand is SFL ⇒ Hence 4th electron will be paired up in t_{2g} orbital ⇒ complex will be 'Low spin'



Electronic configuration → t_{2g}^4 , e_g^0

If, Δ_0 < P.E. : it means effect of ligand is WFL ⇒ Hence 4th electron is enter into e_g orbital ⇒ complex will be 'high spin'

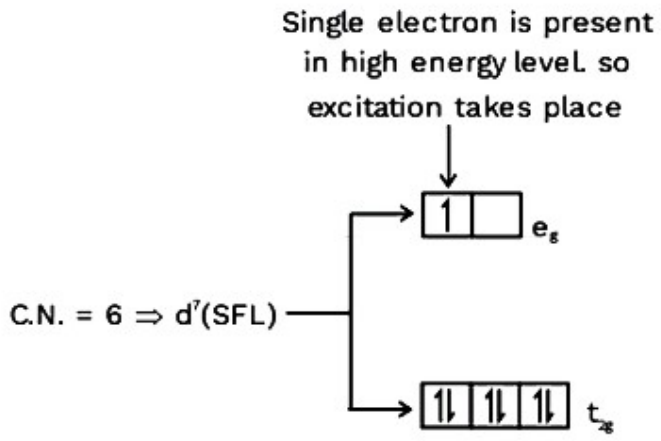


Electronic configuration → t_{2g}^3 , e_g^1

Electronic configuration of d-orbitals in from of t_{2g} and e_g
In C.N. = 6

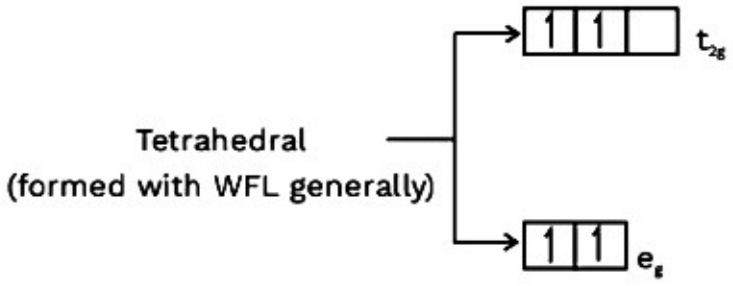
	WFL	SFL
d^0	t_{2g}^0, e_g^0	t_{2g}^0, e_g^0
d^1	t_{2g}^1, e_g^0	t_{2g}^1, e_g^0
d^2	t_{2g}^2, e_g^0	t_{2g}^2, e_g^0
d^3	t_{2g}^3, e_g^0	t_{2g}^3, e_g^0
d^4	t_{2g}^3, e_g^1	t_{2g}^4, e_g^0
d^5	t_{2g}^3, e_g^2	t_{2g}^5, e_g^0
d^6	t_{2g}^4, e_g^2	t_{2g}^6, e_g^0
d^7	t_{2g}^5, e_g^2	t_{2g}^6, e_g^1
d^8	t_{2g}^6, e_g^2	t_{2g}^6, e_g^2
d^9	t_{2g}^7, e_g^2	t_{2g}^6, e_g^3
d^{10}	t_{2g}^8, e_g^2	t_{2g}^6, e_g^4

Transference Explanation :



Q19 **Find the electronic configuration of d^4 tetrahedral ?**

Sol:



So, E.C. will be $\Rightarrow e_g^2, t_{2g}^2$

Q20 **Find the electrons present in 3d orbitals in $[Co(en)_2]^{2+}$?**

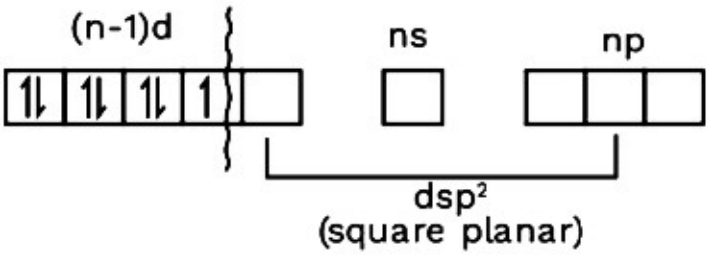
	$d_{x^2-y^2}$	d_{z^2}	d_{xy}	d_{yz}	d_{xz}
(A)	**0**	**1**	**2**	**2**	**2**
(B)	**0**	**2**	**1**	**2**	**2**
(C)	**1**	**1**	**1**	**2**	**2**
(D)	**2**	**1**	**0**	**2**	**2**

Sol: **(B)**

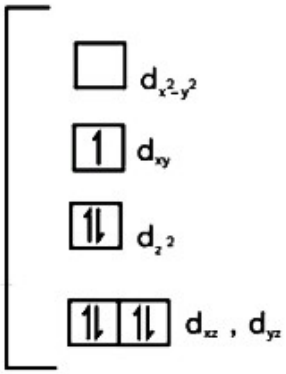
In $[Co(en)_2]^{2+}$

$Co^{2+} \rightarrow d^7$(en is SFL)

$\Rightarrow$



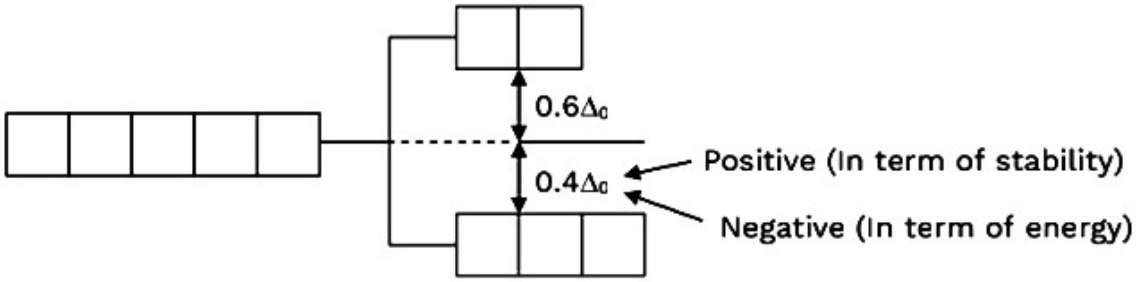
In square planar complex
(As ligand is SFL)
Paring occur and then
high energy orbitals are filled. ⇒



C.F.S.E (Crystal field stabilization energy)
Energy / decrease in energy of system due to crystal field splitting.

CFSE in octahedral complex:
(a) In terms of stability

$$\text{C.F.S.E.} = n_{t_{2g}} \times (0.4\Delta_0) - n_{eg} \times (0.6\Delta_0) - x \times (\text{P.E.})$$



Or

(b) In term of energy:

$$\text{C.F.S.E.} = n_{t_{2g}} \times (-0.4\Delta_0) + n_{eg} \times (0.6\Delta_0) + x \times (\text{P.E.})$$

$n_{t_{2g}} \rightarrow$ no. of electron in t_{2g}

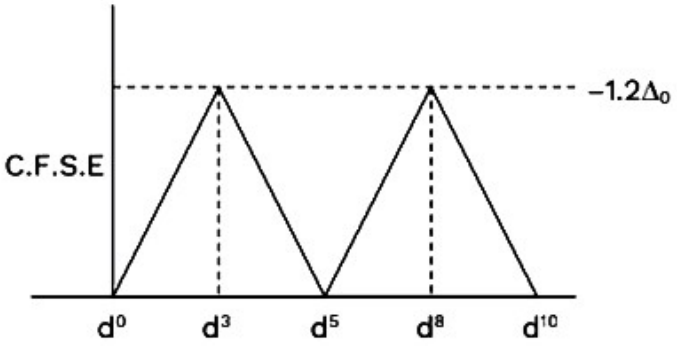
$n_{eg} \rightarrow$ no. of electron in eg

P.E. $\rightarrow$ Pairing Energy

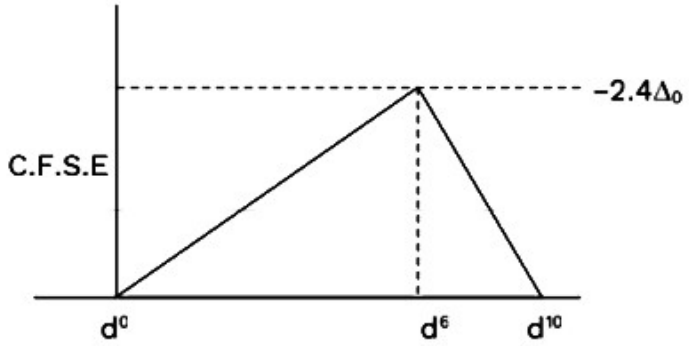
$x \rightarrow$ no. of extra pairs.

	WFL		SFL	
d^n	Electronic configuration	C.F.S.E.	Electronic configuration	C.F.S.E.
d^0	$t_{2g}^0 e_g^0$	0	$t_{2g}^0 e_g^0$	0
d^1	$t_{2g}^1 e_g^0$	$-0.4\Delta_0$	$t_{2g}^1 e_g^0$	$-0.4\Delta_0$
d^2	$t_{2g}^2 e_g^0$	$-0.8\Delta_0$	$t_{2g}^2 e_g^0$	$-0.8\Delta_0$
d^3	$t_{2g}^3 e_g^0$	$-1.2\Delta_0$	$t_{2g}^3 e_g^0$	$-1.2\Delta_0$
d^4	$t_{2g}^3 e_g^1$	$-0.6\Delta_0$	$t_{2g}^4 e_g^0$	$-1.6\Delta_0$ + 1P.E
d^5	$t_{2g}^3 e_g^2$	0	$t_{2g}^5 e_g^0$	$-2.0\Delta_0$ + 2P.E
d^6	$t_{2g}^4 e_g^2$	$-0.4\Delta_0$	$t_{2g}^6 e_g^0$	$-2.4\Delta_0$ + 2P.E
d^7	$t_{2g}^5 e_g^2$	$-0.8\Delta_0$	$t_{2g}^6 e_g^1$	$-1.8\Delta_0$ + 1P.E
d^8	$t_{2g}^6 e_g^2$	$-1.2\Delta_0$	$t_{2g}^6 e_g^2$	$-1.2\Delta_0$
d^9	$t_{2g}^6 e_g^3$	$-0.6\Delta_0$	$t_{2g}^6 e_g^3$	$-0.6\Delta_0$
d^{10}	$t_{2g}^6 e_g^4$	0	$t_{2g}^6 e_g^4$	0

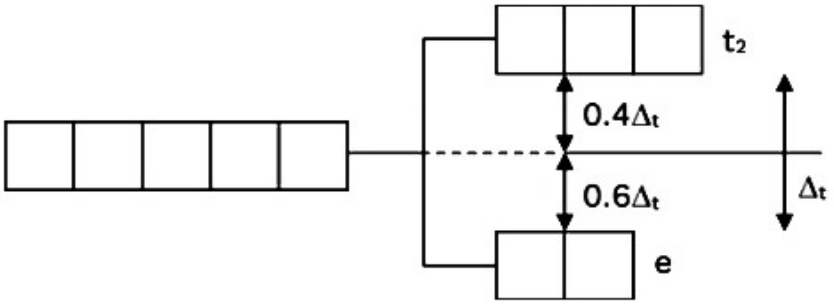
In case of WFL



In case of SFL



Case-2
C.F.S.E. in tetrahedral complexes:

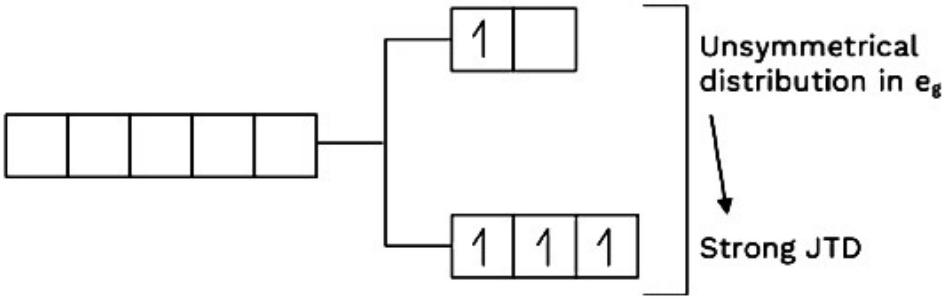


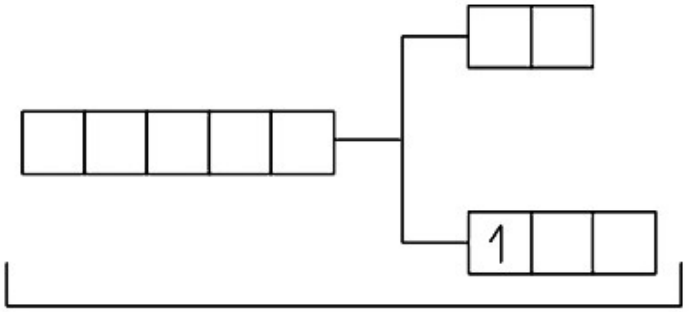
$$C.F.S.E. = n_e \times (0.6\Delta_t) - n_{t_2} \times (0.4\Delta_t)$$

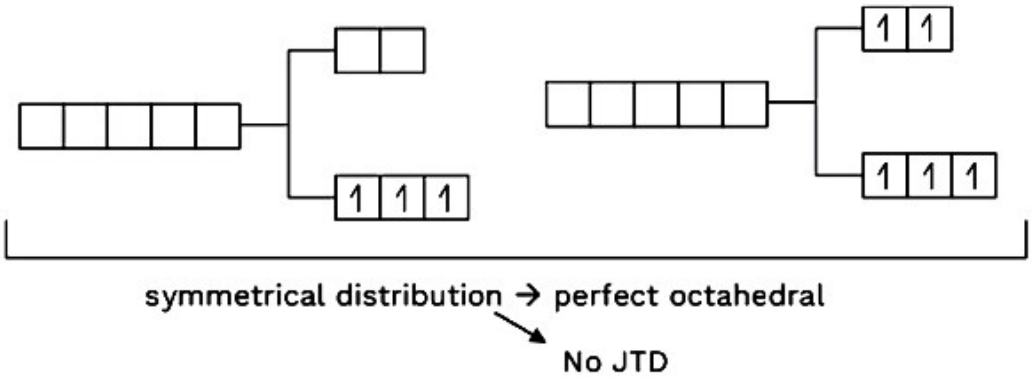
$n_e \rightarrow$ no of electrons in e set
$n_{t_2} \rightarrow$ no of electrons in t_2 set

*→ No need of pairing energy because tetrahedral complexes form in presence of WFL generally.

Jahn Teller Distortion: (JTD)
In octahedral geometry when either t_{2g} or e_g of orbitals are unsymmetrically filled then octahedral geometry remains no more perfect and this is called as Jahn Teller Distortion.







When e_g is unsymmetrically filled then it is called Strong Jahn Teller Distortion (SJTD)
Because here ligands are directly the direction of d-orbitals ($d_{x^2-y^2}, d_{z^2}$)

When t_{2g} set is unsymmetrically filled then it is called weak Jahn Teller distortion (WJTD)

	WFL			SFL		
d^n	t_{2g}	eg	JTD	t_{2g}	eg	JTD
d^1	1	0	W	1	0	W
d^2	2	0	W	2	0	W
d^3	3	0	X	3	0	X
d^4	3	1	S	4	0	W
d^5	3	2	X	5	0	W
d^6	4	2	W	6	0	X
d^7	5	2	W	6	1	S
d^8	6	2	X	6	2	X
d^9	6	3	S	6	3	S
d^{10}	6	4	X	6	4	X
d^0	0	0	X	0	0	X

Example:

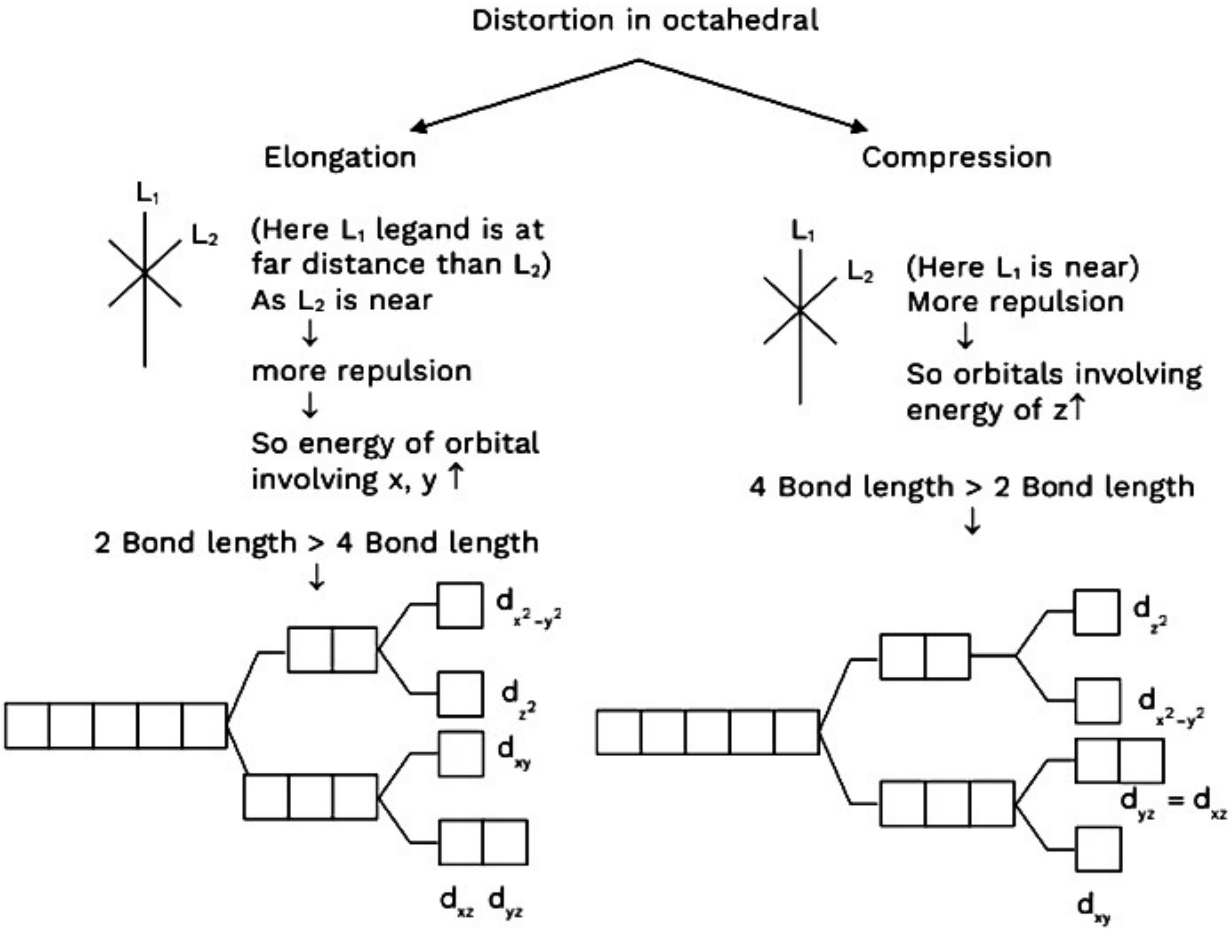
$\left[Cu(H_2O)_6\right]^{2+}$

- Perfect octahedral ×
- Distorted octahedral ✓
- All 6 Cu-O distance are not equal because it is distorted octahedral.

Reason → In the above complex CMI has d^9 configuration

⇒ So, it show strong John teller distortion

⇒ ∴ distorted octahedral.



Q21 **In $\left[Cu(H_2O)_6\right]^{2+}$ has 4-bond length < 2-bond length, then find the orbital in which unpaired electron is present**

(A) $d_{x^2-y^2}$ **(B)** d_{z^2} **(C)** p_x **(D)** d_{xy}

Sol: **(A)**

Colour of Complex:

Colour spectrum → VIBGYOR (390 nm to 700 nm)

Primary colour → R, Y, B

Secondary colour (Formed by mixing Primary colours)

R + Y → O

Y + B → G

R + B → V

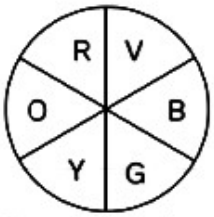
Tertary colour → V + B = I

Colour of compound is usually complementary colour of light absorbed by electron during excitation

Complementary class of R→ G

B → O

Y → V



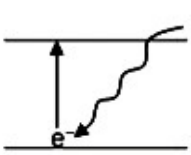
Neutral Colour

Black ⇒ No Colour

White ⇒ All Colour (VIBGYOR)

Grey ⇒ R + G, B + O, Y + V — complementary colour relationship

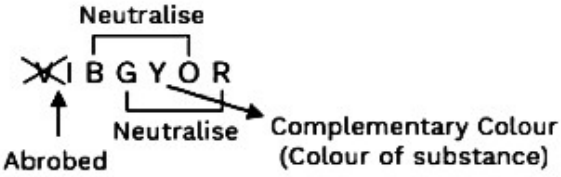
made by mixing complementary colours



If here violet light is abrobed from white light.

↓

Then colour of compound would be yellow (Complementary to violet)



⇒ Complexes show colour due to

d-d transition
(splitting of d-orbital)
In presence of ligand
↓
Occur in an atom
↓
self motivation ⇒ transition ↓
darkness↓

Charge transfer



(potential difference)
↓
Occur between two atoms
↓
External motivated
↓
transition↑
darkness↑

⇒ d–d transition

1. due to splitting

Example:

(a) $CuSO_4$ (anhydrous) → white

$CuSO_4 \cdot 5H_2O$ → Blue

(b) $FeSO_4$ (anhydrous) → White

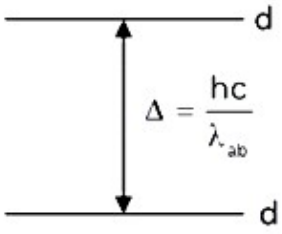
$FeSO_4 \cdot 7H_2O$ → Green

In presence of ligands splitting take place

↓

Electron transition takes place due to different energy level of d-orbital

2. Colour of complex depends on Δ



Where $\Delta = \frac{hc}{\lambda_{ab}}$

λ_{ab} → absorb wavelength

Colour will be complementary colour of λ_{ab}

Example:

If $\left[Ni(NH_3)_6\right]^{2+}$ is blue in colour then absorb wavelength (λ_{ab}) will be orange

Q22 **Which of the following has higher value of λ_{ab} $\underset{I}{\left[Ni(NH_3)_6\right]^{2+}}$, $\underset{II}{\left[Ni(H_2O)_6\right]^{2+}}$**

Sol: We know that $\Delta = \frac{hc}{\lambda_{ab}}$

$\Rightarrow \Delta \downarrow \Rightarrow \lambda_{ab} \uparrow$

$\Rightarrow$ In II complex → H_2O (WFL)
→ Less Δ
→ High λ_{ab}

Q23 **Match them correctly**

Complex	Wavelength absorbed (λ_{ab})
(I) $\left[Co(CN)_6\right]^{3-}$	(A) Violet
(II) $\left[Co(en)_3\right]^{3+}$	(B) Green
(III) $\left[Co(NH_3)_6\right]^{3+}$	(C) Blue
(IV) $\left[Co(H_2O)_6\right]^{3+}$	(D) Yellow

Sol: $\Delta \rightarrow I > II > III > IV$

$\lambda_{ab} \rightarrow \underset{\text{Violet}}{I} < \underset{\text{Blue}}{II} < \underset{\text{Green}}{III} < \underset{\text{Yellow}}{IV}$

So,

$I \rightarrow A$
$II \rightarrow C$
$III \rightarrow B$
$IV \rightarrow D$

How the match colour with complex:

Arrange Complex → 1st step (As per Δ)
→ 2nd step (As per λ_{ab})

If absorbed colour is given (λ_{ab})
↓
Directly match λ_{ab} with colour

If observed colour is given
↓
Find complementary colour and then match λ_{ab} with complementary colour
(∵ here complementary colour will be absorbed)

Q24 **Match the column correctly**

Complex	Colour of Complex
(I) $\left[Ni(en)_3\right]^{2+}$	**(A) Violet**
(II) $\left[Ni(en)_2(H_2O)_2\right]^{2+}$	**(B) Violet-Blue**
(III) $\left[Ni(en)(H_2O)_4\right]^{2+}$	**(C) Blue**
(IV) $\left[Ni(H_2O)_6\right]^{2+}$	**(D) Green**

Sol: (I) → A, (II) → B (III) → C, (IV) → D

	(A)	(B)	(C)	(D)
Colour →	Violet	Violet Blue	Blue	Green
↓	↓	↓	↓	↓
Complementary Colour →	yellow	yellow orange	orange	red

Δ → I > II > III > IV
↓ ↓
(en → SFL) WFL(H_2O)
λ_{ab}

I < II < III < IV

As per complementary Colour → Yellow | Yellow Orange | Orange | Red

colour → Violet (A) | Violet Blue (B) | Blue (C) | Green (D)

Points to remember

In octahedral complex removal of electron from t_{2g} C.F.S.E. ↓

In octahedral complex removal of electron from eg. C.F.S.E. ↑

Example:

In $[Co(Phen)_3]^{2+}$	In $[Fe(Phen)_3]^{2+}$
↓	↓
If CMI oxidises to +3 state	If CMI oxidises to +3 state
↓	↓
$Co^{2+} \rightarrow Co^{3+}$, $d^7 \rightarrow d^6$	$Fe^{2+} \rightarrow Fe^{3+}$, $d^6 \rightarrow d^5$, SFL
↓	↓
e^- remove from e_g	e^- remove from t_{2g}
↓	↓
C.F.S.E.↑	C.F.S.E.↓

d^0, d^{10}, d^5 (WFL) → d–d transition not possible

→ Symmetrical distribution (Half filled)

→ All orbitals are filled

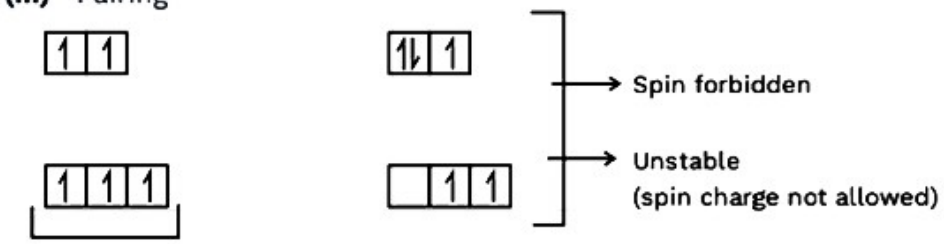
→ No electron for transition

In d^5 (WFL) d—d transition not possible because

(i) After transition symmetry is disturbed

(ii) Number of exchanges ↓

(iii) Pairing



Before transition d^5(WFL)

[After transition d^5 (WFL)]

Q25 **Colour of light absorbed by an aqueous solution of $CuSO_4$?**

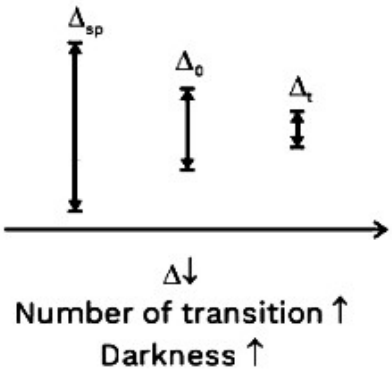
(A) Orange-red **(B) Blue-green**

(C) Yellow **(D) Violet**

Sol: As aq.$CuSO_4$ is blue
So, its complementary colour will be orange/orange-red

We know that

$\Delta_{sp} > \Delta_0 > \Delta_t$



∴ Square planar complexes are generally colourless
Tetrahedral complexes are more intense in comparison to octahedral complexes.

Example :

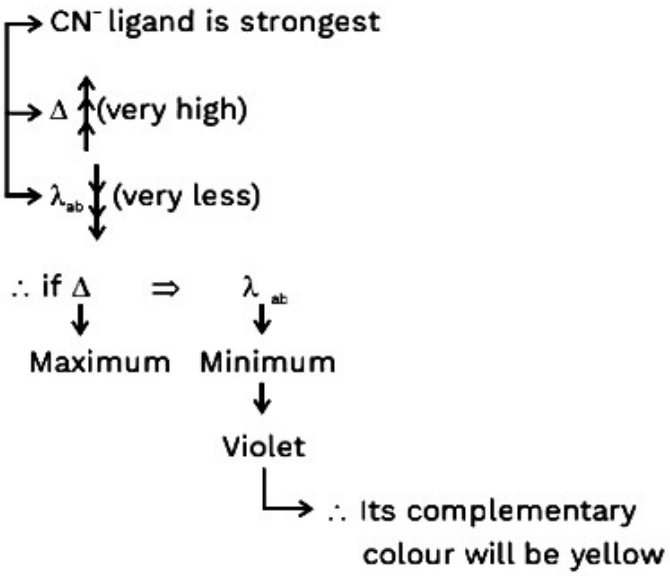
$COCl_4^{2-}$ and $[Co(H_2O)_6]^{2+}$

$COCl_4^{2-}$ → intense blue
$COCl_4^{2-}$ → Tetrahedral (sp^3)

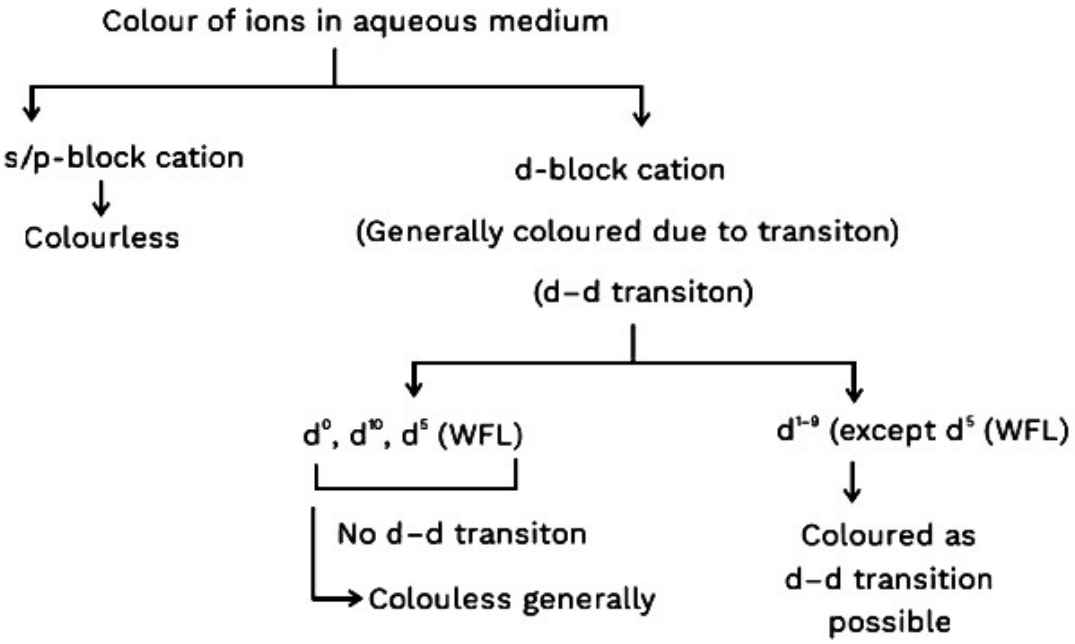
$[Co(H_2O)_6]^{2+}$ → Light pink
$[Co(H_2O)_6]^{2+}$ → Octahedral (∵ C.N. = 6)

Octahedral cyano complexes are generally pale yellow in colour

Reason :



(1) Colour of ions in aqueous medium:



Example :

$[FeF_6]^{3-}$ Colourless

$[CoF_6]^{3-}$ Coloured

$Fe^{3+} \rightarrow d^5$(WFL)

$[M(H_2O)_6]^{x+}$

$d^{1 \text{ or } 9}$	$d^{2 \text{ or } 8}$	$d^{0 \text{ or } 10}$
↓	↓	↓
Blue	Green	Colourless

$d^5_{(WFL)}$
- Fe^{2+}, $Ni^{2+} \rightarrow$ Green
- Cu^{2+}, $Cr^{2+} \rightarrow$ Blue
- Mn^{2+}, $Co^{2+} \rightarrow$ Light pink
- $Fe^{3+} \rightarrow$ Pale yellow

$K_2Cr_2O_7^{2-}|H^+ \xrightarrow{\text{Reducing agent}} Cr^{3+}$ (Green)

(Acidified) → Orange

∴ Pure $Cr_{(aq)}^{3+} \rightarrow$ violet $[Cr(H_2O)_6]^{3+}$ (I)

Acidified $Cr_{(aq)}^{3+} \rightarrow$ Green $[\overset{+3}{Cr}(H_2O)_5X]$ (II)

Where, $X \rightarrow SO_4^{2-}/NO_3^-/Cl^-$

In above complex I and II

$\Delta \rightarrow$ I > II

$\lambda_{ab} \rightarrow$ I < II

yellow — Red (complementary of green)

→ (complementary of violet)

(2) Colour due to charge transfer:

Colour due to charge transfer

(I) Ligand to metal charge transfer (LMCT)	(II) Metal to ligand charge transfer (MLCT)	(III) Metal to Metal charge transfer (MMCT)

Tips and Trick

Remember the colours

Fenni : [Fe^{2+}, Ni^{2+}] green colour

Blue : Cooker [Cu^{2+}, Cr^{2+}] in bazaar or market

Mann [Mn^{2+}], KO [Co^{2+}] pink colour pasand

(i) Ligand to Metal charge transfer :

(Anion) (Cation)

Occur in case of d^0, d^{10}, d^5 (WFL)

↓

If, coloured

Example :

Case of d^0 : (Mn^{7+}, Cr^{6+}, $V^{5+} \rightarrow d^0$)

MnO_4^-	CrO_4^{-2}	VO_4^{-3}
Purple pink	yellow	Pale yellow

$O^{2-} \rightarrow Mn^{7+}$ $O^{2-} \rightarrow Cr^{6+}$ $O^{2-} \rightarrow V^{5+}$

LMCT ↓ (∵ Mn^{7+} attracts electrons fast due to high z_{eff})

Transition ↓

Intensity ↓

Darkness ↓

Case of d^5(WFL) :

FeF_6^{3-}	$[Fe(H_2O)_6]^{3+}$	$[Fe(H_2O)_5OH]^{2+}$	$[Fe(SCN)_6]^{3-}$
Colouless	Pale yellow	Yellow-brown	Blood red
Fe^{3+} ⇠ F^- (d^5-WFL)	← OH_2	← $OH^\ominus$	← $SCN^\ominus$

Polarizability ↑

Case of d^{10} :

Colourless		Yellow-organe
PbI_4^{2-}	HgI_4^{2-}	BiI_4^-
Pb^{2+}	Hg^{2+}	Bi^{3+}

↓

High charge

↓

More electron deficient or More polarizing power

Stability of complexes :

$$M + nL \underset{}{\overset{k_f}{\rightleftharpoons}} ML_n$$

Where, k_f is formation constant

(i) Stability of complexes α k_f

If, k_f data of complexes is ↑ given.

$\Rightarrow$ As, $k_f \uparrow \Rightarrow$ stability ↑

If, k_f data is not given

Points to remember

O ⟶ Oxido

O_2 ⟶ dioxido

O_3 ⟶ trioxido

(ii) Stability/spontaneity :

If negative value of ΔG increases ($-\Delta G\uparrow$)

$\Delta G = \Delta H - T\Delta S$

ΔH ↓ Enthalpy ↓ (C.F.S.E.) $\propto \Delta$

$T\Delta S$ → Entropy (By chelation) → Ring formation

Stability $\propto \Delta$ (∵ Bond strength ↑ → Energy relase ↑ → ΔG ↓)

$\propto$ Ring formation

Chelate Effect :

$[Ni(H_2O)_6]^{2+}$

- 1en, $\Delta n=1$ → $[Ni(en)(H_2O)_4]^{2+} + 2H_2O$, Number of ring = 1
- 2en, $\Delta n=2$ → $[Ni(en)_2(H_2O)_2]^{2+} + 4H_2O$, Number of ring = 2
- 3en, $\Delta n=3$ → $[Ni(en)_3]^{2+} + 6H_2O$, Number of ring = 3
- $edta^{4-}$, $\Delta n=5$ → $[Ni(edta)]^{2-} + 6H_2O$, Number of ring = 5

Where, ΔN = difference in number of moles of product and reactant.

$\Rightarrow$ As number of ring ↑ $\Rightarrow \Delta N \uparrow$

- → $\Delta S \uparrow$
- → $\Delta G \uparrow$
- → Stability ↑

Example of stability order :

(i) $NiCl_4^{2-}$ $PdCl_4^{2-}$ $PtCl_4^{2-}$

3d → 4d → 5d

$\Delta \uparrow$

Stability ↑

(ii) $[Fe(CN)_6]^{3-}$ > $[Fe(CN)_6]^{4-}$

↓ ↓

Fe^{3+} Fe^{2+}

↓

More charge

↓

High Δ

↓

Greater stability

(iii) $[Ni(en)_3]^{2+}$ > $[Ni(NH_3)_6]^{2+}$

↓

Chelation and high Δ

→ More stable

(iv) $[Fe(CN)_6]^{4-}$ > $[Fe(C_2O_4)_3]^{4-}$

$\Delta \downarrow$ chelation ↑

Points to remember

When C.M.I and its coordination number is same (different ligands are present) then stability depends on both

(i) Strength of ligand (Δ)

(ii) Chelation (Number of rings)

Then,

Conclusion (Based on experimental data) → Stability of complex having ligand → CN^- > Chelation > Strength of ligand

Example :

$$\xrightarrow[\text{(In general)}]{\Delta} CN^- > en > NH_3 > edta^{4-} > NCS^- > H_2O > C_2O_4^{2-} > F^-$$

$\therefore$ Stability order of complex having these ligands

$\Rightarrow$ $CN^- > edta^{4-} > \underbrace{en > C_2O_4^{2-}} > NH_3 > NCS^- > H_2O > F^-$

$edta^{4-}$: 5 ring

en, $C_2O_4^{2-}$: 1 ring but en has high Δ

Points to remember

If sp^2 hybridised C-atom is these in ligand forming ring then stability of 6-membered ring > 5-membered ring

Example :

$[Fe(acac)_3]$ > $[Fe(C_2O_4)_3]^{3-}$

$[Fe(acac)_3]$: Form 6-membered ring

$[Fe(C_2O_4)_3]^{3-}$: 5-membered ring

Q26 **Which of the complex will be more stable among $[Ni(en)_2]^{2+}$ and $[Ni(dmg)_2]$**

Sol: Stability $[Ni(en)_2]^{2+}$ < $[Ni(dmg)_2]$

$[Ni(en)_2]^{2+}$: 2-ring

$[Ni(dmg)_2]$: 4-ring

Points to remember

$K_4F_6(CN)_6$ → Obtain yellow colour

VBT say that → d^2sp^3 (Hybridisation)

d^2sp^3 → So coordination number of $K_4F_6(CN)_6$ is octahedral

Then, CFT → say that → Ionic interaction and according to VBT → Octahedral full filled.

Classification of ligands on the basis of bonding pattern:

Classification of ligands
on the basis of bonding pattern

Classical ligands
only donar
F^-, OH_2, OH^-, O^{-2}, NH_3 , NH^{-2}
NH_2^-, CH_3^-, H^-

Non classical donar
as well as acceptor
electron acceptor
metal ligand



π-acid ligand
acid-lewis acid
π-lewis acid in
π fashion
ligands → lewis base

π-acid ligands

Lone pair donor
(σ) donor

Accept e^- in π *
N_2, CO, Na^*
only 14 electron
system after 14 new
electron proceed
into π*

Accept (e^-) in dπ
Pme_3 Phosphorus
gives lone pair and
accept lone pair
only in d orbital

π e^- donor like
Benzene



6π electron

η^6 - benzene



Cyclopentadienyl

η^5 - Cyclopentadienyl

Metal carbonyls : (σ donor- π acceptor synergic bonding)
CO : (Carbon mono oxide) : No. of electron = 14
LUMO (Lowest unoccupied molecular orbital) = π^*
Bond order = 3
donar atom = C

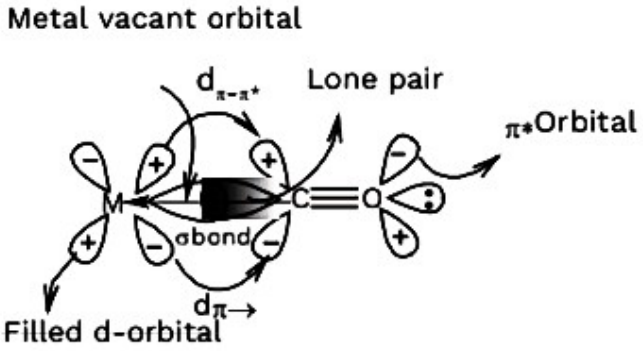
Structure of CO :



Points to remember

We know that electronegative of carbon is less than the oxygen so here donar is carbon.

Bonding in metal carbonyl : It is explained by molecular orbital theory



Here, carbon donates it's lone pair to CMI and formation of σ bond takes place simultaneously metal donates it's electron to π * orbital of CO.

Results of synergic bonding :

Metal – carbon bond order increases due to formation of new bond ie M-C bond order > 1 and M-C bond energy increases.

Due to acceptance of electron in antibonding orbital of CO; C-O bond order decreases ie 2 < B. O_{C-O} < 3.

C – O bond length increases so, C – O bond energy decreases.

As donation tendency of metal increases, then

Number of electrons in anti bonding molecular orbital (ABMO) of CO Increases

C-O bond order decreases

C-O bond length increases

C-O bond energy decreases

If C-O bond length in $Cr(CO)_6$ is x, then comment on CO bond length of given species:

(1) Free Carbon Monoxide

$Cr(CO)_6$ is metal carbonyl. So, bond order is less than 3 (BO < 3) and In free carbon monoxide, bond order of CO = 3

So, CO > Cr $(CO)_6$ (C-O bond order) and C-O Bond length in free CO will be less than X.

(2) $[Cr(CO)_6]^+$

Cr^+ < Cr (electron donar ability)

Electron in ABMO (Anti Bonding Molecular Orbital) of Carbon Monoxide:

$[Cr(CO)_6]^+ < [Cr(CO)_6]$

In $[Cr(CO)_6]^+$ Anti Bonding Molecular Orbital (ABMO) electron is less. So, bond order will be greater and bond length will be less than 'x'

(3) $[Cr(CO)_6]^-$

Cr^- > Cr (electron donor ability)

Electron in ABMO (Anti Bonding Molecular Orbital) of Carbon Monoxide

$[Cr(CO)_6]^- > [Cr(CO)_6]$

So, C-O: Bond order

$[Cr(CO)_6]^- < [Cr(CO)_6]$

C-O: Bond length in $[Cr(Co)_6]^-$ will be greater than X

(4) $[V(CO)_6]^-$

V^- > Cr (electron donor ability)

Electron in ABMO (Anti Bonding Molecular Orbital) of Carbon Monoxide

$[V(CO)_6]^- > [Cr(CO)_6]$

C-O Bond order : $[V(CO)_6]^- < [Cr(CO)_6]$

C-O Bond length in $[V(CO)_6]^-$ will be greater than X.

(5) $[Mn(CO)_6]^+$

Mn^+ < Cr (electron donor ability)

Electron in ABMO of CO : $[Mn(CO)_6]^+ < [Cr(CO)_6]$

C-O Bond Order: $[Mn(CO)_6]^+ > [Cr(CO)_6]$

C-O Bond length: $[Mn(CO)_6]^+$ < x

(6) $[Cr(CO)_5NH_3]$

As NH_3 is Classical ligand, do not accept electrons from metal.

So, in $[Cr(CO)_5NH_3]$ electron is distributed in 5 CO.

and in $Cr(CO)_6$ electron is distributed in 6 CO.

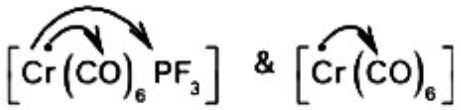
Electron in ABMO of CO: $[Cr(CO)_5NH_3] > [Cr(CO)_6]$

C-O Bond Order: $[Cr(CO)_5NH_3] < [Cr(CO)_6]$

C-O Bond Length: $[Cr(CO)_5NH_3]$ > x

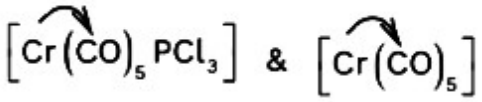
(7) $[Cr(CO)_5PF_3]$

Electron acceptance tendency: $PF_3 > CO > PCl_3$



Electron in ABMO of CO: $[Cr(CO)_6PF_3] < [Cr(CO)_6]$
C-O Bond Order: $[Cr(CO)_6PF_3] > [Cr(CO)_6]$
C-O Bond Length: $[Cr(CO)_6PF_3]$ will be less than X.

(8) $[Cr(CO)_5PCl_3]$

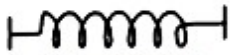


Electron in ABMO of CO: $[Cr(CO)_5PCl_3] > [Cr(CO)_6]$
C-O Bond Order: $[Cr(CO)_5PCl_3] < [Cr(CO)_6]$
C-O Bond Length in $[Cr(CO)_5PCl_3]$ will be greater than X.

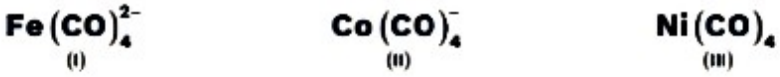
Points to remember

As synergic bonding occurs, bond order decreases, bond length increases and stretching frequency decreases.

Practical evidence of Carbon Monoxide (CO) Bond length (BL)
(Stretching Frequency Method)

Spring:  If Strength of spring ↑ Then required energy to stretch spring will also increases. $E = h\nu$ Where ν = stretching frequency	C ≡ O As like spring As C-O Bond Order ↑ C-O Bonder Energy ↑ To stretch Carbon Monoxide, required energy increases $E = h\nu$ Stretching frequency ↑

Q27 **Compare stretching frequency of CO in given compounds :**



Sol: **Left ⟶ Right**
As –ve charge on central metal ions decreases, donation tendency of CMI decreases so electron in ABMO of CO decreases.
So, C – O Bond Order increases
C – O Bond Energy increases
C - O Stretching frequency increases
So, C – O Stretching frequency increases : (i) < (ii) < (iii)

Q28 **Compare stretching frequency of CO in given compound.**

$[Cr(CO)_6]$ (i) $[Cr(CO)_5NH_3]$ (ii) $[Cr(CO)_4en]$ (iii) $[Cr(CO)_3\ dien]$ (iv)

Sol: As NH_3, en, dien are classical ligands they do not accept electron so on moving from left to right in given compounds, no. of CO ligands decreases. Then electron in ABMO of CO↑

C-O Bond Order ↓

So C - O Stretching frequency decreases

Stretching frequency : I > II > III > IV

Q29 **Compare stretching frequency of CO in given compound.**

$[Cr(CO)_5PF_3]$ I $[Cr(CO)_5PCl_3]$ II $[Cr(CO)_5PMe_3]$ III

Sol: Acceptance tendency → $PF_3 > PCl_3 > PMe_3$

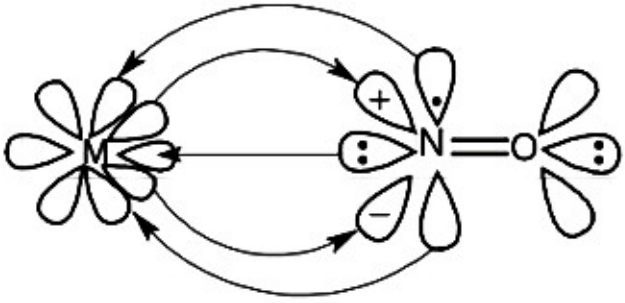
So, electron in ABMO of CO: I < II < III

C–O Bonder order: I > II > III

C–O Stretching frequency: I > II > III

1. Synergy Bonding in Metal Nitrosyl (MNO)

NO: 15 electrons system and (ABMO) also present.



Nitrosyl (NO) acts as a 3 electron donor.

3(CO) = 2NO

(No. of electron in 3 CO is equal to No. of electron in 2 NO)

$3 \times 2 = 6\ (e^-) = 6\ e^-$

i.e. Three 'CO' can be replaced by two 'NO'

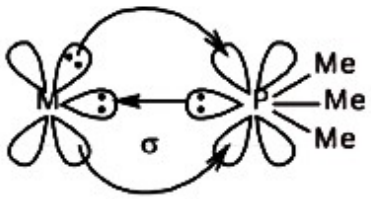
Acceptance tendency : $NO^+ > NO > CO$

Explained by E.N (Electronegativity)

2 **σ donor and d_π acceptor.**

It is occurs in 3rd period and below element

Example : PF_3, PMe_3, PCl_3, R_3As, R_2S



dπ – dπ

No effect on P–C Bond Order but it affect on Bond Length due to Bent's rule.

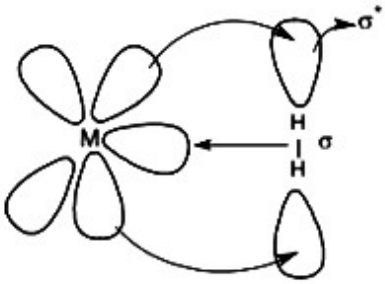
3 **σ donor and σ* acceptor**

M.O.T diagram of H_2:

H_2 □ σ^*_{1s}

H_2 ↿⇂ σ_{1s}

Bonder Order of H_2 < 1

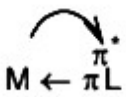


H_2 as a ligand is stable when weak donor metal is present or good acceptor ligands present.

π-donor, π-acceptor

Donor ability : lone pair donor > π donor > σ donor.

They are very weak Lewis Base but in many complexes, they form stable compounds due to synergic bonding.

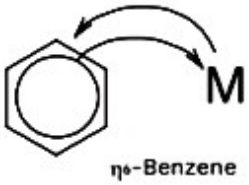


Metal Bond Order ↑. So, stability ↑

Such ligands are written as η^x Prefix.

Where x = number of atoms in donation is called **Hepticity of Ligands**.

1.



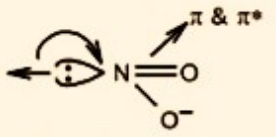
η^6-Benzene

No. of electron in donation = 6

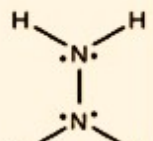
No. of atom in donation = 6

Points to remember

All ligands in which donor atom contains π-bond act as π-acid ligand due to presence of π*.

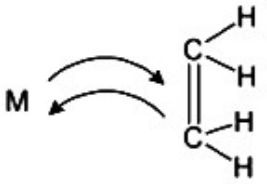


Non Classical



Classical

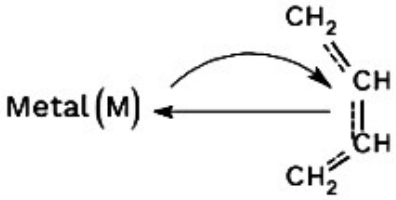
2. **C_2H_4 (ethylene)**



η^2-ethylene/π–Ethylene

Number of electron in donation = 2
Number of atom in donation = 2

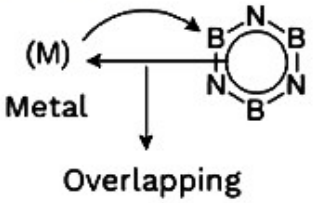
3. **$CH_2 = CH–CH=CH_2$**



η^4 – Butadiene

(π electron cloud collectively share)
Number of atom in donation = 4
Number of electron in donation = 4

4. **$B_3N_3H_6$**

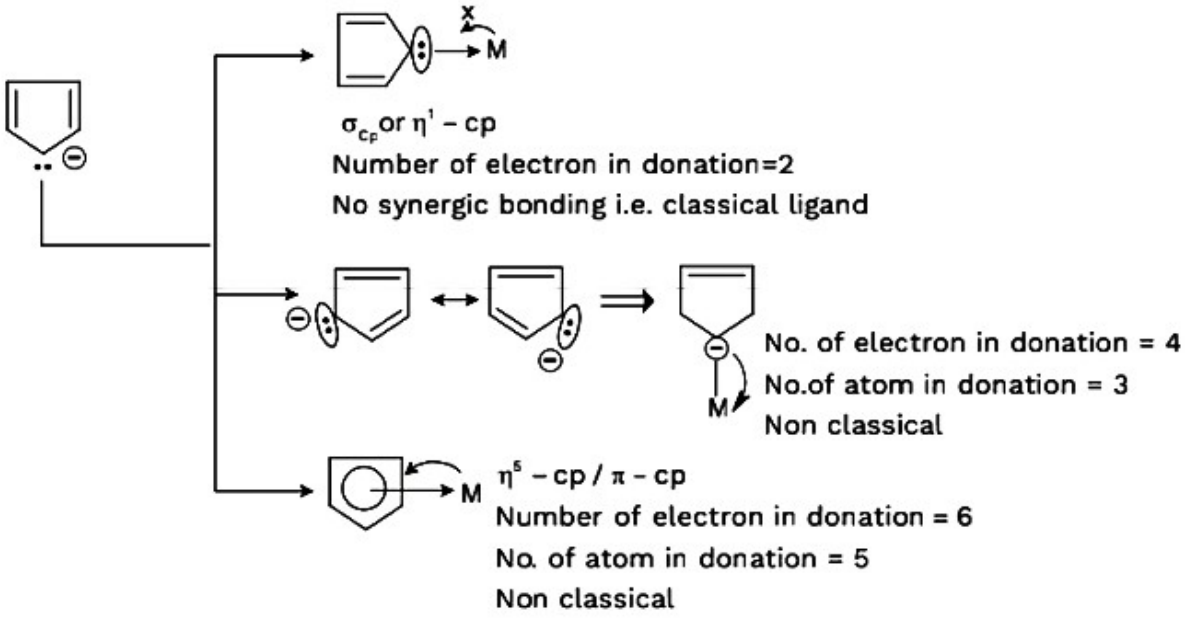


η^6 – Borazine

Overlapping

Number of atoms in donation = 6
Number of electron in donation = 6

5. **C_5H_5 : Cyclopentadienyln (Cp) :**
It shows three types of denticity

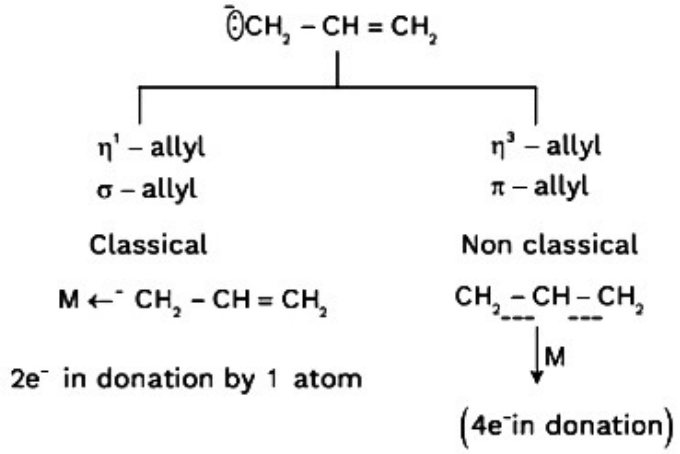


6. **$C_3H_5^-$:**

Double bond eqivatent = 1

two structures are possible.

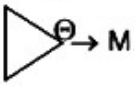
(1) linear structure



Points to remember

When π donor ligands are bonded with central metal ions (CMI) bond length in ligands increases from its expected value due to acceptance of electron in π^* orbital.

(2) **ring structure**



It is classical ligand, No synergic bonding

η^1– cyclopropyl or σ cyclopropyl or cyclipropyl

Comparison of C–C Bond length :

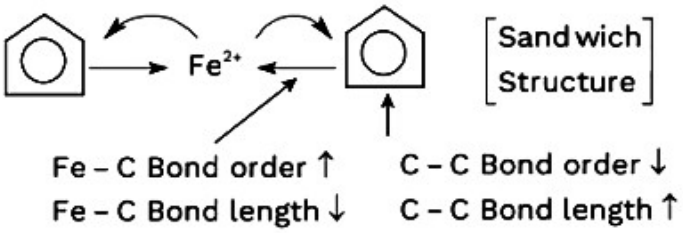
C_2H_4 < $K[PtCl_3(\eta^2-C_2H_4)]$

C_6H_6 < $Cr(C_6H_6)_2$

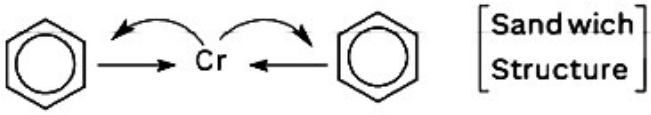
$C_5H_5^-$ < $Fe(C_5H_5)_2$(Ferrocence)

Example :

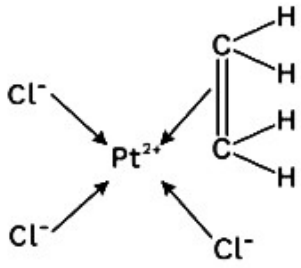
$Fe(C_5H_5)_2$



$Cr(C_6H_6)_2$



$K[PtCl_3(\eta^2-C_2H_4)]$ Ziese's salt

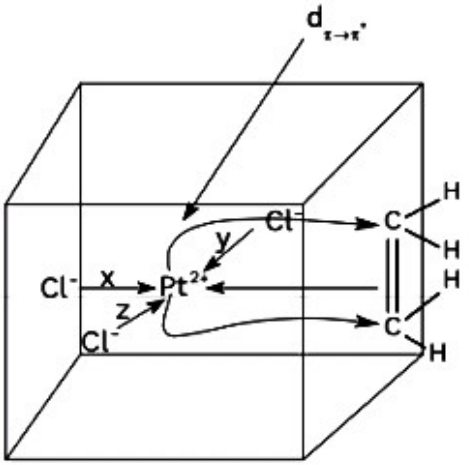


geometry around Pt^{2+} = square planer

$PtCl_3$ plane is perpendicular to C_2H_4 plane

C–C atom is perpendicular to $ptCl_3$ plane.

All hydrogen atoms are present in the same plane but not with C–C atom due to $d_{\pi-\pi^*}$ repulsion on C–H σ bond.

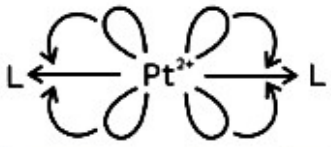


Pt-Cl bond length : x > y = z (due to the trans effect)

"Trans effect" :

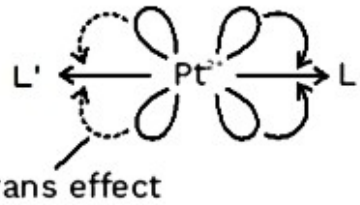
When different ligands are present at trans position (square, planer, octahedral) geometry. Bond length is different from its expected value this is called trans effects.

When strong acceptor present bond length (↑) increases when weak acceptor present bond length decreases.



Equal electron density on both sides.

If acceptance (L > L')



Trans effect

When acceptances increases metal-ligand bond order increases metal-ligand bond length decrease

i.e. $\begin{bmatrix} M-L \text{ Bond order} \uparrow \\ M-L \text{ Bond length} \downarrow \end{bmatrix}$

EAN rule, 18 e^- rule: "Sidgwick EAN rule"

EAN = effective atomic number.

It is like octet rule

Energy central metal ions (CMI) tries to achieve electron equal to the next noble gas to gain stability.

Example : Fe (from 3d series). Next noble gas = Kr (36).

Fe $\longrightarrow$ Kr

3d + 4s + 4p

10 + 2 + 6 = $18e^-$

EAN rule is also called $18e^-$ rule.

$18e^-$ rule calculates balance electron and EAN rule calculates total electron.

Valence electron = $(n-1)d$ ns, np

(Metal follow EAN rule mean 3d series metal EAN will be 36)

For 3d series metal, EAN = 36

For 4d series metal, EAN = 54

For 5d series metal, EAN = 86

$\Rightarrow$ EAN can be calculated by the following relation:

EAN = atomic number of the metal – oxidation state of central metal + number of electrons gained from the ligands.

Ex : $K_3\left[Fe(CN)_6\right] \Rightarrow EAN = 26 - 3 + 2 \times 6$

$= 36$

Q30 **Which of them follow EAN rule?**

Sol:

$K_4[Fe(CN)_6]$	,	$K_3[Fe(CN)_6]$
EAN = 36		EAN = 35
EAN rule follow		EAN rule not follow

Application of EAN rule :

This rule is used to explain the stability of metal carbonyl and π-complex.

Metal carbonyl : Metal carbonyl complex follow EAN rule.

Q31 **Find the value of x, y, and z for?**
$Cr(CO)_x$, $Fe(CO)_y$ & $Ni(CO)_z$

Sol: All the CMI in given complex form 3d series so, EAN will be 36.

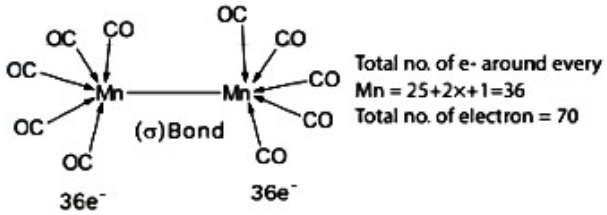
$Cr(CO)_x$	$Fe(CO)_y$	$Ni(CO)_z$
$24 + 2x = 36$	$26 + 2y = 36$	$28+2x = 36$
$x = 6$	$y = 5$	$z = 4$

Example :

(1) $Mn(CO)_5$

EAN = 25 – 0 + 10 = 35 (do not follow EAN rule)

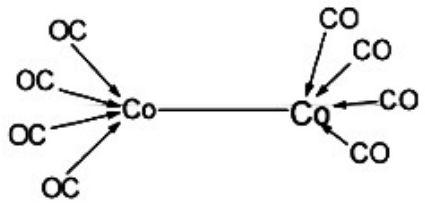
It forms dimer : $Mn_2(CO)_{10}$



(2) $Co_2(CO)_8$:

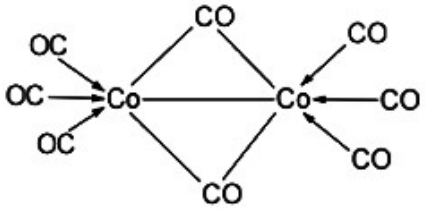
It has two structure.

(i)



Total no. of electron around each Co = 27 + 8 + 1 = 36

(ii)



Total no.of e⁻around Co

$= 27 + 2 \times 3 + 1 + 1 + 1$

$= 36$

Points to remember

If a complex compound does not follow EAN rule, then it will try to achieve stability by two ways:

(1) If EAN > 36, then it release it's electron and act as reducing agent

(2) If EAN < 36, then it gains electron and act as oxidising agent or it forms dimer.

If EAN = 35 ⇒ It forms dimer.

If EAN = 36 ⇒ It forms trimer and so on.

Q32 $Fe_2(CO)_x$ **Find value of x?**

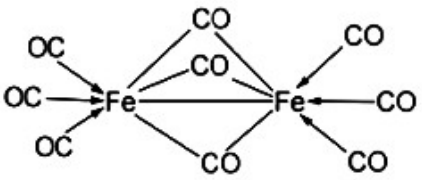
(A) 5 **(B) 8** **(C) 9** **(D) 10**

Sol: **(C)**

$[Fe(CO)_{x/2}]_2$

$26 + \frac{x}{2} \times 2 + 1 = 36$

$x = 9$



Total no of e^- around each = 26 + 3 × 2 + 1 + 1 + 1 + 1 = 36

Isomerism

Structural Isomerism (Complex have different name)	Stereo Isomerism (Prefix cis – Trans and d-l)
1) Ionization Isomerism	1) Geometrical Isomerism
2) Hydrated Isomerism	2) Optical Isomerism
3) Linkage Isomerism	
4) Coordination Isomerism	
5) Ligand Isomerism	
6) Coordination position Isomerism	
7) Polymerization Isomerism	

(1) Ionization Isomerism

$[ML_x]A \longleftrightarrow [ML_{x-1}A]L$

Complex having same Molecular formula but furnishes different - different Counter ions in solution are ionisation Isomers of each other.

Or

When counter ion act as ligands & ligands act as counter ions.

Example :

$[MLx]A \longleftrightarrow [ML_{x-1}A]L$

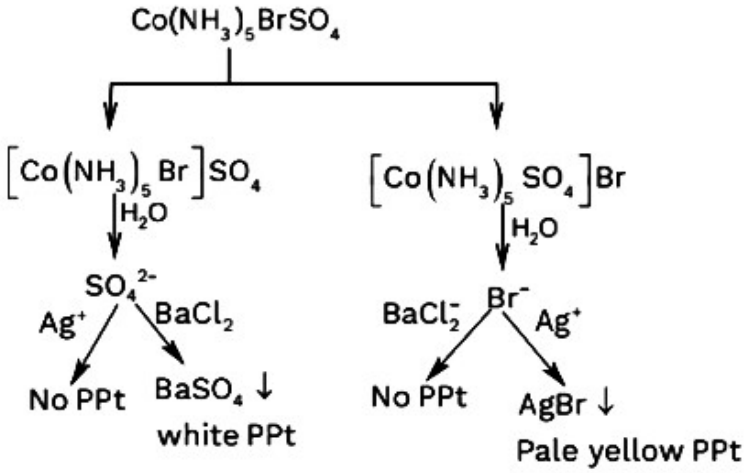
can be distinguish by

(i) Qualitative analysis (1st compound give test of A and 2nd compound give test of L)

(ii) Conductivity measurement.

Example :

$Co(NH_3)BrSO_4$ exist in two forms.



Points to remember

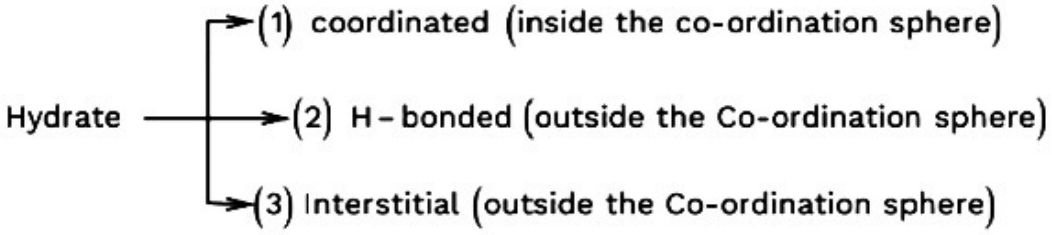
All sulphate $(SO_4)^{2-}$ are soluble except

→ $Hg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}, Pb^{2+}$

Both compounds are ionisation isomer of each other.

(2) Hydrated isomers

⇒ There are three type of hydrate : -



Complexes having same formula but different number of water molecules inside and outside the coordination sphere

Hydrated = H_2O (solvent)

It is also known as solvated isomerism

Complex should not be neutral because in neutral complex there is no existence of interstitial hydrate.

Example :

$(Cr \cdot 6H_2O \cdot Cl_3)$ (Violet colour) have different complex:

(1) $[Cr(H_2O)_6]Cl_3$

(2) $[Cr(H_2O)_6Cl]Cl_2 \cdot H_2O$ } ionic compound

(3) $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$

These are the hydrated isomers of each other :

Conc. H_2SO_4 is dehydrating agent and it removes non-coordinated water.

Hydrated isomer of $Cr \cdot 6H_2O.Cl_3$ are,

$[Cr(H_2O)_6]Cl_3$(A), $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (B)

and $[Cr(H_2O)_4Cl_2]Cl_2.2H_2O$ (C)

Conductivity : A > B > C

Weight loss on reaction with concentrated H_2SO_4 : C > B > A

No weight loss for A as it has no Co-ordinated water

As, No. of Co-ordinated water increases ⇒ weight loss increases

quantitative Analysis : reaction with 1 mole of each complex with excess $AgNO_3$ gives.

A → 3 mole AgCl ↓ PPt

B → 2 mole AgCl ↓ PPt

C → 1 mole AgCl ↓ PPt

Points to remember

$[Cr(H_2O)_3Cl_3] \cdot 3H_2O$ it is neutral so it does not exist.

Points to remember

They can Be distinguish By.

(1) quantitative analysis

(2) Conductivity measurement

(3) Weight loss measurement on reaction with concentrated H_2SO_4.

(3) Linkage Isomerism : This type of isomerism arises when complex contain ambidentate ligands.

Example :

$M \leftarrow C \equiv N$ & $M \leftarrow N \equiv C$

both are linkage isomers of each other :

Ambidentate ligands : $CN^-, NO_2^-, NO_3^-, SCN, NO^-, SO_3^{2-}, S_2O_3^{2-}, C_2S_2O_2$

Q33 **Find No. of linkage isomer of $[Ag(CN)_2]$?**

Sol: As it has ambidentate ligands, it show linkage isomerism.

$[Ag(CN)_2]^-, [Ag(CN)(NC)]^-, [Ag(NC)_2]^-$

Total (3) linkage isomers are possible.

Q34 **Find the linkage isomer of $[Pt(CN)_2(NO_2)_2]^{2-}$.**

Sol: $[Pt(CN)_2(NO_2)]^{2-}$ have two different ambidentate ligands.
Total linkage isomers = 3 × 3 = 9
Total '3' linkage isomers are possible.

(4) Coordination Isomerism : Such type of Isomerism is arises when complex contain. Complex cation and complex anion. Here interchange of ligand takes place of one coordination sphere to other Co-ordination sphere. Charge on complex should not be zero.

Examples :

(i) Coordination isomer of $[Pt(NH_3)_4][PdCl_4]$ are

$[Pt(NH_3)_4Cl][Pd(NH_3)Cl_3]$

$\rightarrow [Pt(NH_3)_2Cl_2][Pd(NH_3)_2Cl_2]$ (do not exist as both are neutral complex)

$[Pt(NH_3)Cl_3][Pd(NH_3)_3Cl]$

$[PtCl_4][Pd(NH_3)_4]$

So, total number of coordination isomerism = 4

(ii) $[Pt(NH_3)_4][Pt(C_2O_4)_2]$

$[Pt(NH_3)_2(C_2O_4)]^0$ $[Pt(NH_3)_2(C_2O_4)]$ (not exist as charge is zero)

So, $[Pt(NH_3)_4][(Pt(C_2O_4)_2]$ It does not show coordination isomerism.

(iii) $[Pt(NH_3)_4][PtCl_4]$

$[Pt(NH_3)_3Cl][PtCl_3NH_3]$

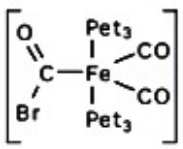
$[Pt(NH_3)_2][PtCl_2(NH_2)_2] \rightarrow$ Not exist

It has only 2 Co-ordinates isomers.

Q35 **In the complex acetyl bromido di carbonyl bis (tri ethyl phosphine) iron (II), the number of Fe-C bonds is**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol: **C**



Q36 **Which of the following complex ions does satisfy the effective atomic number (EAN) rule?**

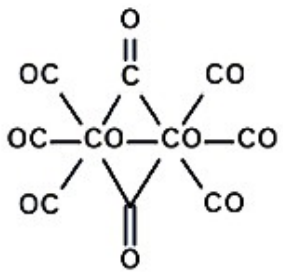
(A) $[Pt(NH_3)_4]^{+2}$ **(B)** $[PtCl_4]^{-2}$

(C) $[PtCl_6]^{-2}$ **(D)** $[Fe(CN)_6]^{-6}$

Sol: **C**

EAN = 78 − 4 + 12 = 86

Q37 **The oxidation state of cobalt in the following molecule is**



(A) 3 **(B) 1** **(C) 2** **(D) 0**

Sol: **D**

Q38 **The number of bridging CO (ligands) and Co-Co- bonds in $Co_2(CO)_8$ respectively.**

(A) 2 and 1 **(B) 2 and 0** **(C) 0 and 2** **(D) 4 and 0**

Sol: **A**

Q39 In $Fe(CO)_5$ the Fe-C bond possesses

(A) ionic character (B) Sigma character

(C) π -character (D) Both sigma and π – character

Sol: D

Q40 $Mn_2(O)_{10}$ is an organometallic compound due to the presence of:

(A) Mn O bond (B) Mn-C bond (C) Mn-Mn bond (D) C-O bond

Sol: B

Q41 Among the following the π -Acid ligand is:

(A) F^- (B) NH_3 (C) CN^- (D) I^-

Sol: C

Q42 The correct pair of orbitals involved in π -bonding between metal and CO in metal carbonyl

(A) d_{xy} and π_x^* (B) d_{xy} and π_x

(C) $d_{x^2-y^2}$ and π_x^* (D) $d_{x^2-y^2}$ and π_x

Sol: A

Q43 The least stable carbonyl as per the bonding considerations should be:
(A) $Cr(CO)_6$ (B) $Mn(CO)_6$ (C) $Fe(CO)_5$ (D) $Ni(CO)_4$

Sol: B

Q44 The C-O bond length is the shortest in
(A) $[Cr(CO)_6]$ (B) $[Mo(CO)_6]$ (C) $[Mn(CO)_6]^+$ (D) $[V(CO)_6]^-$

Sol: C

Q45 Number of bridging CO ligands in $[Mn_2(CO)_{10}]$ is
(A) 0 (B) 2 (C) 4 (D) 6

Sol: A

Q46 If the bond length of CO bond in carbon mono oxide is 1.128Å, then what is the value of CO bond length in $Fe(CO)_5$?
(A) 1.15 Å (B) 1.128 Å (C) 1.72 Å (D) 1.118 Å

Sol: A

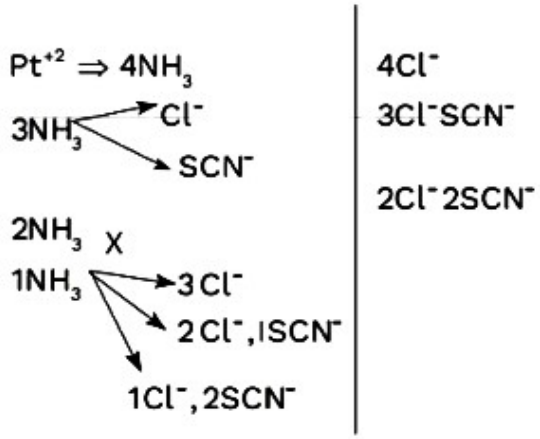
Q47 Among the following metal carbonyl's, the C-O bond order is lowest in
(A) $[Mn(CO)_6]^+$ (B) $[Fe(CO)_5]$ (C) $[Cr(CO)_6]$ (D) $[V(CO)_6]^-$

Sol: D

Coordination isomerism

$\Rightarrow \left[Pt(NH_3)_4\right]\left[PtCl_4(SCN)_2\right]$

Total (9) coordination isomers.



Q48 **Total structural isomers of**

$\left[Cr(NH_3)_6\right]\left[Cr(CN)_6\right]$

Sol: **Coordination isomers** $\begin{bmatrix}\left[Cr(NH_3)_5(CN)\right]\left[Cr(CN)_5NH_3\right] \\ \left[Cr(NH_3)_4(CN)_2\right]\left[Cr(CN)_4(NH_3)_2\right]\end{bmatrix}$

Linkage isomers

$\Rightarrow$ 7 for $[Cr(CN)_6]$

$\Rightarrow$ 12 for $[Cr(NH_3)_5CN]$ $[Cr(CN)_5NH_3]$

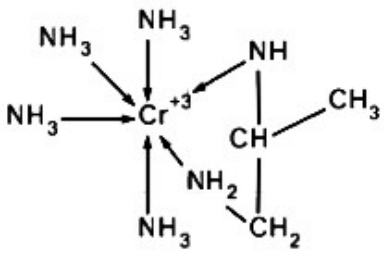
$\Rightarrow$ 15 for $[Cr(NH_3)_4(CN)_2]$ $[Cr(CN)_4(NH_3)_2]$

Total = 34

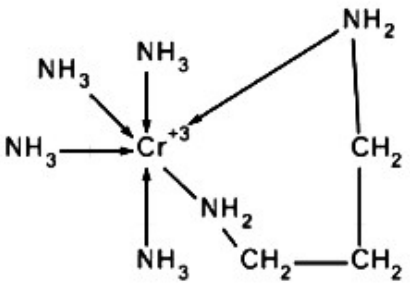
Ligand isomerism :

Complex contains ligand which can show position isomerism can show ligand isomerism

Example:



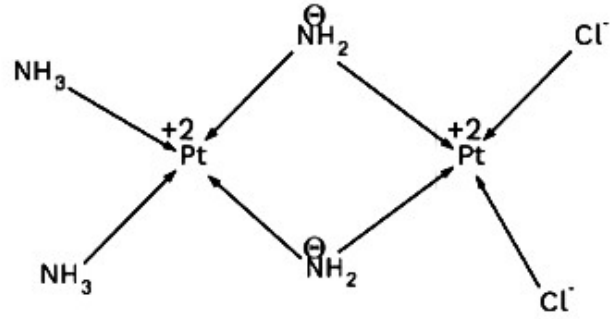
(1,2-position)

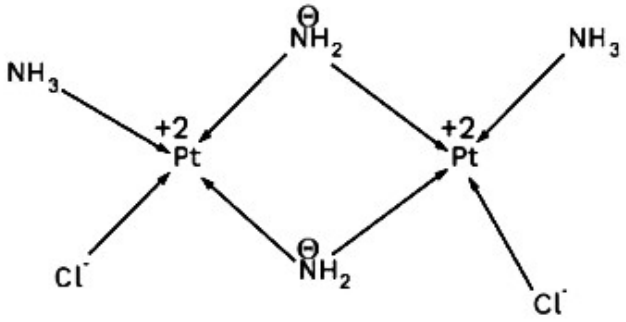


(1,3-position)

Co-Ordination position :

This type of isomerism arises in bi/poly nuclear Complex due to diffirent- diffirent position of group of ligand/ligand with diffirent- diffirent metal ion.





Polymerisation isomerisation :

It is not true isomerism

They have the same empirical formula but different molecular formula.

Example:

$[Pt(NH_3)_2Cl_2]$ 1 : 2 : 2

$[Pt(NH_3)_4][PtCl_4]$ 2 : 4 : 4

They must contain neutral and anionic ligand

Q49 **The complex that can exhibit linkage isomerism is-**

(A) $[Co(NH_3)_5(H_2O)]Cl_3$ **(B)** $[Co(NH_3)_5(NO_2)]Cl_2$

(C) $[Co(NH_3)_5(NO_3)](NO_3)_2$ **(D)** $[Co(NH_3)_5Cl]SO_4$

Sol: **(B)**

Q50 **Which of the following pairs represents linkage isomers?**

(A) $[Co(NH_3)_5NO_2]SO_4$ **and** $[Co(NH_3)_5SO_4]NO_2$

(B) $[PtCl_2(NH_3)_4]Br_2$ **and** $[PtBr_2(NH_3)_4]Cl_2$

(C) $[Cu(NH_3)_4][PtCl_4]$ **and** $[Pt(NH_3)_4][CuCl_4]$

(D) $[Pd(PPh_3)_2(NCS)_2]$ **and** $[Pd(PPh_3)_2(SCN)_2]$

Sol: **(D)**

Q51 **Co-ordination compound** $[Pt(NH_3)_3(SCN)]^+$ **and** $[Pt(NH_3)_3(NCS)]^+$ **are examples of.**

(A) Co-ordination isomerism **(B) Linkage isomerism**

(C) Optical isomerism **(D) Hydrate isomerism**

Sol: **(B)**

Q52 Which isomerism exhibited by $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$?

(A) Ionisation
(B) Linkage
(C) Coordination
(D) Polymerisation

Sol: (C)

Q53 The ionisation isomerism of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is

(A) $[Cr(H_2O)_4(O_2N)]Cl_2$
(B) $[Cr(H_2O)_4Cl_2](NO_2)$
(C) $[Cr(H_2O)_4Cl(ONO)]Cl$
(D) $[Cr(H_2O)_4Cl_2(NO_2)]H_2O$

Sol: (B)

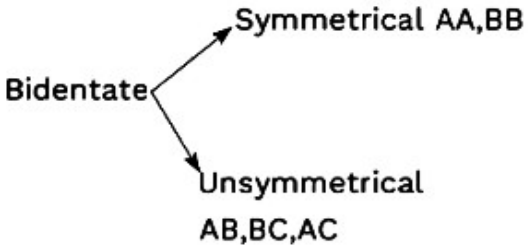
Q54 Type of isomerise which exist between $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ is:

(A) Solvate isomerism
(B) Ionisation isomerism
(C) Linkage isomerism
(D) Coordination isomerism

Sol: (C)

Q55 **Convert complex into simple form**
Central metal ion = M
Monodentate ligand = a,b,c,d.

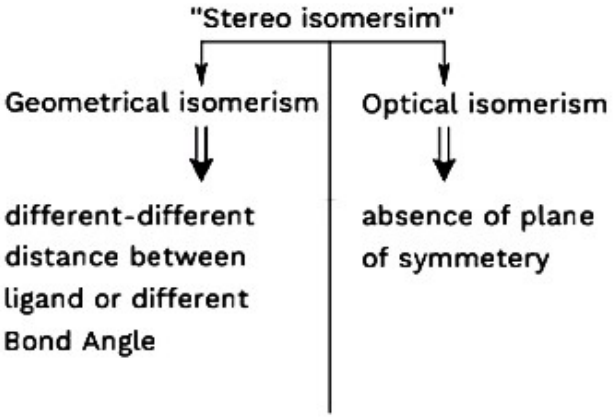
Sol:



Example:

$$\left[Zn(gly)(NH)_3\,H_2O\right],\left[Cr(NH_3)(en)(H_2O)Cl\right]^+$$
$$\Downarrow \qquad\qquad \Downarrow$$
$$\left[M(AB)Cd\right] \qquad \left[M(AA)b_3Cd\right]^+$$

Stereo isomerism :



Geometrical isomerism :
For C.N. = 2 (Coordination number = 2)

b – M – a ⇒ No G.I.
(Linear) ⇒ No O.I.(due to POS)

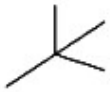
For CN = 3
Trigonal planer



No. G.I (only one type of bond angle)
No O.I (due to POS)

For CN=4

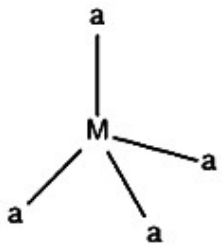
Tetrahedral	Squar planer
NO G.I.	NO O.I.
Can show O.O.I	Can show G.I



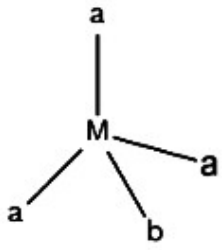


(1) O.I in tetrahedral

(i) Ma_4 type
No. O.I due to POS

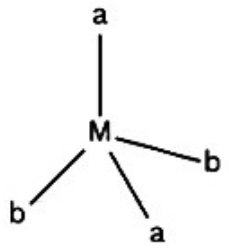


(ii) Ma_3b type
NO.O.I. due to POS



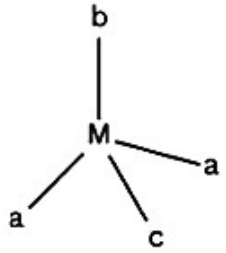
(iii) Ma_2b_2 type

No.O.I. due to POS



(iv) Ma_2bc type

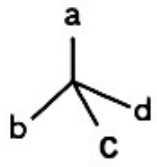
NO.O.I due to POS



(v) Mabcd type

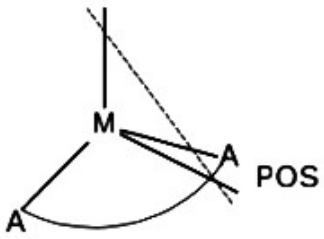
No POS

So two O.I.



(vi) $M(AA)_2$,M (AA)b_2,M(AA)bc,M(AA)(BC),M(AA)(BB)

NO.O.I due to POS



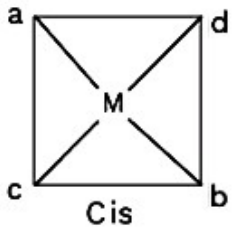
(vii) M(AB)cd,M(AB)(CD),$M(AD)_2$

$\Rightarrow$ 2 O.I. show

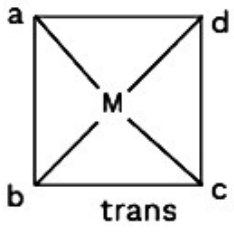
(2) G.I. for square planer :

(i) Mabcd

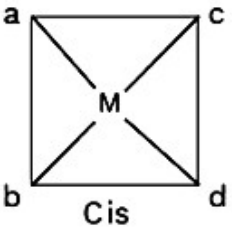
$(a,b)(c,d)$
$(a,c)(b,d)$
$(a,d)(a,c)$



Cis



trans



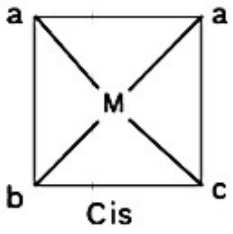
Cis

Example:

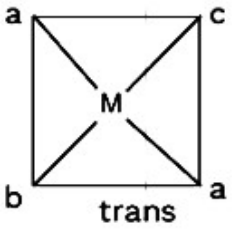
$[Pt(NH_3)(H_2O)(Cl)Br]$

(ii) Ma_2bc

$(a,b)(c,c)$
$(a,b)(a,c)$
2 G.I.



Cis



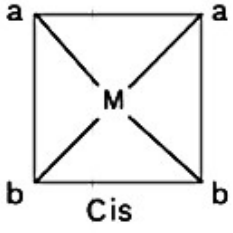
trans

Example:

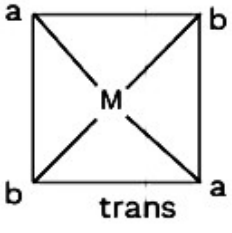
$[Pt(NH_3)_2ClBr]$

(iii) Ma_2b_2

$(a,b)(a,b)$
$(a,a)(b,b)$



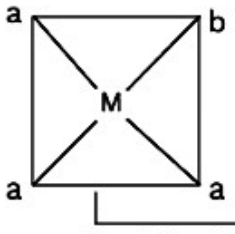
Cis

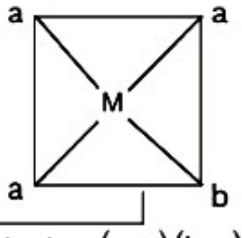


trans

(iv) Ma_3b

No.G.I

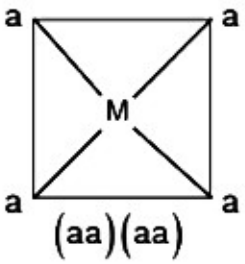




$(a,a)(a,b)$ identical $(a,a)(b,a)$

(v) Ma_4

No.G.I



$(aa)(aa)$

Points to remember

Ligand should be optical in active

(vi) M(AA)bc ⇒ Bidentate ligand always present at cis position.

When symmetrical bidentate ligand present in C.N = 4 (either sp^3 or dsp^2) compound does not show. Stereo isomerism

Q56 **Any draws square of G.I. How many type of G.I in C.N=4**

$M(AB)_2$

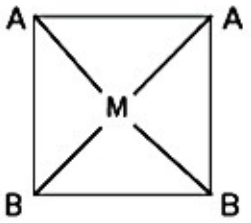
$M(AB)cd$

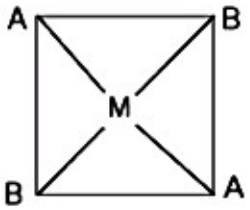
$M(AB)(CD)$

$M(AB)C_2$

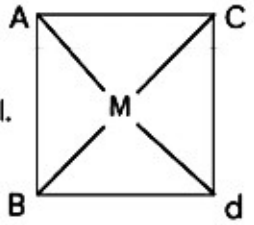
Sol:

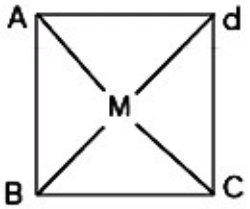
$M(AB)_2 \Rightarrow 2$ G.I.



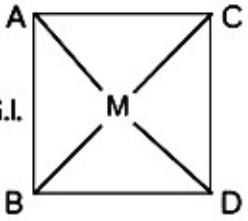


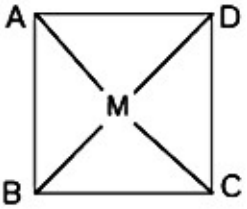
$M(AB)Cd \Rightarrow 2$ G.I.





$M(AB)(CD) \Rightarrow 2$ G.I.

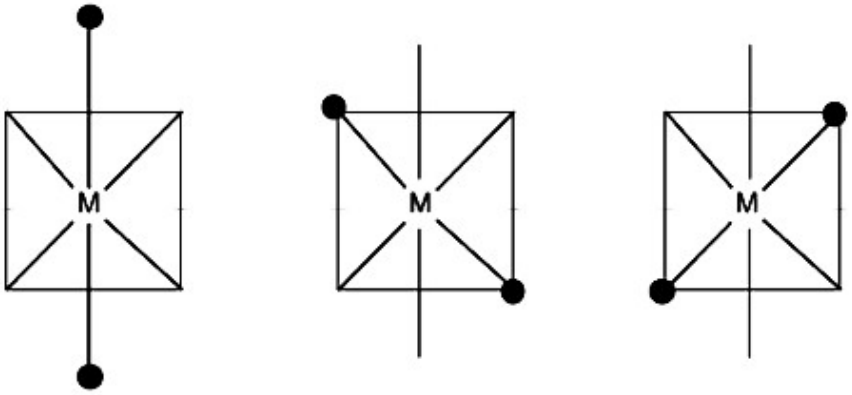




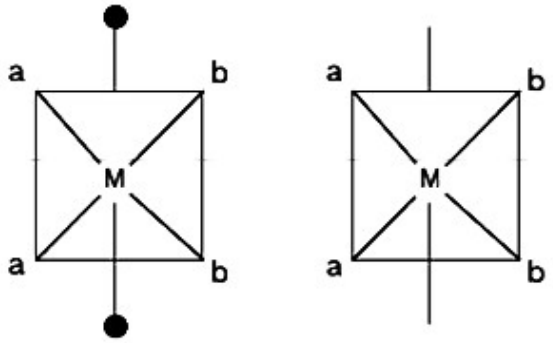
$M(AB)C_2 \Rightarrow$ NO G.I.

For C.N=6 :- (Octahedral Complexes)

(i) When at trans position same ligand present then POS is present

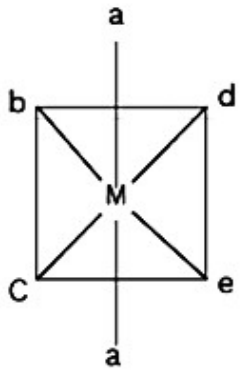


(ii) When two trans must be same



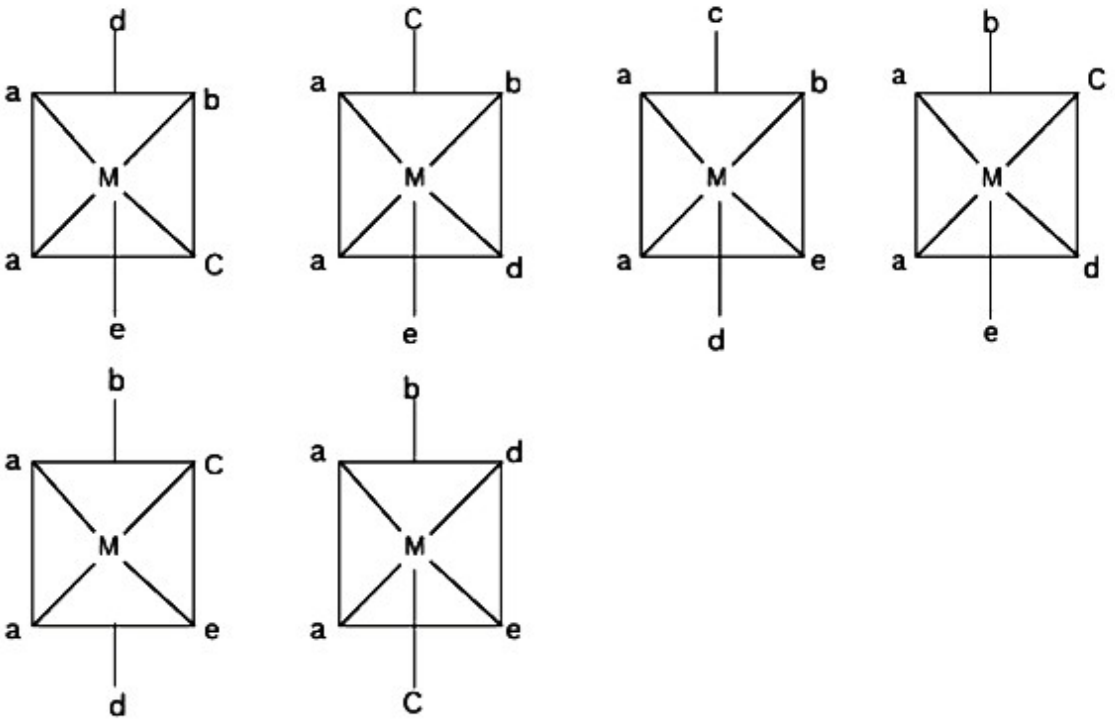
(1) **Ma_2bcde type**

(a,a) trans



3 G.I. optically inactive due to POS present

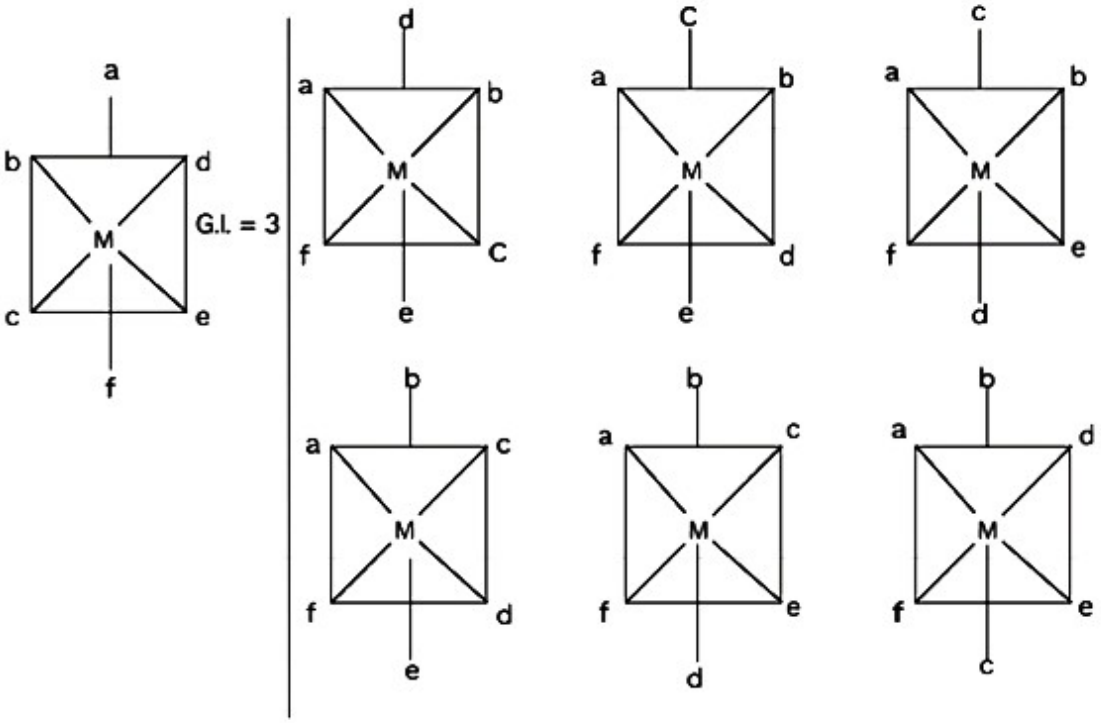
(a,a) Cis



All are optically Active.
And total G.I = 3 + 6 = 9
Optically inactive = 3
Optically Active = 6 x 2 = 12
Optically isomer = 12 + 3 = 15
Stereo isomer = 15

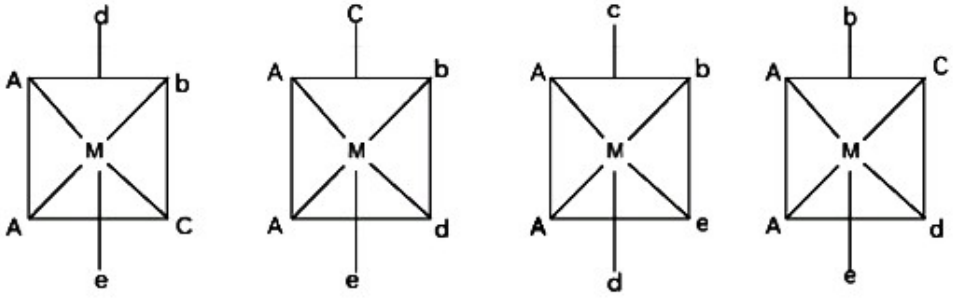
(2) Mabcdef-type

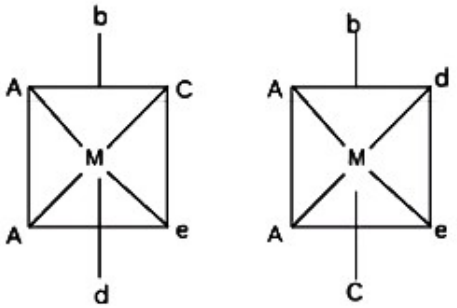
All are optically active (a,f) Cis



total G.I = 12 + 3 = 15
Optical inactive = 0
Optical Active = 15 × 2 = 30
Optical isomer = 30
Sterio isomer = 30

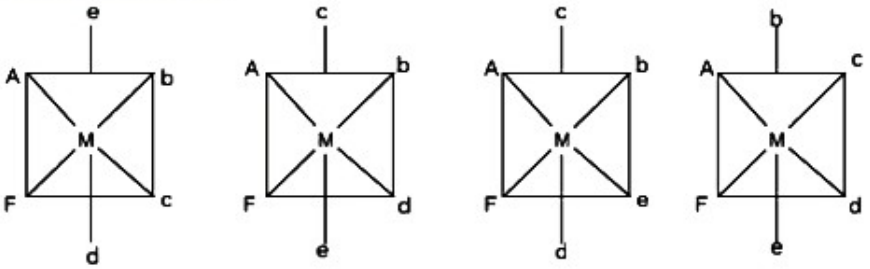
(3) M(AA)b cde type :

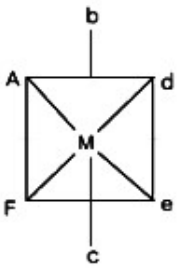




total G.I = 6
Optical isomer = 12
Sterio isomer = 12

(4) M(AF)bcde type

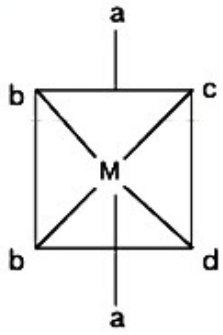




total G.I = 12
Optical isomer = 24
Sterio isomer = 24

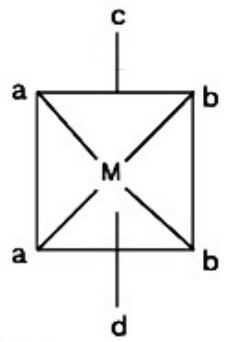
(5) Ma_2b_2cd type

(a,a trans)

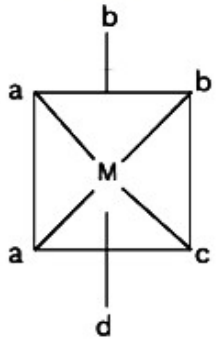


Square planer Mb_2Cd as like Ma_2bc G.I. = 2
Optical inactive (same ligand at trans)

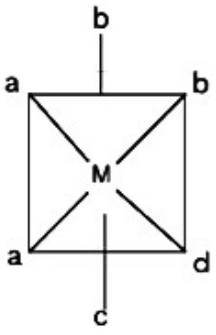
(a,a cis)



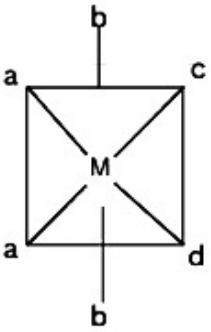
PoS present
optical in active



Obtical Active
(R & S)

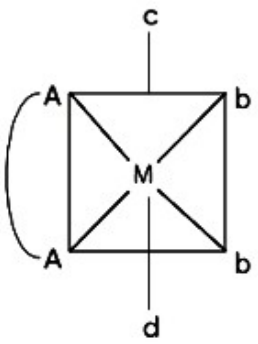


Optical Active
(R & S)

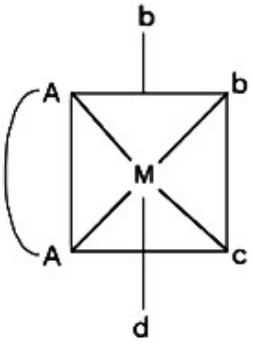


PoS present
optical inactive

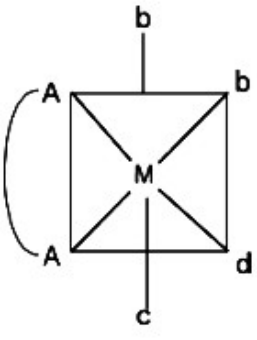
(6) $M(AA)b_2cd$ type



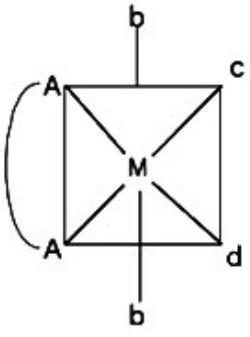
Optically
Inactive



Optically
active



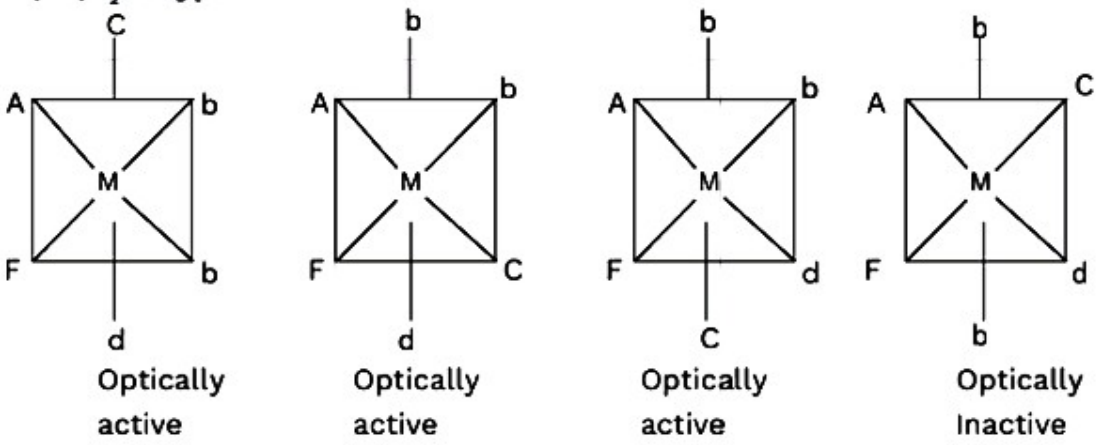
Optically
active



Optically
Inactive

total G.I = 4
Optical inactive = 2
Optical Active = 2 × 2 = 4
Optical isomer = 6
Sterio isomer = 6

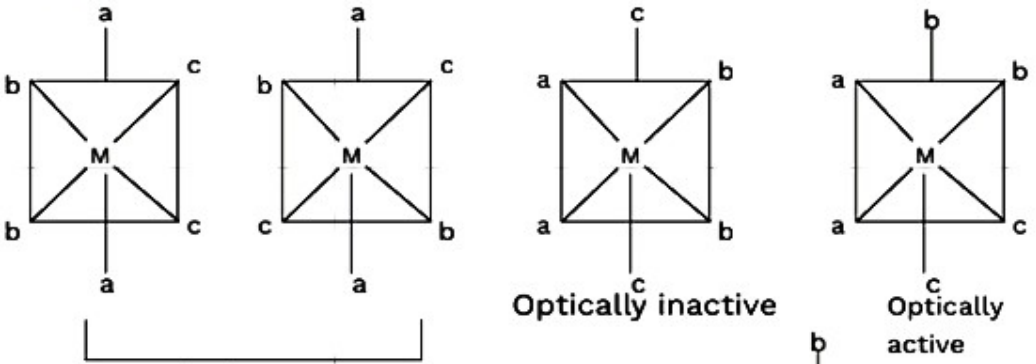
(7) $M(AF)b_2cd$ type



Optically active | Optically active | Optically active | Optically Inactive

total G.I = 1 + 2 + 2 + 2 = 7
Optical inactive = 0 + 0 + 0 + 2 = 2
Optical Active = (1 + 2 + 2) × 2 = 10
Optical isomer = 12
Sterio isomer = 12

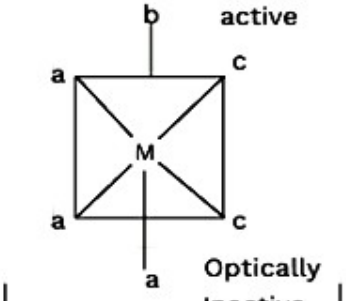
(8) $Ma_2b_2C_2$ type



(a,a) trans
Optically
Inactive
G.I.=2

Optically inactive

Optically active

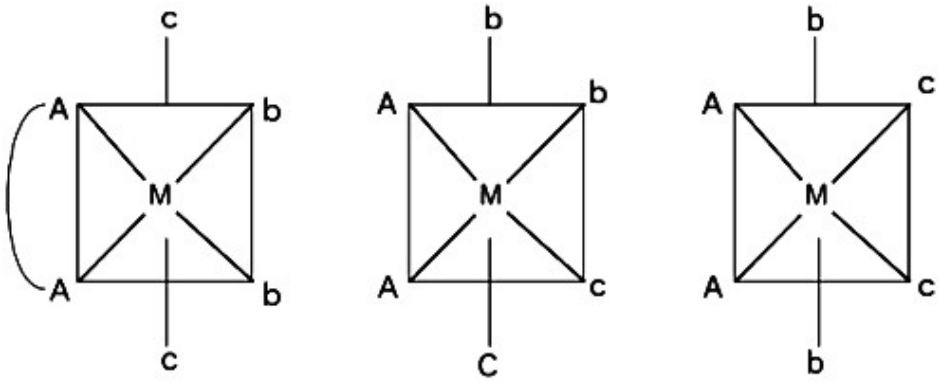


Optically
Inactive

(a, a) Cis

total G.I = 5
Optical inactive = 4
Optical Active = 1 × 2 = 2
Optical isomer = 6
Sterio isomer = 6

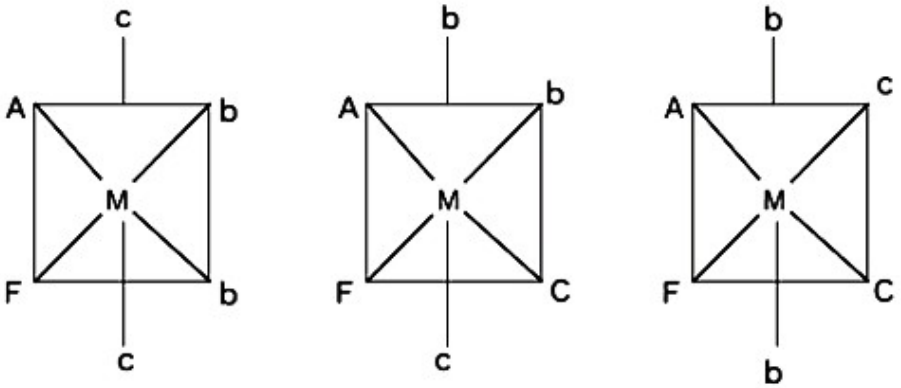
(9) $M(AA)b_2c_2$ type



Optical inactive | Optical active | Optical inactive

total G.I = 3
Optical inactive = 2
Optical Active = 1 × 2 = 2
Optical isomer = 4
Sterio isomer = 4

(10) $M(AF)b_2c_2$ type

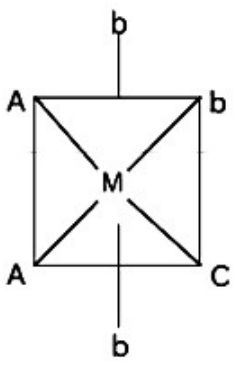


Optically Inactive | Optically active | Optically Inactive

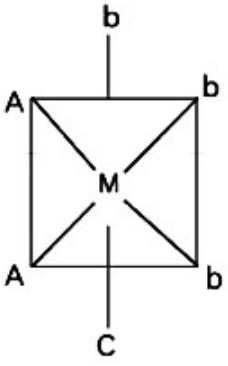
total G.I = 4
Optical inactive = 2
Optical Active = 2 × 2 = 4

Optical isomer = 6
Sterio isomer = 6

(11) M(AA)b_3C type



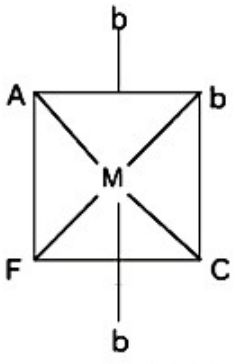
Optical active



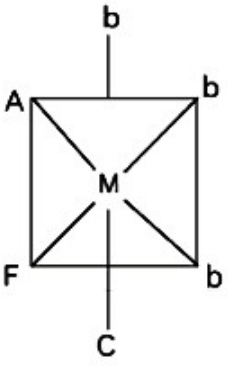
Optical inactive

total G.I = 2
Optical isomer = 0
Stereo isomer = 2

(12) M(AF)b_3C type



Optically Inactive



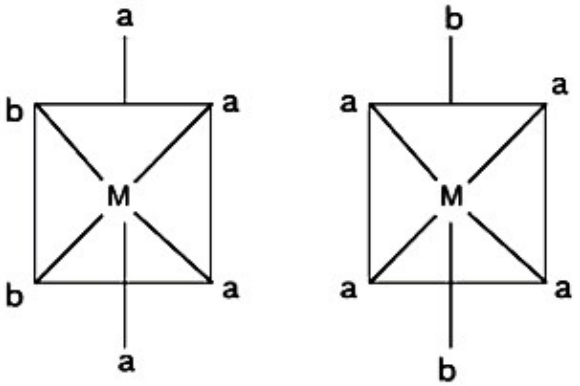
Optically active

total G.I = 3
Optical inactive = 2
Optical Active = 1 × 2 = 2
Optical isomer = 4
Stereo isomer = 4

For optical isomer at least one optically active must be present, if one atleast optically active then in optical isomer, optical active and inactive must be conside which are sterio isomer

If inactive isomer is present then optical isomer not consider and in that case geometrical isomer equal to sterio isomer.

(13) Ma_4b_2 type

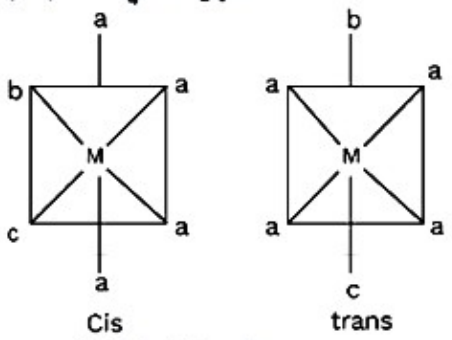


total G.I = 2

Optical isomers = 0

Stereo isomer = 2

(14) Ma_4bC type

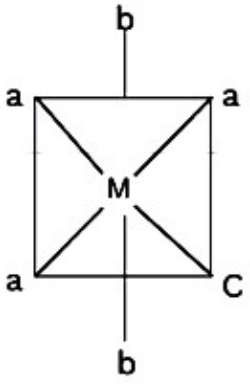


total G.I = 2

Optical isomers = 0

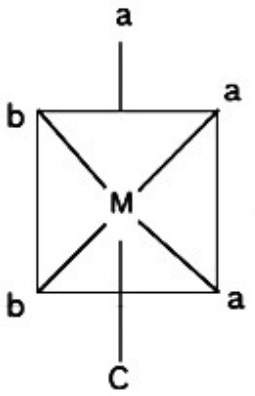
Stereo isomer = 2

(15) Ma_3b_2C type

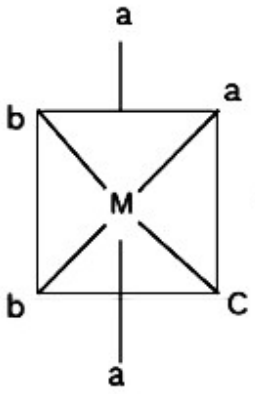


G.I=1

optical inactive



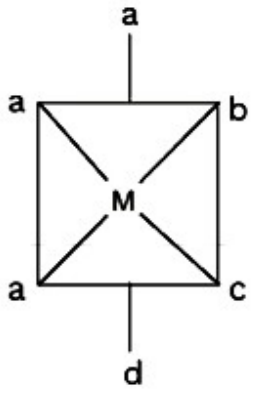
optical inactive



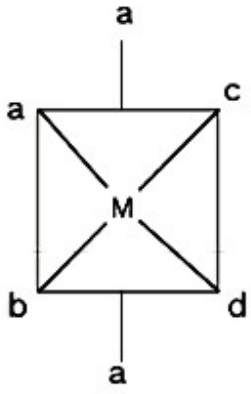
optical inactive

G.I. = 3
optical isomer = 0
Sterio isomer = 3

(16) Ma_3bcd type



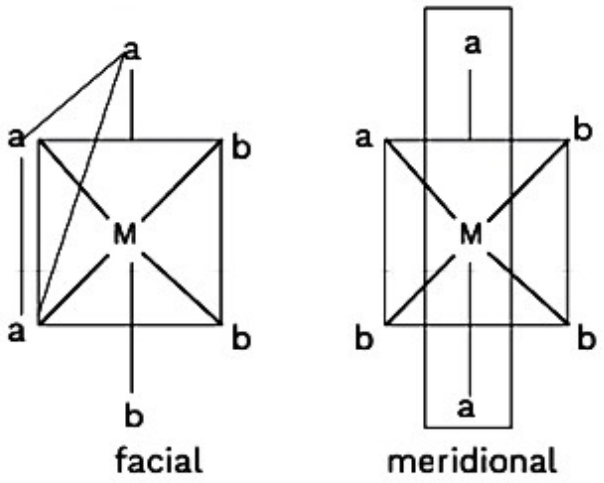
Optically active cis



Optically Inactive
G.I.=3

total G.I = 4
Optical inactive = 3
Optical Active = 1 × 2 = 2
Optical isomer = 5
Sterio isomer = 5

(17) Ma_3b_3 type



fac – mer isomers

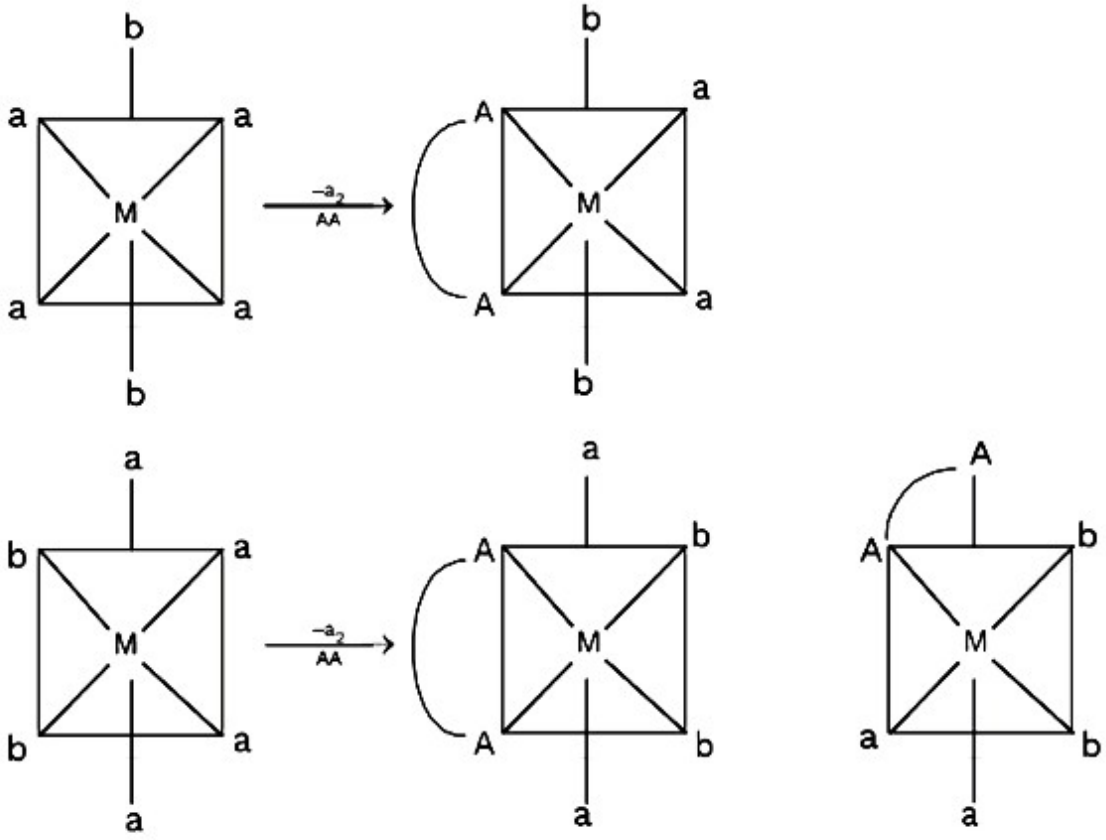
total G.I = 2

Optical isomer = 0

Sterio isomer = 2

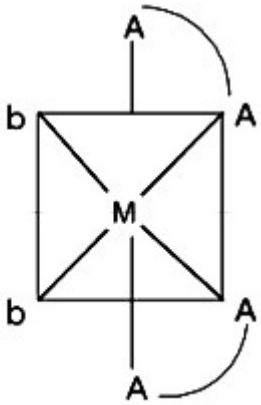
Two bidentate ligand in C.N = 6

$$Ma_4b_2 \xrightarrow[AA]{-a_2} M_{(AA)}a_2b_2$$

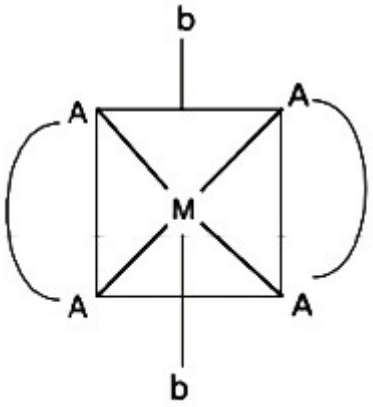


(1) $M(AA)_2b_2$ type

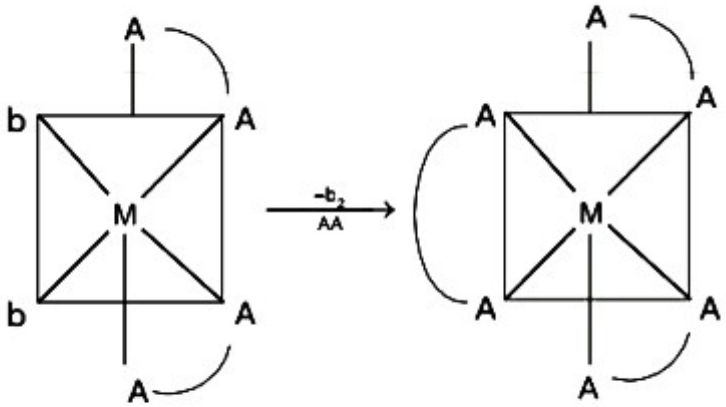
G.I. = 2
Optically inactive = 1
Optically active = $1 \times 2 = 2$
Sterolsomers = 3
Optically isomer=3

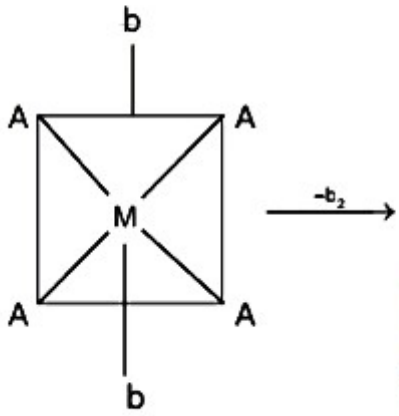


Cis



trans
optical inactive





NO

Points to remember

$M(AA)_3$ always optical active .
Three bidentate & cis isomer two bi-dentate always optical Active in C.N=6
Ma_3b_3 G.I = 3, optically isomerism = NO
$M_{(AB)_3}$ G.I = 2, optically isomerism = $2 \times 2 = 4$

Organometallic compound

When carbon is covalently bonded with metal / B / Si such compound are called organometallic compounds.

Exception
$\left.\begin{matrix} NaCN \\ CaC_2 \end{matrix}\right]$ ionic bond.

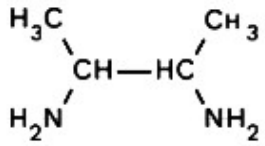
Example:

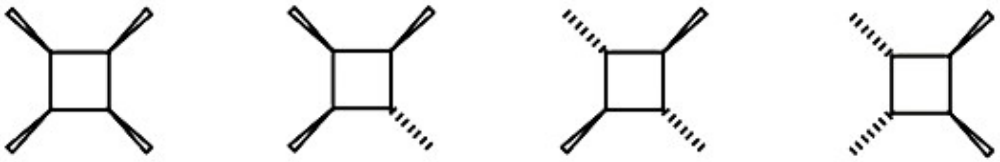
Grignard Reagent (R-mgx)

Ferrocene

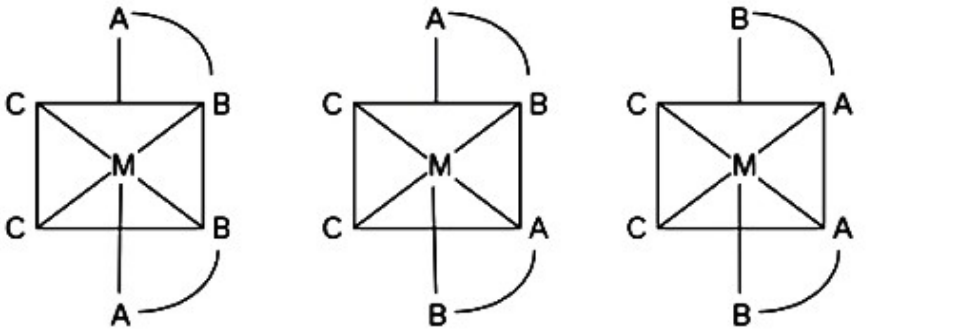
Cyano complex $\left(Ag^{+} \leftarrow CN\right)$

$\left[it(bn)_2\right]$ stereo isomer

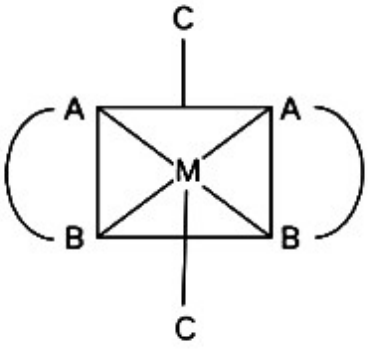
bn = 



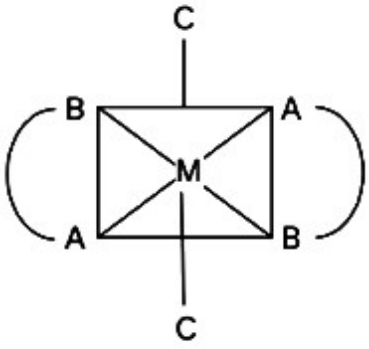
2. $M(AB)_2c_2$ type



Optically active | Optically active | Optically active



Optically Inactive

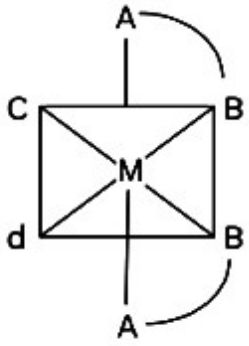


Optically Inactive

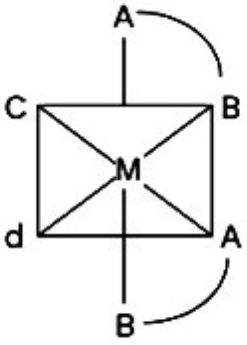
Geometrical isomer = 5
Optical Active = 6
Stereo isomer = 8

Optical inactive = 2
Optical isomer = 8

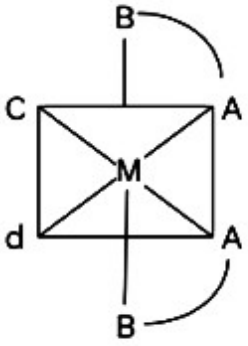
3. $M(AB)_2cd$ type



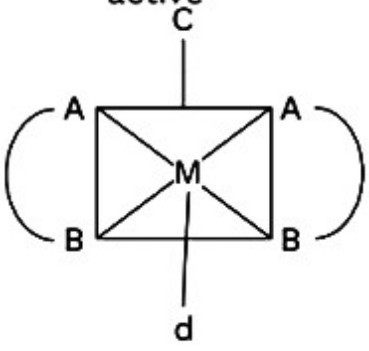
Optically active



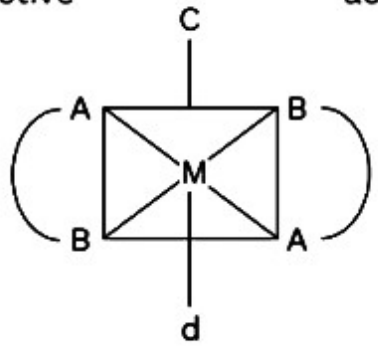
Optically active



Optically active



Optically Inactive

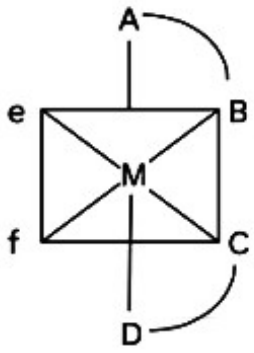


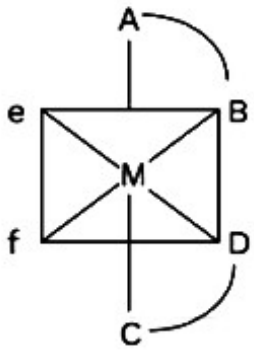
Optically active

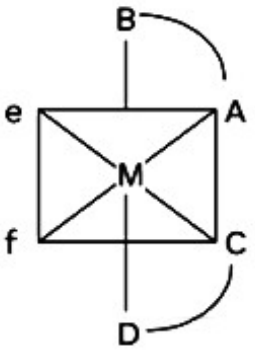
Geometrical isomer = 6
Optical Active = 5 × 2 = 10
Stereo isomer = 11

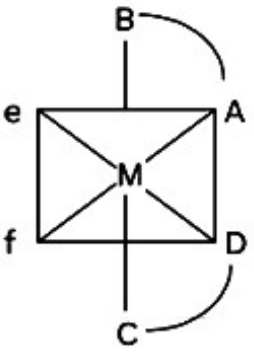
Optical inactive = 1
Optical isomer = 11

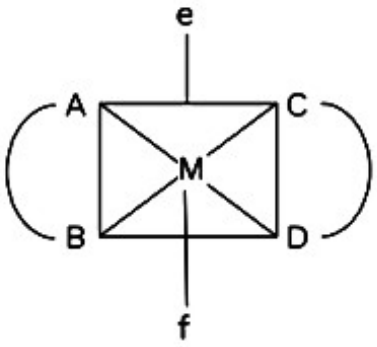
4. $M_{(AB)(CD)ef}$ type

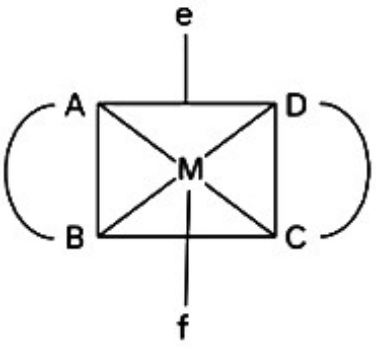








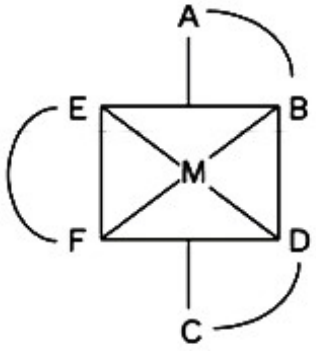




\s

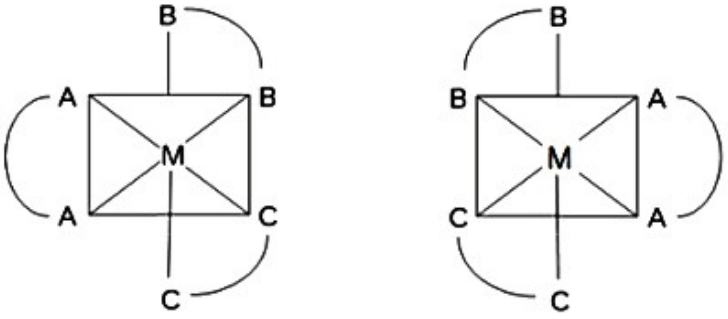
Geometrical isomer = 10
Optical isomer = 20
Stereo isomer = 20

5. M(AB)(CD)(EF) type



Geometrical = 8
Optical isomer = 16
Stereo isomer = 16

6. M(AB)(BB)(CC) type



Geometrical isomer = 0
Optical isomer = 2
Stereo isomer = 2

Application of co-ordination compound

(a) Chlorophyl is a complex of 'mg'

(b) Hardness is a complex 'Fe'

(c) Cobalt Amine $\Rightarrow$ V_{B12} complex of 'Co'

(d) To remove lead poisoning $Ca^{+2}[EDTA]^{-4}$ is used

(e) Hardness estimation $\Rightarrow Na^{+}EDTA$ used

(f) Cis-platin is used as anticancer

(g) Wilkinson catalyst $\left[Rh(pph_3)_3Cl\right]$ is used in hydrogenation of Alkenes

(h) CN^{-} us used in metallurgy in extraction of Au and Ag

(i) CO is used to form volatile $Ni(CO)_4$, used in refining of Ni by Mond's process

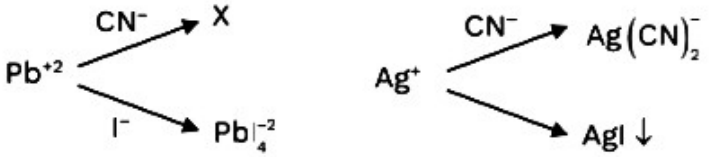
(j) Sodium nitroprusside is used in detection of sulphide ion is Basic medium.

$Na_2\left[Fe(CN)_5NO\right] + S^{-2} \longrightarrow Na_2\left[Fe(CN)_5NOS\right]$ purple colour

(k) Brown ring complex is form in detection of NO_2^{-}/NO_3^{-}

$NO_2^{-}/NO_3^{-} + M_2SO_4 + FeSO_4 \longrightarrow$ Brown Ring complex

(l) Strong field ligand prefer d-block central metal ion to form complex except Hg.
While weak field ligand prefer p-block / Hg central metal ion to form complex



(m) Fischer salt $k_3\left[Co(NO_2)_6\right]$ yellow ppt produced in detection of potassium ligand

MLCT (Metal to ligand Charge Tranfer)
Synergy $\Rightarrow \pi$ Acceptor ligand
CO, NO, $NO^+ \Rightarrow \pi$ donor

Priority order of Colour

1. MLCT
2. $d^5 \Rightarrow$ WFL, d^{10}, $d^0 \Rightarrow$ LMCT
   $d^{1\text{-}9} \Rightarrow$ (except d^5 – WFL) $\Rightarrow$ d-d-transition

Example:
Brown ring Complex

$$\left[Fe(H_2O)_5\,NO\right]SO_4 \Rightarrow \text{MLCT}$$

$$\left[Fe(SCN)_6\right]^{-3}$$

Blood red colour $\Rightarrow$ LMCT
(d^5 – WFL)